

THE FRAMEWORK 1.0

*A Lexicon of Biological Organization:
Aging, Regeneration, and Oncogenesis*

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2026

Preface

The starting point for this monograph was the work of Sui Huang, who described cellular states as stable attractors within high-dimensional regulatory networks. This framework provided a powerful conceptual basis for understanding cellular identity, phenotypic plasticity, and state transitions. It maps the topology of biological possibility — the landscape of states a cell can occupy and the transitions that connect them.

Yet an important question remains. Regulatory models can describe an expanding space of possible trajectories, but they do not always explain why a biological system follows one particular path rather than another. This limitation becomes especially apparent in complex phenomena such as aging, regeneration, adaptation, and oncogenesis, where mechanical, energetic, structural, metabolic, and informational processes interact simultaneously across multiple scales of organization.

The central premise of this monograph is that the accessibility of biological states is determined not only by regulatory architecture, but also by the physical and energetic conditions under which that architecture operates. Cellular and tissue behavior emerges within a framework of constraints imposed by hydrogel organization, energy flux distribution, metabolic resource availability, and the structural properties of living matter. These factors influence which regions of the theoretically accessible state space can actually be reached, maintained, or abandoned.

The Framework presented here does not replace existing biological theories. It introduces a higher-level conceptual structure capable of linking observations from cell biology, tissue mechanics, metabolism, aging research, and pathology within a common analytical framework. Its purpose is to identify recurring organizational patterns, generate testable hypotheses, and provide a coherent language for describing biological dynamics across scales.

At its present stage, the Framework is primarily a conceptual and logical system rather than a completed mathematical theory. Many of its components are working hypotheses — formulated with sufficient precision to be testable, but not yet fully quantified. The underlying principles, however, describe real organizational features of living systems, and future experimental work is expected to allow their progressive formalization.

This monograph contributes to the development of a unified language for biological organization and lays a conceptual foundation for quantitative and probabilistic models of developmental, regenerative, aging, and pathological trajectories.

CHAPTER 1

THE LIVING HYDROGEL

Preface

Classical biology describes the cell as a biochemical reactor — membrane, enzymes, signalling cascades, genes. This description is accurate and productive. But it leaves unanswered one question that seems simple only at first glance: why does the cell not destroy itself spontaneously?

Any molecular system left to its own devices moves toward equilibrium. Proteins aggregate. Polymers condense. Water redistributes. Structure degrades. This is not pathology — it is thermodynamics. And this is precisely where the biochemical description of the cell encounters its limitation: it explains what happens inside the cell, but does not explain why the cell as a physical object does not continuously collapse.

The answer proposed by this chapter is built on three linked propositions.

First. The type of cell death is determined by the hydration state of the cell at the moment of energetic collapse. This is a non-trivial physical dependency: most biochemical descriptions of cell death treat apoptosis and necrosis as different molecular programmes. Here they appear as two physical trajectories of collapse of the same system — a hydrogel with different water content. This reframes experimental questions and opens new metrics for clinical assessment.

Second. The relationship between energy and cellular degradation is formalised not through the concept of 'stress', but through the ratio of two rates: the rate of active renewal of the hydrogel network and the rate of its spontaneous syneresis. While the first exceeds the second — the cell lives. When the balance is disrupted — degradation begins. This is an operational framework: it specifies what to measure. Not simply 'is ATP present', but whether the energy flux is sufficient to maintain structural equilibrium against thermodynamic relaxation.

Third. Syneresis — the spontaneous compaction of a hydrogel with expulsion of water — is not a metaphor, but a physical process with known thermodynamics. Protein aggregation, cytoskeletal condensation, loss of plasticity, fibrosis — phenomena that molecular biology describes with different words and explains through different mechanisms — may physically be the same process occurring in different molecular substrates. Syneresis provides a unified language for their description and a unified thermodynamic logic for identifying connections between them.

This chapter does not have to be read in full to proceed further. But knowing what it contains is essential: the remaining chapters of the monograph use these three dependencies as the physical foundation for analysing oncogenesis, metabolic disease, ageing, and neurodegeneration. Wherever in the subsequent chapters the word 'degradation' appears — behind it stands syneresis. Wherever 'energy deficit' is discussed — behind it stands a disruption of the renewal balance. Wherever a type of cell death is described — behind it stands the hydration state of the hydrogel at the moment of collapse.

Article 1 of 5

Cytoplasmic Gel, Membrane Mechanics, and Physical Modes of Cell Death

A Revised Conceptual Model of Cellular Mechanical Stability

Yuri Dmitrochenko

A Conceptual Biophysical Framework

This paper is part of a conceptual series on hydrogel physics and cellular dynamics.

Abstract

This paper examines the hypothesis that cellular mechanical stability is determined not only by biochemical mechanisms, but by the physical state of the cytoplasmic gel, the membrane-cytoskeletal composite system, and the surrounding extracellular environment.

Using the Laplace equation with physically correct membrane surface tension values — rather than the surface tension of bulk water — it is shown that the erythrocyte membrane generates a characteristic pressure on the order of 100–600 Pa, consistent with experimentally measured micropipette aspiration data. This revised estimate resolves the apparent paradox of the biconcave erythrocyte shape and confirms that shape maintenance cannot be attributed to hydrostatic pressure alone.

It is proposed that cellular shape stability is provided by a distributed viscoelastic composite system: membrane + spectrin cytoskeleton + hydrated cytoplasmic gel. The relationship between cytoplasmic gel stiffness, thermal fluctuations, and the mode of cell death is examined. It is proposed that cytolysis and apoptosis-like degradation represent distinct physical regimes of hydrogel collapse, consistent with the two-trajectory framework developed in the companion papers of this series.

Keywords: *cytoplasmic gel; membrane mechanics; Laplace pressure; viscoelasticity; erythrocyte; thermal fluctuations; syneresis; cell death; hydrogel; spectrin cytoskeleton.*

1. Introduction

Traditionally, the cell is regarded as an aqueous system enclosed within a lipid membrane — a biochemical reactor bounded by a lipid bilayer. However, this model poorly explains several physical properties of living cells: stable non-spherical shape maintenance, capacity for large deformation without rupture, low measured intracellular pressure, resistance to thermal fluctuations, and the existence of controlled, morphologically distinct forms of cell death [1,2].

The erythrocyte is a particularly convenient model for studying these physical properties. It contains no nucleus, exhibits pronounced deformability, maintains a complex biconcave shape under continuous mechanical stress in the circulation, and has a well-characterised membrane-cytoskeletal architecture [3,4]. Its minimal intracellular complexity — essentially a concentrated haemoglobin gel enclosed by a spectrin-actin network anchored to a lipid bilayer — makes it the most experimentally accessible model of cellular mechanical stability.

This paper proposes viewing the cell as a viscoelastic hydrated composite system whose stability is governed by the balance of membrane mechanics, thermal fluctuations, and internal gel organisation. A key aim is to correct a common error in qualitative biophysical treatments: the use of bulk water surface tension in the Laplace equation applied to biological membranes. This correction changes the quantitative estimates by approximately two orders of magnitude and substantially alters the physical interpretation.

This paper forms the physical foundation of a series of conceptual papers developing the hydrogel framework for cellular biology, aging, and genetic disease [5–8].

2. The Laplace Equation and Membrane Mechanics: A Corrected Treatment

2.1 The Laplace Equation

For a spherical surface, the pressure difference across an interface due to surface or membrane tension is described by the Laplace equation:

$$\Delta P = 2T / R$$

where ΔP is the excess pressure, T is the surface or membrane tension, and R is the radius of curvature. For a non-spherical surface with two principal radii of curvature R_1 and R_2 , the Young-Laplace equation generalises to:

$$\Delta P = T \cdot (1/R_1 + 1/R_2)$$

2.2 The Critical Distinction: Membrane Tension vs. Water Surface Tension

A common error in qualitative treatments is the use of the surface tension of bulk water ($\sigma_{\text{water}} \approx 0.072 \text{ N/m}$) as the tension parameter in the Laplace equation applied to biological membranes. This is physically incorrect for two reasons.

First, the plasma membrane is not a water-air interface. It is a lipid bilayer — a two-dimensional fluid with mechanical properties governed by bending rigidity and in-plane tension, not by the cohesive forces of bulk water. Second, the relevant physical parameter for membrane mechanics is the membrane tension T_m , which has units of N/m but reflects the in-plane stress of the bilayer, not the surface energy of a water droplet.

Experimentally measured membrane tension values for the erythrocyte are [3,9,10]:

- Resting (unstressed) membrane tension: $T_m \approx 1\text{--}5 \times 10^{-5} \text{ N/m}$
- At the lytic threshold (micropipette aspiration): $T_{m,\text{lysis}} \approx 5\text{--}10 \times 10^{-3} \text{ N/m}$
- Typical physiological range during deformation: $T_m \approx 10^{-4} \text{ to } 10^{-3} \text{ N/m}$

These values are 1–4 orders of magnitude smaller than the surface tension of bulk water.

2.3 Corrected Laplace Pressure Estimates for the Erythrocyte

Erythrocyte dimensions: diameter 7–8 μm , radius $R \approx 3.5 \times 10^{-6} \text{ m}$. Using the corrected membrane tension values:

At resting membrane tension ($T_m \approx 3 \times 10^{-5} \text{ N/m}$): $\Delta P = 2 \times (3 \times 10^{-5}) / (3.5 \times 10^{-6}) \approx 17 \text{ Pa}$

At physiological deformation tension ($T_m \approx 5 \times 10^{-4} \text{ N/m}$): $\Delta P = 2 \times (5 \times 10^{-4}) / (3.5 \times 10^{-6}) \approx 286 \text{ Pa}$

At lytic threshold ($T_m \approx 7 \times 10^{-3} \text{ N/m}$): $\Delta P = 2 \times (7 \times 10^{-3}) / (3.5 \times 10^{-6}) \approx 4,000 \text{ Pa} \approx 4 \text{ kPa}$

These estimates — 17 Pa to 4 kPa depending on membrane state — are consistent with experimentally measured intracellular pressures in erythrocytes and other cell types [9,10]. They are approximately two orders of magnitude lower than the erroneous estimate of $\sim 41 \text{ kPa}$ that results from using bulk water surface tension.

3. The Erythrocyte Shape Problem

If the erythrocyte were governed solely by membrane tension minimisation — as a liquid droplet is governed by surface tension — it would adopt a spherical shape, since the sphere minimises surface area for a given enclosed volume. Yet the erythrocyte maintains its characteristic biconcave discoid form, with an excess membrane surface area of approximately 40% over the minimum required to enclose its volume as a sphere [3,4].

This paradox is resolved not by invoking anomalously high intracellular pressure, but by recognising that the erythrocyte is not a liquid droplet. It is a pre-stressed composite mechanical system:

Shape stability = membrane bending rigidity + spectrin network elasticity + cytoplasmic gel viscoelasticity

The spectrin-actin network, anchored to the lipid bilayer through ankyrin and band 4.1 protein complexes, provides an elastic restoring force that resists membrane deformation and maintains the biconcave geometry [3,11]. The cytoplasmic haemoglobin gel provides viscous damping that resists rapid deformation and contributes to the overall viscoelastic response.

The 40% excess surface area — the geometric reserve — is not a passive structural surplus. It is an active physical buffer that allows the erythrocyte to deform reversibly as it traverses capillaries narrower than its resting diameter, without generating membrane tensions approaching the lytic threshold [4,12].

4. Membrane Bending Rigidity and the Helfrich Framework

The resistance of the erythrocyte membrane to shape change is governed not primarily by in-plane tension but by bending rigidity κ — the energy cost of membrane curvature. The Helfrich elastic energy of a membrane is [13]:

$$F = \int \left[\frac{1}{2} \kappa (c_1 + c_2 - c_0)^2 + \kappa G \cdot c_1 c_2 \right] dA$$

where c_1 and c_2 are the principal curvatures, c_0 is the spontaneous curvature, κ is the bending modulus, and κG is the Gaussian bending modulus. For the erythrocyte membrane [3,13]:

— Bending modulus: $\kappa \approx 2 \times 10^{-19} \text{ J} \approx 50 \text{ kBT}$

— Shear modulus of spectrin network: $\mu \approx 2\text{--}10 \text{ }\mu\text{N/m}$

— Area compressibility modulus: $KA \approx 450 \text{ mN/m}$

The bending modulus of $\sim 50 \text{ kBT}$ indicates that the membrane is stiff enough to maintain a defined shape against thermal fluctuations (which have energy $\sim kBT$ per mode) but flexible enough to allow the reversible deformations required for capillary transit.

5. Estimation of Cytoplasmic Gel Stiffness

The cytoplasm of the erythrocyte consists of a concentrated haemoglobin solution at approximately 340 g/L — sufficiently concentrated to exhibit gel-like viscoelastic properties rather than simple liquid behaviour [14]. The effective stiffness of this gel contributes to the overall mechanical response of the cell.

From micropipette aspiration experiments, the effective elastic modulus of the erythrocyte cortex is [9,10]:

- Elastic modulus of membrane-cytoskeleton composite: $E \approx 1\text{--}20\text{ Pa}$ (at low deformation rates)
- Effective viscosity of cytoplasm: $\eta \approx 5\text{--}10\text{ mPa}\cdot\text{s}$ (approximately $5\text{--}10\times$ that of water)
- Relaxation time: $\tau \approx 0.1\text{--}1\text{ s}$ (viscoelastic timescale)

These values confirm that the erythrocyte cytoplasm behaves as a soft viscoelastic material — neither a simple liquid nor a rigid solid — consistent with the corrected membrane tension calculations of Section 2.

6. Thermal Fluctuations and Nanostructural Stability

At physiological temperature $T = 310\text{ K}$ (37°C), the thermal energy is:

$$kBT \approx 1.38 \times 10^{-23} \times 310 \approx 4.3 \times 10^{-21}\text{ J} \approx 4.3\text{ zJ}$$

The critical gel stiffness at which thermal fluctuations begin to disrupt structural organisation can be estimated from:

$$E_{\text{crit}} \sim kBT / a^3$$

For structural elements at the scale of spectrin repeats ($a \approx 5\text{ nm}$): $E_{\text{crit}} \sim 34\text{ kPa}$. At cytoskeletal mesh scale ($a \approx 70\text{ nm}$): $E_{\text{crit}} \sim 13\text{ Pa}$. The cytoskeletal network therefore operates near the thermal stability boundary — flexible enough to deform but stiff enough to maintain organisation against Brownian motion.

7. Two Physical Regimes of Cell Death

7.1 Hydropic-Solution Collapse (Swelling Trajectory — Oncosis/Necrosis)

When ATP-dependent ion pump function fails acutely, sodium accumulates intracellularly and osmotic water influx initiates. The cytoplasmic gel undergoes progressive solation: dilution of the polymer network, reduction of viscosity and elasticity, and dissociation of cytoskeletal organisation [5]. Cell volume increases, consuming the geometric reserve; membrane tension rises toward the lytic threshold; at $\Delta P_{\text{lysis}} \approx 3\text{--}6\text{ kPa}$ membrane rupture becomes inevitable. This corresponds to oncosis, cytotoxic oedema, osmotic haemolysis, and necrosis.

7.2 Contractile-Syneretic Collapse (Condensation Trajectory — Apoptosis/Eryptosis)

When energetic failure is gradual and apoptotic signalling activates before the osmotic threshold is crossed, the cytoplasmic gel undergoes syneretic condensation: contraction of the polymer network, extrusion of water, reduction of cellular volume, and progressive densification [6,7]. Cell volume decreases; membrane tension remains low; caspase-mediated cytoskeletal cleavage produces controlled disassembly; apoptotic bodies form. The absence of membrane rupture is the physical basis of the non-inflammatory character of apoptotic death.

7.3 The Kinetic Determinant

The trajectory taken by a cell under energetic stress is determined by the kinetics of ATP depletion relative to the kinetics of apoptotic programme activation [5]. This is not a biological choice but a physical consequence of competing kinetics.

8. Role of the Extracellular Environment

In oedematous tissue, increased free water content reduces the viscosity and elastic modulus of the extracellular space, increasing the probability of osmotic swelling and cytolysis. A structured, viscous extracellular environment — a dense proteoglycan-rich ECM — provides mechanical confinement that may favour the synergetic trajectory. Within the hydrogel framework, the extracellular matrix itself is a hydrogel system subject to syneresis [7,8].

9. Conceptual Synthesis

The revised physical model treats cellular mechanical stability as the result of interactions among five physical elements: (1) membrane bending rigidity ($\kappa \approx 50$ kBT); (2) spectrin network elasticity ($\mu \approx 2\text{--}10$ $\mu\text{N/m}$) with geometric reserve of $\sim 40\%$ excess area; (3) cytoplasmic gel viscoelasticity ($E \approx 1\text{--}20$ Pa; $\eta \approx 5\text{--}10$ mPa·s); (4) thermal fluctuations ($k\text{BT} \approx 4.3$ zJ); and (5) extracellular mechanical environment.

Within this framework, cytolysis corresponds to membrane tension reaching the lytic threshold through osmotic swelling — a solation trajectory. Apoptosis/eryptosis corresponds to cytoplasmic gel syneresis — a condensation trajectory. The two trajectories are not different biological programmes but different physical outcomes of the same energetic failure.

10. Conclusion

This paper has presented a revised physical model of cellular mechanical stability, correcting the common error of using bulk water surface tension in the Laplace equation applied to biological membranes. The corrected estimates — membrane tension 10^{-5} to 10^{-3} N/m, corresponding Laplace pressures of 17 Pa to 4 kPa — are consistent with experimental micropipette aspiration data and resolve the erythrocyte shape paradox.

Cell death under energetic stress follows two physically distinct trajectories: hydroptic-solation collapse when ATP depletion is rapid, and contractile-synergetic collapse when depletion is gradual and apoptotic signalling activates before the osmotic threshold is crossed.

Conflict of Interest: The author declares no conflict of interest.

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Article 2 of 5

The Cell as a Self-Remodelling Hydrogel System

Energy-Dependent Polymer Organisation, Cytoplasmic Remodelling, and Anti-Synergetic Stability

Yuri Dmitrochenko

A Conceptual Biophysical Framework

This paper is part of a conceptual series on hydrogel physics and cellular dynamics.

Abstract

This conceptual framework proposes that the living cell should be interpreted not primarily as a biochemical solution, but as a dynamically maintained hydrogel-like polymer system sustained by continuous energy-dependent remodelling [1,2]. The central hypothesis is that cellular stability depends on coordinated cycles of polymer assembly, partial depolymerisation, hydration control, viscosity regulation, and energetic redistribution [3,4].

Within this framework, intracellular remodelling continuously counteracts gel rigidification, dehydration, polymer collapse, and syneresis-like condensation of the cytoplasmic hydrogel [5,6]. The model proposes that glycogen, glycans, proteins, proteoglycans, cytoskeletal polymers, membranes, and structured water form interconnected anti-synergetic remodelling loops that maintain the dynamic physical state of the cell [7–16].

Cellular aging is interpreted as progressive failure of anti-synergetic remodelling systems [17,18]. Apoptosis represents collapse of energy-supported hydrogel maintenance [19,20]. Regeneration and high-plasticity states represent temporary expansion of anti-synergetic remodelling capacity [21,22].

Keywords: *cytoplasm; hydrogel; syneresis; polymer remodelling; viscoelasticity; cellular aging; macromolecular crowding; anti-synergetic maintenance; Basal Glycolytic Regime; dissipative structure.*

1. Introduction

Classical biology typically interprets the cell as a biochemical reactor or a membrane-bound molecular system. However, modern biophysics has demonstrated that cytoplasm behaves as a crowded, viscoelastic, hydrogel-like, non-equilibrium polymer environment [1,2,3]. Seminal studies by Luby-Phelps established that intracellular diffusion is profoundly restricted compared to aqueous solutions, consistent with a densely packed polymer network [4]. Subsequent work on macromolecular crowding revealed that the high concentration of macromolecules fundamentally alters reaction rates, protein folding, and phase behaviour [5,6].

This framework extends these observations and proposes that life depends on continuous energy-supported anti-synergetic remodelling of intracellular polymer structures. Within this view, metabolism, rheology, diffusion, hydration, and intracellular structure become inseparable aspects of a single dynamic physical process opposing spontaneous gel collapse. The Basal Glycolytic Regime (BGR) — defined as the metabolic state characterised by high glycolytic flux, low reactive oxygen species burden, and maximal generation of biosynthetic precursors for structural renewal — is proposed as the metabolic correlate of maximal anti-synergetic maintenance capacity [26,27].

2. Cytoplasm as a Dynamic Hydrogel

In this model, cytoplasm is interpreted not as a dilute solution, but as a hydrated polymer network composed of glycogen, glycans, proteins, proteoglycans, cytoskeletal polymers, lipid membranes, structured water, ions, and transient polymeric intermediates [1,2,3,7,8]. These components collectively form a viscoelastic, dynamically reorganised, partially phase-separated hydrogel system [1,29,30].

The cytoskeletal network — including actin, microtubules, and intermediate filaments — provides mechanical scaffolding whose dynamics are governed by energy-dependent assembly and disassembly [9,10,11]. A significant fraction of intracellular water exhibits restricted mobility, consistent with organisation around macromolecular surfaces [15].

Particle-tracking microrheology and atomic force microscopy studies have confirmed that cytoplasmic stiffness is actively regulated and depends on metabolic state [31,32]. Cells depleted of ATP show progressive cytoplasmic stiffening — a direct physical demonstration of synergetic drift in the absence of active remodelling [31].

3. Syneresis as a Fundamental Physical Threat

In polymer physics, syneresis refers to the spontaneous contraction of a gel, accompanied by water expulsion, increased density, and progressive rigidification [16,17]. The classical

theoretical framework for gel collapse was established by Tanaka, who demonstrated that polymer networks can undergo discontinuous volume phase transitions [16]. The Flory–Rehner theory of swollen polymer networks provides the thermodynamic basis for understanding how polymer–solvent interactions determine hydration state and gel stability [17].

Syneresis represents the natural thermodynamic tendency of the intracellular gel toward densification, lower mobility, and reduced dynamic organisation. The living cell continuously opposes this tendency through energy-dependent remodelling — which is the central proposition of this framework.

4. Remodelling as an Anti-Syneretic Process

The central hypothesis is that cellular remodelling continuously counteracts intracellular syneresis. Polymer turnover prevents rigidification, redistributes water, restores diffusion dynamics, maintains Brownian mobility, and preserves structural plasticity. The diffusion coefficient of intracellular macromolecules is governed by the Stokes–Einstein relation:

$$D = kBT / (6\pi\eta r)$$

where η is cytoplasmic viscosity and r is the hydrodynamic radius of the diffusing species [4,15]. Remodelling of the polymer network directly regulates η , thereby controlling diffusion accessibility and molecular interaction rates.

A critical thermodynamic principle underlies this framework: polymer assembly requires energy, whereas depolymerisation is energetically favourable or requires substantially less energy. Without continuous energy input, depolymerisation, rigidification, and gel collapse progressively dominate [33].

5. The Glycogen–Glycan Anti-Syneretic Loop

Within the present framework, glycogen acts not only as an energy reserve but also as a hydration reservoir, osmotic stabiliser, rheological regulator, and structural component of the cytoplasmic gel [7,8]. The highly branched architecture of normal glycogen creates a water-retentive polymer network: each granule contains approximately 45,000 glucose residues organised into a spherical structure with a fractal branching pattern that maximises surface area and hydration [7]. Reduction in branching, as occurs in glycogen storage diseases, increases the linear character of the polymer, promoting inter-chain associations, reduced solubility, and condensation [24].

6. Coordinated Anti-Syneretic Remodelling Systems

The cell contains multiple interconnected anti-syneretic loops: glycogen–glycan remodelling (hydration and osmotic regulation [7,8,9]); protein turnover/proteostasis (prevention of aggregation and rigidification [18,19]); cytoskeletal remodelling (mechanical plasticity [10,11,29,30]); membrane dynamics (compartmental organisation [20,21]); proteoglycan turnover (ECM hydration [34,35]); and water redistribution (hydration gradient maintenance [12,13,14,15]).

Hipp et al. demonstrated that age-related decline in proteostasis capacity leads to progressive protein aggregation, consistent with syneretic condensation of the proteinaceous component of the cytoplasmic gel [19].

7. The Basal Glycolytic Regime as Anti-Syneretic Maintenance State

The Basal Glycolytic Regime (BGR) is defined here as the metabolic state characterised by: high glycolytic flux providing ATP for ion pump function and biosynthetic precursor generation; low reactive oxygen species burden protecting structural macromolecules; and generation of nucleotide sugars (UDP-glucose, UDP-GlcNAc) supporting continuous glycan and glycoprotein biosynthesis [26,27].

The spectrum of cellular energy states maps onto anti-syneretic capacity: BGR dominance corresponds to maximal hydrogel self-renewal — characteristic of embryonic cells, stem cells, and activated progenitors. Progressive BGR decline corresponds to progressive dominance of syneresis over remodelling — characteristic of aging, chronic stress, and the preclinical phase of degenerative disease [26,27].

8. Aging as Progressive Syneretic Drift

Within this framework, aging represents the progressive failure of anti-syneretic remodelling. The hallmarks of aging described by Lopez-Otin et al. — including genomic instability, epigenetic alterations, loss of proteostasis, and cellular senescence — may all be interpreted as manifestations of declining anti-syneretic remodelling capacity [17]. Walther et al. demonstrated widespread protein aggregation in ageing *C. elegans*, consistent with progressive syneretic condensation of the cytoplasmic polymer network [36].

9. Apoptosis as Hydrogel Collapse

Apoptosis represents not simply activation of a genetic programme, but the terminal collapse of anti-syneretic maintenance along the contractile-syneretic trajectory [23]. The morphological changes associated with apoptosis — cell shrinkage, membrane blebbing, chromatin condensation, and cytoplasmic fragmentation — are consistent with the physical consequences of gel syneresis [19,20]. Kroemer et al. demonstrated that mitochondrial membrane permeabilisation plays a central role in apoptosis [20]; within the present framework, this event represents the critical failure point of energy supply to anti-syneretic maintenance.

10. Regeneration and High-Plasticity States

Regeneration may represent a temporary expansion of anti-syneretic remodelling capacity. Highly regenerative organisms maintain populations of pluripotent cells with exceptional cytoplasmic plasticity [21], including increased glycogen turnover, cytoplasmic softening, elevated hydration, and enhanced diffusion. Within the BGR framework, regenerative states correspond to re-establishment of BGR dominance in cells or tissues that had partially drifted toward BGR decline [26,27].

11. Discussion

The hydrogel framework proposed here integrates observations from polymer physics, cell biophysics, and molecular cell biology into a unified physical model of cellular homeostasis. The framework connects naturally to the liquid-liquid phase separation literature [38,39]. The dynamic condensates described by Hyman, Brangwynne, and colleagues — stress granules, P-bodies, nuclear speckles — may be understood as locally regulated syneretic events: controlled, reversible phase transitions within the cytoplasmic hydrogel that serve functional purposes. Disease-associated mutations that convert these reversible condensates into irreversible aggregates represent a failure of the controlled syneresis mechanism.

12. Conclusion

The living cell may be interpreted as an energy-supported anti-synergetic self-remodelling hydrogel system whose survival depends on continuous coordinated cycles of polymer renewal, hydration maintenance, and structural reorganisation. The Basal Glycolytic Regime provides the metabolic framework for anti-synergetic maintenance capacity; its progressive decline maps onto the progressive synergetic drift of ageing and disease.

Conflict of Interest: The author declares no conflict of interest.

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The Active Intracellular Hydrogel: Syneresis, Gel Collapse, and Membrane Failure as Two Physically Distinct Levels of Cell Death

The Erythrocyte as a Minimal Model of a Self-Renewing Biological Hydrogel

Yuri Dmitrochenko

A Conceptual Biophysical Framework

This paper is part of a conceptual series on hydrogel physics and cellular dynamics.

Abstract

Cellular death from energy deficit is conventionally described in biochemical terms. This paper proposes a complementary physical framework in which the living cell is interpreted as an actively maintained metastable hydrogel system — a nonequilibrium dissipative structure whose organisation is continuously opposed to thermodynamic relaxation through energy-dependent self-renewal. Cellular life is the continuous maintenance of this self-renewal; cellular death is its irreversible failure.

Three propositions structure the framework. First, the intracellular hydrogel and the plasma membrane represent physically distinct systems that can fail independently. Second, hydrogel destabilisation following energy deficit follows two physically distinct trajectories — hydropic-solation collapse and contractile-syneretic collapse — which correspond to necrosis and apoptosis respectively. Third, the erythrocyte is proposed as the minimal model of a self-renewing biological hydrogel, maintained by a two-component self-renewal system: ATP-dependent ion pump activity and residual mRNA-dependent structural protein synthesis.

Keywords: *intracellular hydrogel; syneresis; gel collapse; membrane failure; erythrocyte; self-renewal; residual mRNA; spectrin; osmotic cytolysis; eryptosis; apoptosis; oncosis; dissipative structure; Basal Glycolytic Regime.*

1. Introduction: The Cell as a Self-Renewing Hydrogel

Classical descriptions of cellular biology represent the cell as a membrane-bound biochemical system. The intracellular environment is not an aqueous solution. It is a densely organised, viscoelastic hydrogel composed of cytoskeletal polymer networks, protein complexes, membraneless organelles formed by phase separation, bound water layers, ion gradients, and active transport systems.

This hydrogel exists far from thermodynamic equilibrium and tends spontaneously toward syneresis — progressive condensation, redistribution of water, reorganisation of the polymer network, and loss of original structure through thermodynamic relaxation. The living cell differs fundamentally: its intracellular hydrogel undergoes continuous energy-dependent self-renewal that continuously opposes thermodynamic relaxation.

Central proposition: Cellular life is the continuous energy-dependent self-renewal of the active intracellular hydrogel. Cellular death is the irreversible failure of that self-renewal. The plasma membrane, while essential, is a boundary condition of this system — not its organisational core.

2. The Erythrocyte as a Minimal Model of a Self-Renewing Hydrogel

2.1 The Classical View and Its Incompleteness

The mature erythrocyte is traditionally described as the paradigmatic example of a cell that has surrendered self-renewal capacity: it is anucleate, lacks mitochondria, cannot undergo apoptosis in the classical sense, and depends entirely on glycolysis for ATP production. Its lifespan of approximately 120 days is described as a simple process of progressive membrane deterioration and eventual phagocytic removal.

This description, while largely correct, is incomplete in a way that is physically significant. The mature erythrocyte is not entirely devoid of protein synthesis capacity. Residual messenger RNA and ribosomal components persist in circulating erythrocytes beyond the reticulocyte stage and support ongoing, low-level translational activity.

2.2 Residual Translational Activity: The Second Component of Erythrocyte Self-Renewal

Residual ribosomal RNA and messenger RNA molecules — including globin mRNA, spectrin mRNA, ankyrin mRNA, and band 3 protein mRNA — persist in the earliest circulating erythrocytes and continue to support translational activity at levels that, while far below those of nucleated cells, are biologically non-trivial. The erythrocyte retains a limited capacity to synthesise the structural proteins of its membrane-cytoskeletal complex: spectrin (alpha and beta chains), ankyrin, band 4.1, and to some degree globin. Their ongoing synthesis represents a genuine self-renewal process: the gradual replacement of oxidatively damaged or mechanically fatigued structural proteins.

2.3 The Two-Component Self-Renewal System

Component 1 — ATP-dependent ion pump activity: Na^+/K^+ -ATPase continuously extrudes sodium, maintaining the pump-leak balance, preventing osmotic water influx, and preserving cell volume within the geometric reserve. This component requires continuous glycolytic ATP production and fails within minutes of glycolytic inhibition.

Component 2 — Residual mRNA-dependent structural protein synthesis: Ongoing low-level translation of spectrin, ankyrin, and related cytoskeletal proteins continuously replaces oxidatively damaged structural components of the hydrogel network.

The 120-day erythrocyte lifespan reflects the gradual, irreversible depletion of residual mRNA — the point at which Component 2 fails completely and the hydrogel network can no longer be replenished.

3. The Physical Architecture: Membrane and Hydrogel as Distinct Systems

3.1 Two Physically Different Levels

The plasma membrane is a two-dimensional fluid lipid bilayer — a dynamic, flexible boundary that separates intracellular from extracellular compartments. The intracellular hydrogel is a three-dimensional viscoelastic network — the primary load-bearing and organisational

structure of the cytoplasm — composed of cytoskeletal polymers, protein condensates, organelles, and bound water layers.

3.2 Reversible Membrane Injury: The Hydrogel as Scaffold

Partial membrane injury may remain reversible when the intracellular hydrogel retains structural and energetic integrity. Membrane repair mechanisms depend critically on the mechanical and energetic support of the intracellular hydrogel. When the hydrogel is intact, membrane injury remains bounded and potentially repairable. When the hydrogel has collapsed, membrane injuries of any size become irreparable.

3.3 Irreversible Hydrogel Collapse: The Point of No Return

The true point of no return in cellular death is not membrane rupture — which is the terminal visible event — but loss of the intracellular hydrogel's capacity for self-renewal. A cell with a transiently perforated membrane but intact hydrogel can survive. A cell with an intact membrane but collapsed hydrogel cannot.

4. Two Trajectories of Hydrogel Destabilisation

4.1 Hydropic-Solution Collapse (Swelling Trajectory)

Component 1 failure → Na^+ accumulation → H_2O influx → hydrogel solation → swelling → geometric reserve exhaustion → cytolysis. This trajectory corresponds to oncosis, cytotoxic oedema, osmotic haemolysis in erythrocytes, and ballooning degeneration in hepatocytes.

4.2 Contractile-Synergetic Collapse (Condensation Trajectory)

Gradual energy deficit + apoptotic signalling → gel condensation → water extrusion → volume decrease → fragmentation → apoptotic bodies. This trajectory corresponds to apoptosis in nucleated cells and eryptosis in erythrocytes.

4.3 The Kinetic Determinant of Trajectory

The trajectory is determined primarily by the kinetics of ATP depletion relative to the kinetics of apoptotic programme activation. Necrosis and apoptosis are not fundamentally different biological processes. They are different physical trajectories of the same event — hydrogel destabilisation — determined by the rate at which energy deficit develops relative to the capacity of the cell's pre-lytic death programme to execute before the osmotic threshold is crossed.

5. Clinical Models

5.1 Glucose-6-Phosphate Dehydrogenase Deficiency

In G6PD deficiency, the pentose phosphate pathway is impaired, reducing NADPH production and allowing oxidative stress to accumulate. This produces a dual failure: Component 1 failure (oxidative impairment of Na^+/K^+ -ATPase and reduced glycolytic ATP) and Component 2 failure (oxidative damage to residual mRNA and ribosomes, reducing translational capacity for spectrin and ankyrin replacement).

5.2 Preeclampsia and HELLP Syndrome

In preeclampsia, erythrocyte Component 1 failure arises through external glucose competition mediated by human placental lactogen. Rising placental lactogen concentrations induce maternal insulin resistance that redirects glucose toward the fetoplacental unit. Erythrocytes experience reduced glucose substrate availability, declining ATP production, and impaired Na^+/K^+ -ATPase function. Experimental studies confirm increased osmotic fragility in erythrocytes from women with severe preeclampsia — a direct quantitative measurement of consumed geometric reserve.

6. Syneresis, Geometric Reserve, and Osmotic Fragility as Biophysical Metrics

Ageing of the erythrocyte can be reinterpreted as progressive dominance of syneresis over self-renewal: as residual mRNA is depleted and Component 2 self-renewal capacity falls, syneretic tendencies accumulate — manifesting as progressive membrane microvesiculation, spectrin network reorganisation, reduction of deformability, and increased osmotic fragility.

The osmotic fragility test measures the haemolysis curve of an erythrocyte population as a function of decreasing extracellular osmolarity. Within the two-component self-renewal framework, osmotic fragility is a direct quantitative measurement of residual geometric reserve — the distance between the current hydrogel state and the critical volume at which membrane tension reaches the lytic threshold.

7. Connection to the Basal Glycolytic Regime Framework

BGR dominance: both components functional — maximal hydrogel self-renewal — maximum geometric reserve — maximum distance from lytic threshold. This corresponds to embryonic cells, stem cells, activated progenitors, and regenerating tissues. BGR collapse: both components fail — rapid hydrogel destabilisation — trajectory determined by collapse kinetics — corresponds to acute ischaemia, severe oxidative crisis, severe metabolic competition.

8b. Proteolysis as a Component of Living Hydrogel Maintenance

8b.1 Proteolytic Enzymes as Remodelling Tools, Not Only Death Executors

Within the hydrogel framework, caspases, matrix metalloproteinases, calpains, and cathepsins must be reinterpreted. In the living cell, these enzymes participate in continuous intracellular hydrogel remodelling: selective degradation of damaged, misfolded, or structurally fatigued proteins creates space for renewal of the polymer network. Most proteolytic enzymes do not require ATP for their catalytic activity. When cellular energy fails, proteolysis continues at near-normal rates while protein synthesis, structural renewal, membrane repair, and active osmotic stabilisation progressively collapse. Cellular viability depends not on the absence of proteolysis but on the dynamic equilibrium between ATP-independent proteolytic degradation and ATP-dependent structural renewal.

8b.2 Hydration State as the Physical Determinant of Proteolytic Outcome

High-water state — hydrophilic proteolytic amplification: proteolysis under these conditions generates peptides and amino acids — osmotically active molecules that further elevate intracellular osmotic pressure. A positive feedback loop emerges: increased hydration → reduced gel viscosity → accelerated proteolytic fragmentation → more osmotic particles → more water influx → gel liquefaction → cytolysis.

Low-water state — synergetic proteolytic condensation: limited proteolysis under these conditions enables the partially degraded hydrogel to reorganise into a denser, more condensed structure. Water is extruded rather than accumulated.

8b.3 Caspases as Directors of Organised Synergetic Collapse

The caspase cascade in apoptosis is not merely a destruction programme. It is a controlled synergetic remodelling programme: caspases selectively cleave cytoskeletal proteins — lamin, spectrin, gelsolin, ICAD — at defined sites, producing controlled disassembly of the hydrogel network that enables condensation without catastrophic swelling.

9. Discussion

This paper, together with the two companion papers on osmotic cytolysis and preeclampsia/HELLP, constitutes a three-level physical account of cell death from energy deficit. Level 1 (this paper): the cell is a self-renewing hydrogel with two distinct failure levels and two collapse trajectories; the erythrocyte is the minimal experimental model. Level 2 (companion paper on osmotic cytolysis): the universal cascade from ATP↓ to cytolysis. Level 3 (companion paper on preeclampsia): BGR collapse in specific maternal cell populations as the unified pathogenetic mechanism.

10. Conclusion

The living cell is an actively maintained metastable hydrogel system. The plasma membrane and the intracellular hydrogel are physically distinct systems. Membrane injury may remain reversible while the hydrogel is preserved. Hydrogel collapse is irreversible regardless of membrane state.

Hydrogel destabilisation follows two physically distinct trajectories — hydropic-solation collapse and contractile-synergetic collapse — determined by the kinetics of ATP depletion relative to pre-lytic programme activation. The erythrocyte is the minimal model of a self-renewing biological hydrogel with a two-component self-renewal architecture: its 120-day lifespan reflects Component 2 depletion; its osmotic fragility reflects geometric reserve; and its haemolysis is the terminal physical event of hydrogel collapse.

Conflict of Interest: The author declares no conflict of interest.

11. Proposed Experimental Framework

11.1 Experiment 1: Proteolytic Enzyme Spectrum in Intact Erythrocytes

Hypothesis: Mature erythrocytes contain a defined pool of free intracellular proteolytic enzymes — including MMPs, calpains, and caspases — that constitute Component 2b of the active hydrogel maintenance system. Protocol: Fresh human blood is washed three times in isotonic PBS. Washed erythrocytes are subjected to controlled osmotic haemolysis. The haemolysate is assayed for MMP activity (gelatin zymography), calpain activity (casein zymography), and caspase-3/8 activity (fluorogenic substrate assay).

11.2 Experiment 2: Time-Course of Metabolic and Structural Deterioration at 38°C

Hypothesis: Accelerated metabolic demand at febrile temperature will deplete the glycolytic substrate pool and progressively impair the two-component hydrogel maintenance system.

Protocol: Washed erythrocytes (haematocrit 5%) are suspended in isotonic glucose-free PBS and incubated at 38°C. Samples are collected at T=0, 2, 4, 8, 12, 24, and 48 hours. Measured parameters: (1) intracellular glycogen and ATP; (2) osmotic resistance; (3) erythrocyte morphology by SEM. The model predicts a specific temporal order: glycogen and ATP decline precedes measurable changes in osmotic resistance, which precede morphological transformation.

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Genetic Diseases as Manifestations of Hydrogel Pathology?

Progressive Syneresis, Structured Hydration Failure, and Polymer-Coherence Disorders

Yuri Dmitrochenko

A Conceptual Biophysical Framework

This paper is part of a conceptual series on hydrogel physics and cellular dynamics.

Abstract

Genetic diseases are traditionally interpreted as consequences of defective proteins, abnormal enzymes, disrupted signalling pathways, or metabolic insufficiency. However, many hereditary disorders demonstrate striking physical commonalities: progressive rigidity, loss of tissue plasticity, intracellular aggregation, altered hydration, osmotic instability, impaired viscoelasticity, condensation phenomena, and structural degeneration.

This paper proposes a conceptual framework in which many genetic diseases may be reinterpreted as disorders of hydrogel coherence and structured hydration. Within this model, genetic defects are proposed to destabilise specific components of coupled hydrogel-maintenance loops, progressively shifting living systems toward pathological syneresis: condensation, rigidity, loss of viscoelastic plasticity, impaired diffusion, and altered water organisation [3,4].

Keywords: *hydrogel coherence; syneresis; glycogen storage disease; mucopolysaccharidoses; protein aggregation; neurodegeneration; fibrosis; structured hydration; proteostasis; viscoelasticity; Basal Glycolytic Regime.*

1. Introduction: Beyond Molecular Defects

Despite diverse molecular origins, many genetic diseases converge onto remarkably similar physical manifestations: progressive tissue rigidity, intracellular aggregation, altered cellular hydration, osmotic instability, condensation phenomena, impaired deformability, and loss of structural plasticity [7,8]. This convergence suggests that molecular defects may funnel into common physical failure states of living matter.

2. Living Matter as a Structured Hydration System

Biological systems are not simply aqueous solutions of macromolecules; they are dynamically organised hydrogels in which water exists in multiple physical states — bulk, interfacial, and bound — regulated by a continuous investment of metabolic energy [11,12]. Proteins provide structural scaffolding; glycans and proteoglycans regulate hydration shells and ECM viscoelasticity; glucose metabolism maintains the energetic basis for hydrogel dynamics; glycogen functions as a reserve enabling rapid carbohydrate mobilisation; and proteolysis provides continuous entropic remodelling pressure that prevents pathological condensation [13,14,15].

3. Syneresis as a Universal Degenerative Trajectory

Progressive syneresis of intracellular hydrogel networks may manifest as: aggregation of proteins into insoluble condensates; collapse of the glycan-rich pericellular coat; stiffening of the nuclear lamina; compaction of chromatin; and reduction of cytoplasmic fluidity [21,22]. At the tissue level, syneresis of the extracellular matrix manifests as fibrosis — the replacement of dynamic, hydrated ECM with dense, poorly remodelled collagen networks [23,24].

4. Glycogen Disorders as Carbohydrate Hydrogel Pathology

4.1 Structural Functions of Glycogen Beyond Energy Storage

Glycogen acts not only as an energy reserve but also as a hydration reservoir, osmotic stabiliser, rheological regulator, and structural component of the cytoplasmic gel [14,28]. Reduction in branching increases the linear character of the polymer, promoting inter-chain associations, reduced solubility, and condensation — a transition analogous to the gelation and syneresis observed in poorly branched polysaccharide gels [30].

4.2 Lafora Disease

Lafora disease is an autosomal recessive progressive myoclonic epilepsy caused by mutations in EPM2A or EPM2B [31]. The hallmark pathology is the accumulation of Lafora bodies — intraneuronal inclusions of poorly branched, hyperphosphorylated polyglucosan. Nitschke et al. demonstrated that abnormal glycogen chain length pattern — not hyperphosphorylation per se — is critical to Lafora body formation [32], consistent with the physical prediction that reduced branching transforms a soluble, dynamic hydrogel particle into an insoluble, rigid aggregate.

4.3 Adult Polyglucosan Body Disease

APBD is caused by deficiency of glycogen branching enzyme (GBE1) and is characterised by accumulation of polyglucosan bodies in neurons, astrocytes, and peripheral nerves [34]. APBD demonstrates that poorly branched glycogen polymers are inherently condensation-prone, sharing morphological and biochemical features with Lafora bodies despite different enzymatic origins.

4.4 Glycogen Storage Disease Type IV (Andersen Disease)

GSD-IV is caused by biallelic mutations in GBE1 producing severe branching enzyme deficiency, resulting in systemic accumulation of amylopectin-like polysaccharide [36]. The progressive fibrosis of liver and other organs in GSD-IV may reflect dual hydrogel pathology:

intracellular condensation of poorly hydrated glycogen polymers and secondary activation of fibrogenic responses to chronic cellular damage [37].

5. Mucopolysaccharidoses as Hydration-Structuring Failure

The mucopolysaccharidoses (MPS) are caused by deficiency of enzymes responsible for glycosaminoglycan (GAG) catabolism, resulting in accumulation of partially degraded GAGs in lysosomes and the ECM [38]. In MPS disorders, partially degraded GAGs may have altered charge distributions impairing their ability to organise structured water layers; they may compete with normal GAGs for binding sites on proteoglycan core proteins; and their lysosomal accumulation may impair the normal cycling of ECM components [40,41].

6. Protein Aggregation Diseases as Scaffold-Coherence Failure

Neurodegenerative diseases characterised by protein aggregation — including Alzheimer disease, Parkinson disease, Huntington disease, and amyotrophic lateral sclerosis — represent pathological condensation of intracellular scaffold networks [44,45]. The normal cytoplasm maintains proteins in dynamic equilibrium through active processes including chaperone activity, the ubiquitin-proteasome system, autophagy, and ATP-dependent disaggregation machinery [46,47]. Tau pathology demonstrates the hydrogel relevance of cytoskeletal scaffold condensation: hyperphosphorylated tau undergoes liquid-liquid phase separation followed by progressive solidification into neurofibrillary tangles [51], impairing axonal hydrogel dynamics and disrupting transport.

7. Fibrosis as Extracellular Hydrogel Condensation

Normal ECM is a dynamically remodelled hydrogel: collagen fibres are continuously synthesised and degraded by matrix metalloproteinases (MMPs), proteoglycans provide hydration and viscoelasticity, and fibronectin and laminin organise cell-matrix adhesion [54]. In fibrosis, the equilibrium is disrupted in favour of net accumulation. Progressive collagen deposition creates a denser, stiffer ECM with reduced water content and altered mechanosignalling that itself promotes further fibrogenic activation through integrin and YAP/TAZ pathways [55,56] — a positive feedback loop characteristic of synergetic collapse in polymer networks [57].

8. Aging as Progressive Hydrogel Destabilisation

Aging represents the progressive failure of anti-synergetic remodelling. The hallmarks of ageing — increased cellular stiffness, reduced cytoplasmic fluidity, progressive ECM stiffening, accumulation of intracellular aggregates — are precisely the features predicted by progressive syneresis of biological hydrogel systems [26]. Extracellular matrix ageing reflects hydrogel condensation processes: collagen cross-linking increases with age through enzymatic and non-enzymatic (glycation-mediated) mechanisms, progressively rigidifying connective tissues [60].

9. The Basal Glycolytic Regime as an Anti-Synergetic State

The Basal Glycolytic Regime (BGR) represents the metabolic regime characteristic of cells that maintain maximal hydrogel self-renewal capacity [61,62]. Glycolysis generates not only ATP but also metabolic intermediates (glucose-6-phosphate, UDP-glucose, nucleotide sugars) that feed glycan biosynthesis pathways [63]. Within this interpretation, BGR is a universal anti-synergetic maintenance state that prioritises the preservation of hydrogel coherence.

10. A Unified Interpretation of Genetic Hydrogel Pathology

Glycogen storage diseases become disorders of carbohydrate hydrogel architecture. Mucopolysaccharidoses become disorders of glycan hydration organisation. Protein aggregation diseases become disorders of scaffold coherence. Fibrosis becomes extracellular hydrogel condensation. Neurodegeneration becomes the culmination of progressive intracellular syneresis.

This unified perspective generates testable predictions: physical parameters — cytoplasmic stiffness, organelle mobility, molecular diffusion rates, water organisation by NMR relaxometry — should show characteristic signatures across disease categories that correlate with clinical severity.

11. Discussion: Implications and Limitations

Several important limitations must be acknowledged. The framework is currently largely conceptual: the specific physical measurements required to validate its predictions across disease categories have not been systematically assembled. The causal relationships proposed require experimental testing in each disease context. The framework risks overgeneralisation: not all features of all genetic diseases may be productively interpreted through the hydrogel lens.

12. Conclusions

Glycogen storage diseases, mucopolysaccharidoses, protein aggregation diseases, fibrotic conditions, and neurodegenerative disorders may all reflect specific modes of hydrogel coherence failure, producing overlapping physical manifestations despite diverse molecular origins. Aging may represent the slow systemic version of the same process. The Basal Glycolytic Regime is proposed as a key anti-syneretic maintenance state that sustains cellular hydrogel coherence.

Conflict of Interest: The author declares no conflict of interest.

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Article 5 of 5

Experimental Verification of the Hydrogel Pathology Hypothesis

A Framework for Testing Structured Hydration Failure, Syneresis, and Polymer-Coherence Collapse in Genetic Disease

Yuri Dmitrochenko

A Conceptual Biophysical Framework

This paper is part of a conceptual series on hydrogel physics and cellular dynamics.

Abstract

This paper proposes an experimental framework for testing the hypothesis that a subset of genetic diseases may represent disorders of hydrogel coherence and structured hydration rather than exclusively molecular or energetic defects. The paper defines the positive and negative applicability domains of the framework, formulates seven experimentally measurable predictions with explicit falsification criteria, proposes candidate experimental systems, and addresses the limits of the model.

Keywords: *hydrogel coherence; syneresis; experimental verification; viscoelasticity; NMR relaxometry; atomic force microscopy; glycogen branching; mucopolysaccharidoses; proteostasis; fibrosis; falsification; Basal Glycolytic Regime.*

1. Introduction

Many genetic diseases display convergent physical manifestations despite distinct molecular origins: rigidity, intracellular aggregation, altered hydration, impaired deformability, condensation phenomena, fibrosis, progressive loss of plasticity, and degeneration [5,6]. The hydrogel pathology hypothesis proposes that some hereditary diseases are fundamentally disorders of hydrogel coherence arising from progressive syneresis of intracellular or extracellular polymer systems [1–4].

The present paper does not attempt to prove the hypothesis directly. Instead, it establishes the logical applicability limits of the framework, formulates experimentally testable predictions with explicit falsification criteria, and proposes a strategy for validation or refutation. A hypothesis that cannot be falsified is not a scientific hypothesis.

2. Applicability Domain of the Hydrogel Pathology Framework

The framework appears specifically applicable only when disease directly disrupts a hydrogel-maintenance loop: glycan architecture, glycogen structure, proteostasis, extracellular matrix remodelling, or polymer-remodelling systems responsible for maintaining structured hydration.

Three applicability tiers are distinguished. Primary direct applicability: glycogen branching disorders (Lafora, APBD, GSD-IV); mucopolysaccharidoses (Hurler, Hunter, Sanfilippo); proteostasis disorders (Alzheimer, Parkinson, ALS, Huntington); ECM remodelling disorders (fibrosis, OA, disc degeneration). Secondary indirect: mitochondrial disorders (ATP depletion impairs all anti-syneretic maintenance). Negative domain: DNA repair disorders (BRCA, mismatch repair); coagulation disorders (haemophilia).

3. Core Hydrogel Pathology Chain

Defective enzyme or remodelling system → Substrate accumulation and/or product deficiency → Failure of hydrogel-maintenance component → Local syneresis → Loss of viscoelastic coherence → Progressive structural degeneration.

This sequence generates a critical and testable prediction: physical degeneration — measurable as altered viscoelasticity, changed water organisation, or increased stiffness — may precede complete energetic collapse.

4. Positive Test Categories

4.1 Glycogen Branching Disorders

Candidate diseases: Lafora disease, Adult Polyglucosan Body Disease (APBD), Glycogen Storage Disease Type IV. Proposed mechanism: Reduced glycogen branching decreases surface hydration per glucose unit, increases inter-chain van der Waals contacts, promotes aggregation, and generates poorly soluble polyglucosan bodies [9,10]. Nitschke et al. demonstrated that abnormal glycogen chain length pattern — rather than hyperphosphorylation — is the critical determinant of Lafora body formation [11].

Hydrogel prediction: Disease severity should correlate with altered intracellular viscoelasticity, reduced water mobility by NMR relaxometry, increased cytoplasmic stiffness by AFM, impaired macromolecular diffusion by FCS, and progressive intracellular syneresis — independently of total cellular ATP content.

4.2 Lysosomal Storage Diseases — Mucopolysaccharidoses

Candidate diseases: Hurler syndrome (MPS I), Hunter syndrome (MPS II), Sanfilippo syndrome (MPS III), Morquio syndrome (MPS IV), Maroteaux–Lamy syndrome (MPS VI). Hydrogel prediction: Affected tissues should demonstrate altered water organisation by NMR diffusion-weighted imaging, altered ECM rheology by indentation measurements, and progressive matrix condensation correlating with GAG accumulation burden.

4.3 Proteostasis Disorders

Candidate diseases: Alzheimer disease, Parkinson disease, amyotrophic lateral sclerosis, Huntington disease. Hydrogel prediction: Aggregation-prone cells should show increased cytoplasmic stiffness by AFM and optical tweezers, reduced viscoelastic relaxation time, and progressive intracellular syneresis correlating with aggregation burden and clinical severity.

4.4 Extracellular Matrix Remodelling Disorders

Candidate diseases: Hepatic fibrosis, pulmonary fibrosis, osteoarthritis, intervertebral disc degeneration, systemic sclerosis. Hydrogel prediction: Fibrotic tissues should display reduced water mobility by NMR, increased stiffness by AFM indentation, altered viscoelastic relaxation spectra, and reduced swelling capacity — all correlating with degree of fibrosis.

5. Indirect or Secondary Hydrogel Pathology: Mitochondrial Disorders

Mitochondrial disorders produce ATP depletion that impairs all anti-synergetic maintenance processes simultaneously [19]. Hydrogel disruption in these disorders is indirect rather than primary: the genetic defect does not disrupt a specific hydrogel-maintenance component but rather the energy supply to all such components. Physical changes should track global energetic state rather than specific polymer characteristics.

6. Negative Control Categories

6.1 Haemophilia as Primary Negative Control

Haemophilia A (factor VIII deficiency) and haemophilia B (factor IX deficiency) produce bleeding and coagulation defects through mechanisms entirely unrelated to polymer-maintenance loops [20]. The framework therefore predicts that erythrocytes and fibroblasts from haemophilic patients should show normal osmotic fragility, normal cytoplasmic stiffness by AFM, normal intracellular water mobility by NMR, and normal viscoelastic relaxation — indistinguishable from healthy controls.

6.2 Isolated Channelopathies

Ion channel disorders — including long QT syndrome and isolated channelopathies without structural degeneration — disrupt ion transport without primary disruption of polymer-maintenance loops. The hydrogel framework predicts no primary hydrogel pathology signatures in these conditions.

7. Core Experimental Predictions

Seven experimentally testable predictions, each with explicit falsification criteria:

Prediction 1 (Glycogen branching disorders): Osmotic fragility and intracellular water mobility (NMR). Falsified if fragility unchanged independently of ATP; NMR indistinguishable from controls.

Prediction 2 (Lafora disease; GSD-IV): Viscoelastic coherence (microrheology) prior to energetic collapse. Falsified if rheological changes appear only after complete ATP depletion.

Prediction 3 (BGR vs. oxidative metabolic states): NMR relaxation parameters (T1, T2) of intracellular water. Falsified if no NMR difference between BGR-dominant and oxidative states.

Prediction 4 (Poorly branched glycogen in vitro): Hydration, rigidity, solubility, condensation tendency. Falsified if poorly branched glucan behaves identically to normal glycogen in solution.

Prediction 5 (Neurodegenerative disease models): Cytoplasmic stiffness (AFM); diffusion (FCS). Falsified if stiffness and diffusion uncorrelated with aggregation burden.

Prediction 6 (Fibrotic tissue): Water mobility (NMR); stiffness; viscoelastic relaxation. Falsified if fibrotic tissue water mobility indistinguishable from normal ECM.

Prediction 7 — negative (Haemophilia): Cytoplasmic stiffness; osmotic fragility; NMR. Falsified if positive hydrogel signatures found in haemophilia — this would challenge specificity of the framework.

Prediction 2 — that viscoelastic changes precede energetic collapse — is the most critical single test of the framework.

8. Proposed Experimental Systems

Human erythrocytes (G6PD deficiency; glycogen branching variants): Ektacytometry; osmotic fragility; micropipette aspiration; NMR relaxometry; intracellular viscosity. Advantage: minimal complexity; quantitative physical metrics; established protocols.

Patient-derived fibroblasts (MPS I, II, III; GSD-IV; Lafora disease): AFM stiffness mapping; particle-tracking microrheology; FCS; lysosomal NMR.

iPSC-derived neurons (Alzheimer, Parkinson, ALS, Huntington): Cytoplasmic stiffness (AFM); organelle mobility (SPT); phase-separation assays; FCS.

In vitro glycogen models (normal vs. poorly branched): Dynamic light scattering; NMR; rheometry; turbidity; water activity. Advantage: pure physical characterisation independent of cellular complexity.

Decellularised ECM / tissue slices (fibrotic liver, lung, cartilage): AFM; indentation rheology; NMR water mobility; swelling assays.

9. Falsification Criteria

The hydrogel pathology framework would be significantly weakened by any of the following: hydrogel-associated physical changes occur only after complete energetic collapse; poorly branched glycogen polymers in vitro demonstrate properties identical to normally branched glycogen; aggregation diseases show no measurable viscoelastic alterations in living cells prior

to cell death; hydrogel markers fail to correlate with disease severity; negative-control diseases demonstrate hydrogel signatures identical to primary hydrogel pathologies.

10. Conceptual Interpretation

The framework proposes that molecular lesions converge onto a limited number of physical degeneration trajectories — intracellular syneresis, extracellular matrix condensation, hydrogel rigidification, hydration collapse, and loss of viscoelastic coherence [4]. The framework supplements rather than replaces molecular biology. Molecular biology provides the specificity — which enzyme, which tissue, which polymer component is disrupted; hydrogel physics provides the physical interpretation of the convergent outcome.

11. Conclusion

The hydrogel pathology hypothesis appears specifically applicable to diseases directly disrupting glycogen architecture, glycan remodelling, proteostasis, extracellular matrix turnover, or polymer-maintenance systems responsible for preserving structured hydration. Seven experimentally testable predictions have been formulated, each with explicit falsification criteria.

If experimentally supported, the framework may provide a unifying physical interpretation of multiple degenerative diseases, a bridge between glycobiology and polymer physics, and a thermodynamic language for structured hydration pathology that complements — and does not replace — the molecular language of classical genetics.

Conflict of Interest: The author declares no conflict of interest.

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CHAPTER 2

TISSUE ORGANIZATION AND ONCOGENESIS

Preface

In the first chapter of this monograph, the living cell was considered as an active hydrogel system that continuously opposes thermodynamic relaxation through energy-dependent remodeling. Within this framework, aging, degeneration, and cell death emerge not only as consequences of molecular damage, but also as manifestations of a shifting balance between structural renewal and the universal tendency toward syneresis.

However, a living organism is not merely a collection of isolated cells. Cells exist within tissues, and tissues form complex systems of mechanical, metabolic, and informational interactions. Consequently, processes such as regeneration, fibrosis, adaptation, aging, and oncogenesis cannot be fully understood at the level of the individual cell alone.

This chapter extends the framework from the cellular to the tissue level of biological organization. Its central premise is that tissues do not occupy a single fixed condition, but rather exist within a dynamic space of accessible states. The probability of transitions between these states is determined by the interaction of structural constraints, metabolic organization, and the availability of energetic resources.

To describe these processes, a set of interconnected concepts is introduced: the Energetic Window, the Regeneration Pendulum, Temporal Tissue Loops, and a dual-flux energetic architecture composed of Oxidative Functional Flux (OFF) and the Basal Glycolytic Regime (BGR). Together, these concepts provide a unified language for describing tissue plasticity, regeneration, and pathological state transitions.

The logic of this chapter develops progressively. The first article examines the reactivation of embryonic-associated programs during regeneration, inflammation, and tissue remodeling, introducing the core concepts of the framework. The second article provides their energetic basis through the interaction of OFF and BGR and describes tissue behavior as probabilistic transitions within ontogenetic state space. The third article analyzes the consequences of prolonged energetic-window expansion at the level of tissue architecture and examines the transition from organized three-dimensional structures to infiltrative two-dimensional configurations. The fourth article introduces the topological dimension of the framework, linking oncogenesis to the loss of intermediate ontogenetic states and the disruption of local connectivity between tissue and cellular clone. The final article integrates the preceding concepts into a unified model of cellular clone autonomization and its possible developmental trajectories.

Taken together, the chapter describes a continuous progression from physiological regeneration and tissue plasticity toward states of increasing structural and functional autonomy.

The ideas presented here remain working hypotheses. They are intended to serve as tools for interpretation, sources of testable predictions, and a foundation for future quantitative models of tissue remodeling, regeneration, aging, and oncogenesis.

Article 1 of 5

Embryonic-Associated Programs Outside Pregnancy and Cancer: Energetic Window Expansion, the Regeneration Pendulum, and Temporal Tissue Loops

Yuri Dmitrochenko

A Conceptual Biophysical Framework

Series: Embryonic Programs in Non-Oncological Tissue States | Paper 1 of 5

Abstract

Embryonic and fetal proteins are traditionally associated with embryogenesis, pregnancy, and malignancy. However, numerous physiological and pathological conditions outside both pregnancy and cancer demonstrate partial reactivation of embryonic-associated programs. In this paper — the first in a series — we summarize evidence that fetal proteins, developmental pathways, and partial ontogenetic reversion occur in a broad range of non-malignant contexts, including regeneration, chronic inflammation, fibrosis, tissue remodeling, immune conflict, wound healing, and high-turnover stem-cell niches.

We propose that these phenomena are linked to expansion of the tissue energetic window associated with structural remodeling and increased metabolic flux. A conceptual framework is introduced in which recurrent coupling between cell death, matrix metalloproteinase (MMP) activation, extracellular matrix remodeling, vascular expansion, and partial ontogenetic reversion generates oscillatory tissue dynamics termed the Regeneration Pendulum. Within this framework, tissues undergoing repeated cycles of destruction and repair transiently revisit earlier developmental-like functional states, forming Temporal Tissue Loops.

Keywords: *embryonic-associated programs; oncofetal antigens; fetal proteins; matrix metalloproteinases; tissue remodeling; regeneration pendulum; temporal tissue loops; energetic window; fibrosis; chronic inflammation*

1. Introduction

Embryonic and fetal proteins are commonly regarded as markers of embryogenesis or malignancy. Classic examples include fetal hemoglobin (HbF), alpha-fetoprotein (AFP), carcinoembryonic antigen (CEA), embryonic extracellular matrix proteins, and developmental signaling pathways such as WNT, Notch, and Hedgehog. The prevailing clinical interpretation treats their reappearance in adult tissues either as a normal product of ontogenesis or as an aberrant signal of neoplastic transformation.

However, increasing evidence demonstrates that such programs also emerge in multiple non-malignant adult tissues under conditions of remodeling, regeneration, and chronic stress. The partial reactivation of developmental gene networks in healing skin, regenerating liver, inflamed intestinal mucosa, and fibrotic lung suggests that the association between embryonic programs and cancer may be more contextual than intrinsic.

In this article, we propose that embryonic-associated programs emerge predominantly in tissues operating within expanded energetic windows characterized by increased metabolic flux, extracellular matrix disruption, high cellular turnover, structural instability, or active regeneration. We introduce the Regeneration Pendulum and Temporal Tissue Loops as conceptual tools for understanding how recurrent cycles of tissue destruction and repair generate oscillatory reactivation of developmental programs across a wide range of biological contexts.

2. Embryonic-Associated Programs Outside Pregnancy and Cancer

2.1 Regeneration and Wound Healing

During tissue repair, adult tissues frequently reactivate developmental pathways otherwise suppressed in homeostatic conditions. In skin wound healing, re-epithelialization involves upregulation of embryonic fibronectin isoforms (EDA and EDB domains) and Tenascin-C,

alongside glycolytic metabolic reversion. [1,5,6] Liver regeneration following partial hepatectomy involves transient reactivation of fetal hepatocyte gene expression programs, including AFP and embryonic metabolic enzymes. [7,8] Skeletal muscle repair recapitulates aspects of embryonic myogenesis, including sequential expression of embryonic and neonatal myosin heavy chain isoforms. [10,11] Intestinal regeneration following epithelial injury involves WNT pathway hyperactivation mirroring embryonic crypt morphogenesis. [12,13] Post-infarction cardiac remodeling activates fetal gene programs including embryonic myosin heavy chain isoforms and fetal metabolic shifts toward glycolysis. [14,15]

2.2 Fibrosis

Fibrotic tissues exhibit partial reactivation of developmental fibroblast programs. In liver cirrhosis, activated hepatic stellate cells acquire a myofibroblast phenotype with fetal-like expression of smooth muscle actin, collagen subtypes, and developmental matrix-remodeling enzymes. [16,17] Pulmonary fibrosis involves persistent activation of developmental fibroblast states including embryonic-like TGF- β signaling and fetal matrix composition. [18,19] The partial but persistent nature of developmental program reactivation in fibrosis — unlike the transient reactivation in acute regeneration — suggests that fibrosis may represent a dysregulated exit from the regenerative oscillatory cycle rather than simply a failure of repair.

2.3 Chronic Inflammation

Chronic inflammatory states frequently exhibit developmental-like tissue programs. In rheumatoid arthritis, synoviocytes display sustained WNT pathway activation and partial embryonic fibroblast characteristics. [22,23] Inflammatory bowel disease involves repeated cycles of mucosal injury and repair in which intestinal stem cell compartments are repeatedly activated, with transient reversion toward embryonic-like crypt developmental states. [24,25]

2.4 Immune Conflict States

Immune-mediated tissue injury creates conditions favoring recurrent repair activation. Autoimmune reactions combine recurrent cell death, inflammatory cytokine signaling, matrix remodeling, vascular adaptation, and metabolic stress in a manner that cyclically activates developmental programs in affected tissues. [38] Chronic graft rejection generates sustained partial embryonic program reactivation through persistent tissue damage-repair cycles. [39] These states demonstrate that the trigger for developmental program reactivation need not be intrinsic tissue damage: extrinsic immune-mediated injury can equally establish the conditions required for energetic window expansion.

2.5 High-Turnover Adult Niches

Certain adult tissues naturally maintain partial persistence of developmental programs even under homeostatic conditions. Intestinal crypts maintain active WNT-dependent stem cell signaling and elevated cellular turnover. [12,25] Bone marrow hematopoietic niches sustain Notch, Hedgehog, and WNT signaling required for stem cell self-renewal. [34,35] Hair follicles cycle through phases of growth and regression in which stem cell populations repeatedly reactivate developmental transcriptional programs. [36] Neural progenitor niches maintain partial embryonic-like metabolic and signaling states to support adult neurogenesis. [37]

3. Energetic Window Expansion

3.1 Conceptual Definition

We propose that the emergence of embryonic-associated programs is linked to expansion of the tissue energetic window (ΔE_{window}) — the range of metabolic and structural conditions permitting elevated tissue plasticity and remodeling. In stable mature tissues, this window is narrow: structural constraints, low metabolic flux, and tight intercellular coupling limit the range of accessible cellular states. Expansion of this window is associated with increased vascular permeability and angiogenesis, extracellular matrix degradation, elevated nutrient and oxygen flux, reduced mechanical coupling between cells, and release of matrix-bound growth factors.

3.2 Relationship to the Epigenetic Landscape

From a developmental biology perspective, the energetic window concept resonates with Waddington's epigenetic landscape model. [29] In the embryo, cells occupy a high-energy state with access to multiple differentiation trajectories; developmental commitment progressively narrows the accessible state space. Our proposal suggests that tissue remodeling events can transiently re-expand this state space in adult tissues — not returning cells to full pluripotency, but restoring access to adjacent developmental-like attractor states.

3.3 Metabolic Correlates

Embryonic cells characteristically rely on glycolytic metabolism even under aerobic conditions — a pattern analogous to the Warburg effect in cancer. [31] Adult regenerating tissues display analogous metabolic shifts toward glycolysis during active repair phases. [32] This metabolic reversion may not merely correlate with developmental program reactivation but may be mechanistically upstream of it: glycolytic intermediates serve as epigenetic substrates (e.g., acetyl-CoA for histone acetylation) that regulate chromatin accessibility and gene expression.

4. Matrix Metalloproteinases and Structural Destabilization

Matrix metalloproteinases (MMPs) are zinc-dependent endopeptidases classically recognized for extracellular matrix degradation in wound healing and cancer invasion. Within the present framework, MMPs play a broader systems-level role: they transiently widen the tissue energetic window by reducing structural constraints. Increased MMP activity leads to ECM degradation, increased vascular access, altered mechanical coupling between cells, and enhanced tissue permeability. [26,27] Critically, MMP activation also releases matrix-sequestered growth factors including VEGF, FGF, and TGF- β , which directly activate developmental signaling pathways in resident cells. [28]

The temporal dynamics of MMP activity are therefore critical: acute MMP activation followed by TIMP-mediated inhibition characterizes normal wound healing. Persistent or recurrent MMP activation — as seen in chronic inflammation and fibrosis — maintains the tissue in an expanded energetic window and thereby sustains developmental program reactivation beyond its physiologically appropriate duration. [27,28]

5. The Regeneration Pendulum

5.1 Model Description

We propose that many remodeling tissues exhibit oscillatory dynamics driven by recurrent coupling between tissue destruction and repair — the Regeneration Pendulum:

Cell death → MMP↑ → $\Delta E_{\text{window}} \uparrow$ → Plasticity↑ → Repair → Structural Instability →
Cell death

Cell death releases damage-associated molecular patterns (DAMPs) and activates inflammatory signaling. MMP-mediated matrix remodeling widens the tissue energetic window. The widened window permits partial developmental program reactivation and enhanced cellular plasticity. Repair and growth restore tissue mass and structure. Restoration of structural complexity generates renewed mechanical stress and vulnerability to recurrent damage.

5.2 Oscillatory Dynamics

The Regeneration Pendulum can be formally represented as a limit cycle in phase space defined by energetic window width (ΔE) and tissue structural integrity (S). Under physiological conditions of acute wound healing, the system traverses a brief excursion from the stable resting state and returns. Under chronic inflammatory conditions, the system orbits persistently around the limit cycle. Under fibrotic conditions, the system attempts to escape the limit cycle by irreversibly increasing structural rigidity, mechanically interrupting the oscillation.

5.3 Amplitude Determinants

The degree of developmental program reactivation during each oscillatory cycle is determined by: severity of the initiating damage signal; baseline tissue MMP activity; vascular responsiveness; and epigenetic accessibility of developmental gene loci.

6. Temporal Tissue Loops

6.1 Definition

Repeated oscillatory remodeling generates Temporal Tissue Loops: recurrent transient revisitations of earlier developmental-like functional states by adult tissues undergoing cyclic destruction and repair. This concept does not imply literal return to embryogenesis. Rather, tissues repeatedly and transiently activate developmental signaling programs, fetal metabolic states, embryonic-associated extracellular matrix components, and progenitor-like repair phenotypes.

6.2 Dynamic Regimes

Within this framework, regeneration, chronic inflammation, fibrosis, and immune conflict represent different dynamic regimes of the same underlying oscillatory system. Acute wound healing: brief dampened oscillation, shallow loop depth — return to homeostasis. Successful regeneration: transient wide excursion, moderate loop depth — new homeostatic state. Chronic inflammation: sustained limit cycle, moderate-deep loop depth — persistent oscillation. Fibrosis: interrupted oscillation, deep pre-interruption loop — mechanical restabilization. High-turnover niches: low-amplitude steady oscillation — constitutive developmental state.

6.3 Relationship to Cancer

Cancer may represent a state in which the Regeneration Pendulum undergoes a bifurcation: rather than oscillating between destruction and repair while maintaining tissue identity, the system escapes into an unbounded growth attractor in which developmental program reactivation is no longer transient but constitutive, and structural restabilization never occurs. [40,41] Critically, the present framework emphasizes that cancer is only one of several possible dynamic outcomes of persistent oscillatory remodeling. Fibrosis and chronic inflammation represent alternative stable or semi-stable attractor states that share developmental program reactivation without requiring malignant transformation.

7. Structural Interpretation

Within this framework, different tissue states correspond to different positions and dynamics within the energetic landscape. Stable mature tissues: narrow energetic windows with strong structural constraints and minimal developmental program expression. Acutely regenerating tissues: temporarily widened windows with transient developmental reactivation. Chronically inflamed tissues: persistent oscillatory widening with recurrent developmental program activation. Fibrotic tissues: attempts at mechanical restabilization through irreversible matrix deposition.

Fibrosis may function as an emergency stabilization mechanism that terminates persistent regenerative oscillations by mechanically enforcing structural rigidity. This reframes fibrosis not simply as a failure of repair but as a terminal attractor state that the system converges toward when normal dampening of the Regeneration Pendulum fails. [16,17,18]

8. Discussion

8.1 Unifying Framework

The proposed framework offers a unifying interpretation of embryonic-associated protein expression across diverse non-malignant biological contexts. Rather than treating each instance of fetal protein reactivation as a distinct unexplained phenomenon, the energetic window model situates them within a common dynamic: the widening of structural and metabolic constraints that permits adult tissues to access adjacent developmental-like states.

8.2 Distinction from Malignancy

The model provides a principled basis for distinguishing developmental program reactivation in non-malignant contexts from that in cancer. In physiological regeneration and chronic inflammatory states, ontogenetic reversion is: partial rather than complete; transient rather than constitutive; spatially constrained to regions of active remodeling; coupled to structural restabilization mechanisms; and subject to normal tissue architecture controls. In cancer, these constraints are systematically violated. [40,41]

8.3 Limitations and Future Directions

The present framework is conceptual and requires formal quantitative development. Open questions include: the molecular and epigenetic determinants of loop depth; the mechanisms coupling structural restabilization to oscillation dampening; the relationship between the energetic window and chromatin accessibility dynamics; and whether temporal tissue loops can be directly detected through longitudinal biomarker profiling in chronic inflammatory tissues.

9. Conclusion

Embryonic-associated proteins and developmental programs occur widely outside both pregnancy and malignancy. Their emergence correlates with tissue remodeling, regeneration, chronic inflammation, immune conflict, and expanded energetic windows — conditions unified by elevated metabolic flux, extracellular matrix disruption, and structural instability.

We propose that recurrent coupling between cell death, MMP activation, energetic window expansion, and partial developmental program reactivation generates oscillatory tissue dynamics — the Regeneration Pendulum — within which tissues repeatedly transiently revisit earlier developmental-like functional states, forming Temporal Tissue Loops. The critical variable distinguishing these regimes is not the presence or absence of developmental program reactivation, but its temporal structure: transient, oscillatory, or constitutive.

Conflict of Interest: The author declares no conflict of interest.

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Basal Glycolytic Regime, Energetic Window Modulation, and Bidirectional Cell-State Transitions

Yuri Dmitrochenko

A Conceptual Biophysical Framework

Series: Embryonic Programs in Non-Oncological Tissue States | Paper 2 of 5

Abstract

Classical models of cellular differentiation describe ontogenesis as a predominantly unidirectional developmental trajectory driven by genetic and epigenetic programming, in which differentiated cellular states are viewed as progressively stabilized endpoints. However, accumulating evidence from regeneration, chronic inflammation, fibrosis, stem-cell niches, and malignant transformation demonstrates that partial developmental reactivation remains possible in adult tissues under specific conditions. [41,4,7,8]

The present paper proposes a systems-level mechanistic framework in which ontogenetic state transitions are interpreted as stochastic movements within a dynamic cellular state landscape. A dual-flux energetic architecture is introduced, comprising two partially interconnected metabolic regimes: Oxidative Functional Flux (OFF) — a high-power oxidative regime supporting specialized tissue function at the progressive cost of oxidative molecular wear; and Basal Glycolytic Regime (BGR) — a low-dissipation glycolytic-recycling regime associated with reduced reactive oxygen species, preservation of genomic integrity, regenerative capacity, and developmental plasticity. [6,13,14]

Within this framework, differentiation, regeneration, fibrosis, aging, and oncogenesis emerge as distinct dynamic regimes of a common nonequilibrium biological system governed by energetic-window modulation and the balance between OFF and BGR.

Keywords: *dual-flux energetics; Oxidative Functional Flux; Basal Glycolytic Regime; energetic window; stochastic ontogenesis; cell-state transitions; Warburg effect; tissue plasticity; regeneration; fibrosis; aging; oncogenesis*

1. Introduction

Developmental biology has traditionally represented cellular differentiation as a hierarchical, predominantly irreversible process in which pluripotent embryonic cells progressively lose developmental potential while acquiring stable, specialized identities. Waddington's epigenetic landscape remains the canonical metaphor: a ball rolling downhill through a branching valley system, with each bifurcation representing a commitment decision that is not subsequently reversed. [1]

This deterministic model has been productively extended by systems biology, which represents cellular identity as an attractor state within a high-dimensional gene regulatory network. [3,30,31] However, numerous biological phenomena challenge the strict irreversibility assumption: tissue regeneration, induced pluripotency, stem-cell activation, chronic inflammatory remodeling, fibrosis, embryonic gene reactivation in adult tissues, and malignant transformation all demonstrate that cells can exit and partially re-enter earlier developmental attractor states. [4,40,41]

Paper 1 of this series catalogued the empirical scope of this phenomenon and proposed the energetic window — together with the Regeneration Pendulum and Temporal Tissue Loops — as the unifying explanatory framework. The present paper provides the mechanistic substrate: what metabolic architecture permits adult tissues to maintain access to developmental-like states without constitutive dedifferentiation, and how can cell-state transitions be formally modelled as stochastic processes governed by energetic and structural constraints?

2. Ontogenesis as a Stochastic Dynamical System

2.1 Conceptual Foundations

Within the proposed framework, cellular states are represented as partially connected attractor states within a multidimensional biological state space. Classical deterministic models predict that once a cell reaches a differentiated attractor, it remains there indefinitely in the absence of oncogenic mutation. The present framework replaces this with a stochastic model in which transitions are always possible but occur with probabilities that depend sensitively on local energetic and structural conditions.

2.2 Formal State-Transition Model

Let $S(i)$ denote a cellular ontogenetic state, where i indexes position along an ontogenetic trajectory from totipotent ($i = 0$) toward fully differentiated ($i = N$). Transitions between neighboring states occur probabilistically:

$$S(i) \rightarrow S(i+1) \text{ [forward differentiation, probability } P_{\text{forward}}(i)] \dots (2.1)$$

$$S(i) \rightarrow S(i-1) \text{ [partial dedifferentiation, probability } P_{\text{reverse}}(i)] \dots (2.2)$$

The net direction of ontogenetic movement at any given state is determined by the ratio $P_{\text{forward}} / P_{\text{reverse}}$, which is itself a function of the energetic window width and prevailing metabolic regime. The full transition probability matrix:

$$P(i \rightarrow j) = f(E, M, C, O, G, S) \dots (2.3)$$

where E = energetic-window width; M = prevailing metabolic regime (OFF or BGR dominance); C = extracellular matrix constraints; O = oxygen availability; G = glucose and substrate availability; S = structural and inflammatory stress signals. Transitions are predominantly local ($|i - j| = 1$).

2.3 Noise and Stochastic Variability

Stochastic gene expression noise plays a functional role in enabling cell-state transitions. [28,29] In a narrow energetic window, noise is insufficient to drive cells out of their current attractor states. As the energetic window broadens, barrier heights decrease and the same level of stochastic noise becomes sufficient to generate transitions. This provides a mechanistic explanation for why energetic-window expansion is both necessary and sufficient for partial ontogenetic reversion, without requiring specific dedifferentiation signals.

3. Energetic Window: Formalization

3.1 Definition and Physical Analogy

The energetic window ΔE_{window} is defined as the range of metabolic and structural conditions permitting access to alternative cellular attractor states. In physical terms it is analogous to the

thermal energy available to surmount energy barriers in a potential landscape: wider windows permit access to more alternative states; narrower windows confine the system to its current attractor.

$$\Delta E_{\text{window}} = f(V, \Lambda, F, O_2, \text{ROS_burden}, C_{\text{rigidity}}) \dots(3.1)$$

where V is vascular permeability and perfusion, Λ is MMP activity, F is metabolic flexibility, O_2 is local oxygen tension, ROS_burden is cumulative reactive oxygen species load, and C_{rigidity} is ECM mechanical rigidity.

3.2 Narrow and Expanded Window States

Narrow energetic window — characterizes stable mature differentiated tissues. Features: strong ECM structural constraints; low metabolic flexibility; stable intercellular coupling; restricted chromatin accessibility; low plasticity; dominant Oxidative Functional Flux; high structural rigidity; low MMP activity. Transition probability toward dedifferentiated states approaches zero. [24,26]

Expanded energetic window — characterizes embryonic tissues, regenerating zones, chronic inflammatory foci, and cancer. Features: ECM remodeling; increased vascular permeability; elevated nutrient flux; reduced structural rigidity; elevated MMP activity; chromatin accessibility expansion; dominant BGR; elevated inflammatory signaling. Transition probabilities toward developmental-like states increase substantially. [9,7,8]

3.3 Energetic Window and Chromatin State

The molecular substrate of energetic-window width is, at least in part, the chromatin accessibility landscape. Glycolytic activity generates acetyl-CoA that sustains histone acetylation and maintains open chromatin configurations at developmental regulatory regions. [15,27] Conversely, oxidative metabolism is associated with progressive chromatin compaction at developmental loci through oxidative modification of histone-modifying enzymes. This metabolic–epigenetic coupling provides a direct mechanistic link between BGR dominance and energetic-window expansion.

4. Dual-Flux Energetic Architecture

4.1 Overview

The proposed model introduces two partially interconnected energetic regimes that together govern tissue-level dynamics. These regimes are not mutually exclusive at the single-cell level — most cells employ both simultaneously in varying proportions — but their relative dominance at the tissue level determines the position of the tissue within the ontogenetic state landscape.

4.2 Oxidative Functional Flux (OFF)

Oxidative Functional Flux represents the high-power oxidative metabolic regime characteristic of mature, specialized, differentiated tissues. Its principal biological functions include muscular contraction, cardiac output, neuronal signaling, ion transport, and secretory activity. Biochemically, OFF is characterized by mitochondrial oxidative phosphorylation, high ATP yield per glucose molecule, elevated oxygen consumption, and proportionally elevated reactive oxygen species generation as a metabolic byproduct. [19,6]

The energetic cost of OFF-dominated metabolism is progressive oxidative molecular wear: DNA oxidative lesions, protein carbonylation, lipid peroxidation, and mitochondrial membrane damage accumulate over time, progressively narrowing the energetic window. [16,17]

OFF $\rightarrow$ ΔE_{window} narrowing $\rightarrow$ differentiated state stabilization $\rightarrow$ reduced plasticity

4.3 Basal Glycolytic Regime (BGR)

Basal Glycolytic Regime represents a low-dissipation, glycolytic-recycling energetic regime associated with preservation rather than maximal functional output. BGR is characterized by aerobic glycolysis even under normoxic conditions — the Warburg-like metabolic configuration observed in embryonic cells, stem cells, and actively regenerating tissues. [6,11,13,14]

BGR metabolic features include: elevated glucose uptake; lactate production via aerobic glycolysis; reduced reliance on mitochondrial oxidative phosphorylation; reduced reactive oxygen species burden; and generation of biosynthetic precursors supporting cellular proliferation and matrix synthesis. [6,10,36,37]

Critically, BGR supports maintenance and expansion of the energetic window through multiple mechanisms: glycolysis-derived acetyl-CoA supports histone acetylation [15,27]; reduced ROS burden preserves epigenetic flexibility [18]; AMPK pathway activation under mildly reduced ATP promotes autophagy and cellular renewal [20]; lactate recycling supports neighbouring cell metabolic diversity [10,37].

BGR $\rightarrow$ ΔE_{window} expansion $\rightarrow$ increased plasticity $\rightarrow$ access to developmental-like states

4.4 Interaction Between OFF and BGR

In adult tissues, OFF and BGR coexist in proportions that reflect the functional demands and regenerative state of the tissue. The balance is regulated through AMPK/mTOR signaling, HIF-1 α -mediated metabolic switching, mitochondrial biogenesis dynamics, and substrate availability. [20,12,19]

OFF dominance $\rightarrow$ narrow ΔE_{window} $\rightarrow$ differentiation stabilization ... (4.1)

BGR dominance $\rightarrow$ broad ΔE_{window} $\rightarrow$ developmental plasticity ... (4.2)

OFF/BGR balance $\rightarrow$ determines $P(i \rightarrow j)$ across the ontogenetic landscape ... (4.3)

5. Biological Interpretation of Dynamic Regimes

The dual-flux framework permits a unified interpretation of major biological tissue states as distinct dynamic regimes of the same underlying system.

5.1 Embryogenesis

Embryogenesis corresponds to BGR dominance, broad energetic windows, maximal plasticity, and high transition probabilities across the entire ontogenetic state space. Embryonic cells exhibit aerobic glycolysis, minimal mitochondrial activity, low reactive oxygen species burden, and globally open chromatin configurations permitting access to any developmental trajectory. [11,13,14,15]

5.2 Mature Differentiated Tissue

Mature tissue corresponds to OFF dominance, narrow energetic windows, structural rigidity, and low transition probabilities. The tissue is stabilized within its differentiated attractor, and developmental programs are epigenetically silenced. The cost of this stability is progressive accumulation of oxidative damage through sustained OFF activity. [17,26]

5.3 Regeneration

Regeneration corresponds to a transient shift toward BGR dominance triggered by tissue injury. The shift in metabolic regime widens the energetic window, increases chromatin accessibility at developmental loci, and transiently elevates transition probabilities toward developmental-like states — producing the Temporal Tissue Loops described in Paper 1. Resolution of the injury signal reverses the OFF/BGR balance, re-narrows the energetic window, and returns the tissue to a differentiated attractor state, completing one cycle of the Regeneration Pendulum.

5.4 Chronic Inflammation

Chronic inflammation corresponds to a persistent oscillatory OFF/BGR mixed regime driven by recurrent cycles of tissue injury, repair, and re-injury. The energetic window remains constitutively broadened relative to mature tissue but does not fully stabilize in either the narrow (mature) or maximally broad (embryonic/cancer) configuration. The result is sustained partial developmental program activation and persistent Temporal Tissue Loop dynamics. [41,22,8]

5.5 Fibrosis

Fibrosis represents an emergency mechanical stabilization mechanism. When persistent oscillatory remodeling fails to dampen, the tissue attempts to forcibly narrow the energetic window through irreversible extracellular matrix deposition, progressive structural rigidification, and replacement of metabolically flexible parenchymal cells with collagen-secreting myofibroblasts. [21,22,23] This narrows the energetic window mechanically, terminating the oscillatory dynamics at the cost of permanent loss of tissue plasticity.

5.6 Aging

Aging may be interpreted as the slow, irreversible dominance of Oxidative Functional Flux over Basal Glycolytic Regime. Progressive accumulation of oxidative molecular damage — DNA lesions, protein carbonylation, mitochondrial dysfunction — progressively rigidifies the chromatin and extracellular matrix landscapes, mechanically and epigenetically narrowing the energetic window. [16,17,18] The aged tissue landscape becomes increasingly trapped within differentiated attractor states with progressively diminishing regenerative capacity.

5.7 Cancer as Stabilized Energetic-Window Expansion

Cancer may correspond to a pathological stabilization of an expanded energetic window in which BGR-like metabolic configurations are constitutively maintained (consistent with the Warburg effect), structural restabilization mechanisms fail, developmental programs remain constitutively active, and the tissue escapes normal architecture controls. [5,6,25] Within the Regeneration Pendulum framework, cancer represents a dynamical bifurcation: the system escapes the limit cycle and enters an unbounded growth attractor. The transition to cancer requires not merely BGR activation — which occurs transiently in all regenerating tissues —

but the failure of mechanisms that normally terminate BGR dominance and restore OFF dominance following repair completion.

6. The Regeneration Pendulum: Dual-Flux Mechanistic Interpretation

Paper 1 introduced the Regeneration Pendulum as an oscillatory tissue-level dynamic. The present paper provides its mechanistic substrate in dual-flux terms. The complete oscillatory cycle:

Cell death → DAMP release → inflammatory signaling → MMP activation ... (6.1a)

MMP activation → ECM remodeling → $\Delta E_{\text{window}} \uparrow$... (6.1b)

$\Delta E_{\text{window}} \uparrow$ → BGR induction → chromatin accessibility $\uparrow$ → developmental programs $\uparrow$... (6.1c)

Repair/growth → tissue mass restoration → OFF restoration → ECM restabilization ... (6.1d)

ECM restabilization → $\Delta E_{\text{window}} \downarrow$ → BGR suppression → differentiated state restabilization ... (6.1e)

7. Aging and the Progressive Dominance of Oxidative Functional Flux

The aging phenotype emerges as the inevitable long-term consequence of sustained OFF dominance in post-mitotic and slowly dividing adult tissues. Accumulation of ROS-derived molecular damage over decades produces: mitochondrial dysfunction and reduced respiratory efficiency; progressive chromatin compaction at developmental and regenerative gene loci; increased ECM crosslinking and mechanical stiffening; and reduced vascular responsiveness to regenerative stimuli. [16,17,18,24]

An important corollary: interventions that partially restore BGR accessibility in aged tissues — caloric restriction, NAD⁺ repletion, mitochondrial-targeted antioxidants, senolytics — may widen the energetic window and partially restore regenerative capacity. [17,20]

8. Discussion

8.1 Synthesis with Paper 1

The dual-flux framework of this paper provides the mechanistic substrate for the empirical observations organized in Paper 1. The catalogue of contexts in which embryonic-associated programs appear outside pregnancy and cancer maps directly onto contexts predicted by the dual-flux model to exhibit BGR induction and energetic-window expansion.

8.2 Relationship to Existing Frameworks

The present model is explicitly indebted to several prior conceptual frameworks: Waddington's epigenetic landscape [1], Prigogine's nonequilibrium thermodynamics [2], Kauffman's attractor model [3], the Warburg effect literature [6,12], and Takahashi and Yamanaka's induced pluripotency [4]. The dual-flux model extends these frameworks by linking them through a common energetic-window currency; providing bidirectional transition dynamics absent from classical Waddington models; integrating metabolic regime as a primary determinant of ontogenetic state accessibility; and connecting individual cell-state dynamics to tissue-level oscillatory phenomena.

8.3 Limitations

The present framework is conceptual rather than quantitatively predictive in its current form. The probability transition function $P(i \rightarrow j)$ is schematic; its explicit parameterization from measurable biological quantities remains to be developed. The OFF/BGR balance is a simplification: metabolic regulation is considerably more heterogeneous at the single-cell level. The framework does not yet specify the threshold conditions distinguishing damped regenerative oscillations from pathological sustained cycling or oncogenic bifurcation.

9. Conclusion

Cellular differentiation is most productively interpreted not as a strictly irreversible deterministic program, but as a stochastic dynamical process whose dominant direction emerges from the progressive modulation of energetic and structural constraints within a changing tissue landscape. The dual-flux energetic architecture — Oxidative Functional Flux and Basal Glycolytic Regime — provides the metabolic substrate governing this landscape. OFF dominance narrows the energetic window and stabilizes differentiated states. BGR dominance expands the energetic window and increases the probability of transitions toward developmental-like attractor states.

Conflict of Interest: The author declares no conflict of interest.

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Article 3 of 5

Loss of Tissue Integration and the Transition from 3D to 2D Organization in Malignancy:

A Conceptual Morphodynamic Model Based on Hydatidiform Mole, Glandular Carcinoma, and Energetic Window Dynamics

Yuri Dmitrochenko

A Conceptual Biophysical Framework

Series: Embryonic Programs in Non-Oncological Tissue States | Paper 3 of 5

Abstract

The preceding papers established that embryonic-associated programs are reactivated through energetic-window expansion and the oscillatory dynamics of the Regeneration Pendulum, and that the dual-flux energetic architecture governs cell-state transition probabilities. [41,42] The present paper addresses the morphodynamic consequences of pathological stabilization of an expanded energetic window: the progressive dissolution of organized three-dimensional tissue architecture and its replacement by infiltrative, planar, two-dimensional-like growth configurations.

A conceptual morphodynamic model is proposed in which malignant transformation is interpreted as a progressive loss of integration between a cellular clone and the multicellular organism. The structural signature of this process is the transition from closed, polarized, geometrically constrained 3D tissue organization toward open, infiltrative, functionally autonomous 2D-like growth — driven mechanistically by constitutive BGR dominance, sustained energetic-window expansion, loss of ECM structural feedback, and failure of structural restabilization. Two biological systems are examined as paradigmatic models: hydatidiform mole and trophoblastic neoplasia; and glandular epithelial malignancy.

Keywords: *morphodynamics; 3D-to-2D transition; tissue integration; hydatidiform mole; trophoblastic neoplasia; glandular carcinoma; energetic window; Basal Glycolytic Regime; epithelial-mesenchymal transition; collective invasion; malignant transformation*

1. Introduction

The classical framework for understanding cancer describes it in terms of molecular events: oncogene activation, tumor suppressor loss, genomic instability, and the stepwise accumulation of driver mutations. [8,19] An alternative and complementary tradition interprets malignancy as a failure of tissue organization and multicellular integration — a perspective in which the key

event is not the mutation itself but the loss of tissue-level regulatory constraints that normally suppress aberrant cellular behavior. [2,22,6]

Three-dimensional culture systems demonstrate that the same cells which grow invasively on flat surfaces reorganize into morphologically normal glandular structures when provided with appropriate extracellular matrix contexts — and revert to malignant behavior when these contexts are disrupted. [15,3] The geometry of tissue architecture is therefore not merely a reflection of cellular behavior but an active regulator of it.

The present paper asks: what are the morphodynamic consequences of the oncogenic bifurcation at the tissue level? We propose that the structural corollary of constitutive energetic-window expansion is a progressive 3D-to-2D morphodynamic transition: the dissolution of organized, geometrically constrained, polarized three-dimensional tissue architecture and its replacement by infiltrative, planar, autonomously expanding two-dimensional-like growth.

2. Tissue as a Spatially Constrained Communication and Mechanical System

2.1 The Regulatory Role of Three-Dimensional Architecture

Normal multicellular tissue architecture represents far more than a spatial arrangement of cells. It constitutes a regulatory system embedded in three-dimensional space: an integrated network of mechanical forces, chemical gradients, cell-cell contacts, and matrix-cell communication that continuously constrains and coordinates cellular behavior. [6,7,38] Components include: apical-basal polarity; basement membrane; tight and adherens junctions mediating contact inhibition; vascular organization; and tensional homeostasis. Within this framework, tissue architecture is not merely a consequence of cellular differentiation but a cause of its maintenance.

2.2 Three-Dimensional Architecture as Energetic-Window Narrowing

Within the dual-flux framework of Paper 2, normal three-dimensional tissue architecture functions as a mechanism for maintaining a narrow energetic window. The structural constraints of organized tissue — basement membrane integrity, ECM mechanical stiffness, tight intercellular coupling, restricted vascular permeability — directly suppress the variables that expand the energetic window. This creates a bidirectional coupling between morphology and metabolism: organized 3D architecture narrows the energetic window by imposing structural OFF-favoring constraints, while a narrow energetic window stabilizes organized 3D architecture. Conversely, disruption of 3D architecture expands the energetic window — a positive feedback loop central to the 3D-to-2D morphodynamic transition in malignancy. [14,16]

2.3 Communication Redistribution in Malignancy

A critical feature of malignant transformation is that it does not represent a simple loss of cellular communication but a redistribution of communicative coupling. Cells within a malignant clone maintain — and often strengthen — internal coordination: collective invasion, tumor angiogenesis, stromal remodeling, and metabolic cooperation all require sophisticated intercellular communication. [4,34,36] What is lost is not communication per se but compatibility with the shared regulatory language of the host tissue — responsiveness to polarity signals, contact inhibition, basement membrane anchoring, and systemic architectural cues. [33,26]

3. The 3D-to-2D Morphodynamic Transition: Structural Mechanics

3.1 Normal Three-Dimensional Glandular Organization

Normal glandular epithelial tissues represent highly organized three-dimensional systems. Acini, ducts, and glandular crypts are characterized by: luminal organization with apical-surface facing the lumen; apical-basal polarity; geometric closure of epithelial sheets around a central lumen; basement membrane enclosure; and coordinated cell cycle regulation coupled to architectural integrity. [26,27,6] The three-dimensional architecture is not incidental: it is a consequence and a perpetuator of polarized epithelial identity.

3.2 Progression: Dysplasia to Invasive Carcinoma

Malignant progression in glandular epithelia follows a reproducible morphodynamic trajectory. Stage 1 — Dysplasia: loss of polarity and luminal disorganization; BGR induction focal and partial, energetic window widened locally. Stage 2 — Carcinoma in situ: full loss of normal polarity while basement membrane integrity is initially preserved; energetic window constitutively expanded within the intraepithelial compartment. Stage 3 — Invasive carcinoma: basement membrane breach, MMP-mediated degradation eliminates the primary architectural constraint, massively expanding the energetic window; cells transition to infiltrative single-cell, cord, sheet, and nest configurations. Stage 4 — Advanced infiltrative growth: replacement of all three-dimensional organizational features by autonomous infiltrative expansion; dominant growth pattern sheets, cords, and single-cell infiltration. [5,24,16,17,18,13]

3.3 Energetic-Window Mechanics of the Transition

Oncogenic BGR stabilization $\rightarrow \Delta E_{\text{window}}$ constitutively expanded ... (3.1)

ΔE_{window} expansion $\rightarrow$ MMP upregulation $\rightarrow$ ECM degradation $\rightarrow$ structural constraint $\downarrow$... (3.2)

Structural constraint $\downarrow \rightarrow$ further ΔE_{window} expansion (positive feedback) ... (3.3)

Positive feedback $\rightarrow$ progressive 3D $\rightarrow$ 2D architectural dissolution ... (3.4)

The positive feedback between structural constraint loss and energetic-window expansion is the key destabilizing mechanism. In normal regeneration, this feedback is terminated by structural restabilization. In the oncogenic regime, constitutive BGR self-reinforcement prevents restabilization, and the system undergoes progressive architectural dissolution in a forward-accelerating manner.

4. Hydatidiform Mole as a Morphodynamic Paradigm

4.1 Structural Features and the Absent Embryonic Integration

Complete hydatidiform mole provides an exceptionally illuminating morphodynamic model because it separates two components that are normally coupled: embryonic structural integration and trophoblastic invasive and angiogenic capacity. In complete mole, the former is absent or severely impaired — embryonic development fails — while the latter is hyperactive. [11,12,10] The molar lesion consists of swollen, avascular chorionic villi surrounded by hyperplastic trophoblastic tissue, with strong maternal vascular recruitment but without the formation of a fetal-placental circulatory unit.

4.2 Progression to Invasive Mole and Choriocarcinoma

The morphodynamic trajectory of trophoblastic disease follows the 3D-to-2D dissolution sequence in accelerated form. Hydatidiform mole: villous three-dimensional architecture preserved, albeit abnormally — swollen villi maintain their three-dimensional closed form. Invasive mole: villi invade the myometrium, breaching the uterine architectural boundary; three-dimensional villous organization partially preserved within the invasive compartment but spatial constraint is lost. Choriocarcinoma: villous architecture disappears entirely; the tumor consists of sheets and infiltrative networks of cytotrophoblast and syncytiotrophoblast — a maximally 2D-like configuration in which no vesicular or closed three-dimensional structure is present. [11,10]

4.3 The Trophoblastic-Embryonic Axis as a Model for Integration

The normal trophoblast-embryo system illustrates a general principle: invasive and expansive capacity requires architectural integration to remain regulated. Hydatidiform mole demonstrates what happens when this integration is absent: the same invasive and metabolic capacities, without architectural context, produce unregulated expansion. Malignancy is fundamentally a failure of integration, not merely of proliferative control. [22,2,40]

5. Glandular Epithelial Malignancy: Loss of Geometric Closure

5.1 Architecture as Identity

Normal human mammary epithelial cells, when cultured within reconstituted basement membrane, self-organize into polarized acini with luminal organization, growth arrest, and differentiation markers — despite being removed from their *in vivo* tissue context. When malignant cells are placed in the same three-dimensional matrix context, acquisition of integrin-blocking agents can partially reverse their malignant morphology. [15,3] Within the dual-flux framework, organized 3D geometry actively maintains OFF dominance by providing structural constraints that prevent energetic-window expansion.

5.2 EMT as Molecular Mediator of the 3D-to-2D Transition

At the molecular level, the 3D-to-2D morphodynamic transition is substantially mediated by epithelial-mesenchymal transition (EMT) — a coordinated reprogramming event in which epithelial cells lose polarity, junction proteins, and basement membrane attachment while acquiring mesenchymal motility, ECM-remodeling capacity, and resistance to anoikis. [5,29] EMT is directly connected to the dual-flux framework: EMT induction is associated with HIF-1 α activation and metabolic shift toward glycolysis; Snail, Twist, and ZEB family transcription factors simultaneously remodel the chromatin accessibility landscape in ways consistent with energetic-window expansion. [5,30,31] MMP upregulation — a key effector of both EMT and energetic-window expansion — mediates basement membrane degradation and further expands the energetic window. [24,16]

5.3 Collective Invasion as Partial 3D-to-2D Transition

An important intermediate state in the 3D-to-2D transition is collective invasion, in which groups of cells invade as cohesive units maintaining some cell-cell contacts and partial epithelial organization while penetrating surrounding stroma. [4,36] Collective invasion represents a partial uncoupling from three-dimensional architectural constraints. As the energetic window expands further, collective invasion grades into single-cell and sheet-like infiltration, completing the 3D-to-2D transition.

6. Unified Morphodynamic Framework: Synthesis Across Papers 1–3

Paper 1 described how tissues undergoing regenerative remodeling form Temporal Tissue Loops — normally bounded: the pendulum oscillates and returns to a stable differentiated attractor. Paper 2 identified the oncogenic bifurcation as the dynamic regime in which BGR stabilization prevents this return. The present paper describes what this transition looks like at the morphological level: the constitutively expanded energetic window progressively dissolves the three-dimensional structural constraints, generating the 3D-to-2D morphodynamic transition.

BM = basement membrane; BGR = Basal Glycolytic Regime; OFF = Oxidative Functional Flux.

7. Malignancy as a Change in the Level of Biological Integration

Cancer is best characterized as a progressive transition in the level of biological integration — from participation in organism-level spatial and regulatory architecture toward predominantly clone-centered organization. [22,2,40] This transition has several distinguishable components: energetic autonomy through constitutive BGR dominance; architectural independence through loss of basement membrane anchorage and polarity; signaling incompatibility through progressive loss of responsiveness to organism-level restraining signals; and vascular autonomy through development of autonomous angiogenic signaling.

8. Discussion

8.1 Synthesis With Preceding Papers

Paper 1 established the empirical scope. Paper 2 provided the mechanistic architecture. The present paper closes the loop by characterizing the structural, tissue-level manifestation of the oncogenic bifurcation: the 3D-to-2D morphodynamic transition. Together, the three papers provide a multi-scale account of a single biological phenomenon: the progressive loss of integration between a cellular clone and the multicellular organism.

8.2 Implications for Morphological Grading

Standard histopathological tumor grading systems — the Gleason system for prostate cancer, WHO/ISUP grading for urothelial and renal tumors — are fundamentally based on the degree of architectural preservation. The present framework provides a mechanistic interpretation: histological grade reflects the degree of 3D-to-2D morphodynamic transition, which in turn reflects the degree of constitutive energetic-window expansion and BGR stabilization.

8.3 Therapeutic Implications

If the 3D-to-2D transition is a structural consequence of BGR self-stabilization and energetic-window expansion, then interventions aimed at re-narrowing the energetic window — restoring OFF/BGR balance, imposing structural constraints through matrix manipulation, or blocking the MMP-mediated positive feedback loop — may have morphodynamic consequences beyond individual molecular targets. [15,24,17]

9. Conclusion

The transition from organized, polarized, geometrically constrained three-dimensional tissue architecture toward infiltrative, planar, two-dimensional-like growth is proposed as a fundamental morphodynamic signature of malignancy. This transition is the structural corollary of constitutive BGR dominance and energetic-window expansion — the oncogenic bifurcation of the Regeneration Pendulum identified in Paper 2. Cancer, within this framework, is most precisely characterized as a progressive transition from organism-level integration to clone-centered autonomy — a change in the level of biological integration manifest at every scale: molecular (BGR stabilization), cellular (developmental attractor accessibility), and morphological (3D-to-2D architectural dissolution).

Conflict of Interest: The author declares no conflict of interest.

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Loss of Intermediate Ontogenetic States and Disruption of Local Markov Connectivity:

A Conceptual Model of Structural Decoupling Between Cellular Clones and Tissue

Yuri Dmitrochenko

A Conceptual Biophysical Framework

Series: Embryonic Programs in Non-Oncological Tissue States | Paper 4 of 5

Abstract

The preceding papers established that embryonic-associated programs are reactivated through energetic-window expansion and dual-flux metabolic dynamics (Papers 1–2), and that pathological stabilization of an expanded energetic window generates a progressive 3D-to-2D morphodynamic transition at the tissue level (Paper 3). [41,42,43] The present paper introduces a complementary topological framework, proposing that ontogenesis is best represented not as a continuous epigenetic landscape but as a locally connected Markov system in which transitions are structurally constrained to occur between adjacent developmental states.

Within this framework, the integrity of intermediate ontogenetic states is structurally essential: their loss or inaccessibility ruptures local developmental connectivity, forcing non-local transitions that lack compatible regulatory intermediates, metabolic synchronization, chromatin-transition architecture, and tissue-level signaling continuity. Several novel corollaries are developed: embryonic protein expression may represent an attempt to reconstruct lost ontogenetic bridges; immune surveillance may include a morphodynamic recognition component; and the tumor microenvironment may emerge as a compensatory autonomous developmental system.

Keywords: *local Markov connectivity; intermediate ontogenetic states; ontogenetic decoupling; developmental topology; embryonic protein reconstruction; immune morphodynamic recognition; tumor microenvironment; developmental state space; malignancy; tissue integration*

1. Introduction

Classical models of cellular differentiation share a common implicit assumption: developmental transitions occur along continuous or quasi-continuous trajectories, with cells moving progressively between states that are energetically and molecularly compatible. [1,11,12] This continuity assumption carries a corollary that has received insufficient attention: if transitions must be locally continuous, then the integrity of intermediate states is structurally essential. The loss of an intermediate state does not merely reduce the number of available transitions — it ruptures connectivity between states that were previously linked through that intermediate.

The present paper addresses three questions: topologically, what happens when an intermediate state is lost? Immunologically, what does the immune system recognize in a cellular clone that has undergone non-local ontogenetic transition? And reconstructively, why do malignant cells

express embryonic proteins — is this passive developmental regression or an active attempt to reconstruct lost intermediate states?

2. Ontogenesis as a Locally Connected Markov System

2.1 Formal Definition

We represent the ontogenetic state space as a discrete set of cellular developmental states:

$$S = \{ S_0, S_{-1}, S_{-2}, \dots, S_{-k} \} \dots(2.1)$$

where S_0 denotes the terminal differentiated state, S_{-1} the immediately preceding developmental configuration, and deeper negative indices correspond to progressively earlier ontogenetic states. Transitions occur as a Markov process. The system is locally connected:

$$P(S_0 \rightarrow S_{-1}) \gg P(S_0 \rightarrow S_{-2}) \gg P(S_0 \rightarrow S_{-3}) \dots(2.2)$$

This local suppression reflects three distinct constraints: regulatory compatibility — intermediate states express the molecular machinery required to enter the next state; metabolic synchronization — each transition requires a specific energetic configuration that must be established progressively; and chromatin accessibility sequencing — the epigenetic landscape opens and closes in a sequential order that cannot be bypassed. [13,18,21]

2.2 Local Connectivity and the Energetic Window

The local Markov connectivity model is directly compatible with the energetic-window framework of Papers 1–2. Energetic-window expansion increases transition probabilities across the ontogenetic state space, but preferentially for locally adjacent transitions. The pathological feature of oncogenesis is not that the energetic window is wide enough to permit non-local jumps directly, but rather that progressive intermediate state loss forces the system to attempt non-local transitions despite their low intrinsic probability.

2.3 Normal Regeneration as a Locally Bounded Loop

Within the local Markov framework, normal regeneration is formally represented as a bounded local excursion:

$$S_0 \rightarrow S_{-1} \rightarrow S_0 \text{ (physiological regeneration, damped) } \dots(2.3)$$

The loop passes through a single intermediate state and returns. The intermediate state S_{-1} acts as a structural bridge: it is both the target of the regenerative excursion and the platform from which return to S_0 becomes locally probable. Its integrity is therefore the structural basis for the reversibility of the Regeneration Pendulum. [41]

3. Loss of Intermediate Ontogenetic States: Mechanisms and Consequences

3.1 Mechanisms of Intermediate State Loss

3.1.1 Epigenetic Landscape Erosion. Progressive loss of epigenetic information — through DNA methylation drift, histone modification decay, and chromatin accessibility erosion — can eliminate the epigenetic configurations that define intermediate states. Yang et al. (2023) demonstrated that epigenetic information loss is a primary driver of mammalian aging, with tissues progressively losing access to developmental epigenetic states. [3,19]

3.1.2 Chronic Inflammatory Disruption. Persistent oscillatory remodeling generates recurrent cycles of ECM degradation and incomplete repair. Each incomplete repair cycle may fail to fully reconstitute the structural and signaling environment that defines the intermediate state. [41,39,24]

3.1.3 Metabolic Asymmetry Between States. Intermediate states may require specific metabolic configurations that become unavailable due to mitochondrial dysfunction, substrate limitation, or sustained OFF dominance. [16,17,42]

3.1.4 Senescent Cell Accumulation. Senescent cells may themselves represent lost intermediate states: cells that have irreversibly exited the normal differentiation trajectory and can no longer serve as transitional platforms. Their accumulation reduces the density of accessible intermediate states. [33,37]

3.2 Topological Consequences: Connectivity Rupture

When intermediate state S_{-1} is lost, the local Markov chain is ruptured:

$$S_0 \rightarrow S_{-1} \text{ (} S_{-1} \text{ inaccessible) } \dots (3.1)$$

The system may still attempt transition toward developmental-like states under conditions of energetic-window expansion, but is now forced to attempt a non-local jump:

$$S_0 \rightarrow S_{-2} \text{ (non-local, low-probability, regulatory-mismatch) } \dots (3.2)$$

This non-local transition lacks the regulatory preparation that S_{-1} would have provided: transcriptional incompatibility; chromatin mismatch; metabolic discontinuity; and signaling incompatibility. The result is a cellular state that is ontogenetically displaced: it occupies a developmental-like configuration without the regulatory, metabolic, and signaling context that normally accompanies that configuration. [10,14,34]

3.3 The Disconnection Criterion

Under normal conditions with intact intermediate connectivity:

$$S_{-1}^T \rightarrow S_0^T \text{ (locally connected, tissue-compatible return) } \dots (3.3)$$

After loss of S_{-1} , the clone in state S_{-2} faces:

$$S_{-2}^U \rightarrow S_0^T \text{ (disconnected: no locally permissible return path) } \dots (3.4)$$

Disconnection is self-reinforcing: loss of intermediate connectivity prevents reintegration, which prevents recovery of the intermediate state, which consolidates the disconnection. [34,14,40]

4. Ontogenetic Decoupling and Progressive Loss of Tissue Architecture

The ontogenetic structural decoupling provides a topological interpretation of the 3D-to-2D morphodynamic transition described in Paper 3. A clone that has undergone ontogenetic decoupling is not merely displaced in developmental state space — it is displaced in a way that makes it architecturally incompatible with the surrounding tissue. The intermediate state S_{-1} is also the state in which cells express the surface molecules, adhesion proteins, and mechanosensory machinery that integrate them into three-dimensional tissue architecture. [43,5,8]

Decoupling proceeds in distinguishable stages: partial connectivity (equivalent to dysplasia); connectivity rupture (equivalent to in situ carcinoma); full decoupling (equivalent to invasive carcinoma); and metastatic colonization (topological transplantation).

5. Embryonic Protein Expression as Ontogenetic Bridge Reconstruction

5.1 Two Interpretations

The energetic-window framework interprets embryonic program activation as a direct consequence of BGR dominance expanding chromatin accessibility at developmental gene loci — a passive consequence of the metabolic state. The present topological framework proposes a complementary and active interpretation: embryonic protein expression in decoupled clones may represent a reconstructive attempt — the clone's effort to recreate the intermediate ontogenetic states it has lost, in order to re-establish local Markov connectivity with surrounding tissue.

5.2 The Reconstruction Hypothesis

Embryonic proteins — embryonic fibronectin isoforms, developmental matrix components, oncofetal signaling ligands — are precisely the molecular components that characterize and define intermediate developmental states. Their expression by the clone can be interpreted as an attempt to reconstruct, locally and autonomously, the molecular environment of the missing S_{-1} .

$$S_{-2} \rightarrow [\text{express embryonic } S_{-1} \text{ proteins}] \rightarrow \text{attempt to reconstruct } S_{-1} \text{ bridge} \rightarrow S_0^T \dots (5.1)$$

The reconstruction is necessarily incomplete: the proteins of S_{-1} can be expressed, but the tissue-level architectural context of S_{-1} cannot be reconstructed by a single clone in isolation. The result is a partial, unstable, and architecturally incoherent facsimile of the intermediate state.

5.3 Implications for the Heterogeneity of Embryonic Protein Expression

The reconstruction hypothesis predicts that embryonic protein expression should be highest at the invasive front of the tumor — where the clone is most actively attempting to interface with the surrounding normal tissue state space — and lower in the tumor core, where the clone has abandoned the reconstruction attempt and established an autonomous microenvironment.

6. The Tumor Microenvironment as a Compensatory Autonomous Developmental System

Once ontogenetic structural decoupling is complete, the clone can no longer access the regulatory, trophic, and architectural signals it requires from the surrounding tissue state space. The solution the clone adopts is the construction of an autonomous microenvironment: a locally generated facsimile of the developmental niche that the missing intermediate states would have provided. This construction involves: ECM remodeling; autonomous angiogenesis (VEGF and FGF-driven neovascularization); inflammatory loop self-maintenance (clone-secreted cytokines); and stromal reprogramming (cancer-associated fibroblast induction, immune cell recruitment and education, pericyte remodeling). [20,30,39,31,8,29]

The tumor microenvironment can therefore be understood as the clone's best available approximation of the missing S_{-1} — an autonomous construction that provides some of the

molecular signals of the intermediate state but lacks the tissue-level coherence and architectural integration that the genuine intermediate state would have provided.

7. Immune Recognition as Detection of Ontogenetic Discontinuity

7.1 The Standard Model and Its Limitations

Standard models of tumor immune surveillance focus on antigenic recognition: the immune system detects and eliminates cells expressing non-self antigens generated by somatic mutation, viral integration, or post-translational modification. [28,4] However, many tumor cells do not express detectably foreign antigens, yet are still recognized and eliminated; and many cells that do express mutant antigens are not eliminated. Antigen-based recognition alone is insufficient to explain the full scope of anti-tumor immune activity.

7.2 The Danger Model and Developmental Signals

Matzinger's danger model proposed that immune activation is triggered not only by non-self antigens but by damage-associated molecular patterns (DAMPs). [26] The present framework extends this concept into the developmental domain: the immune system may also respond to ontogenetic discontinuity signals — molecular patterns indicating violations of normal developmental topology. Candidate signals include: embryonic proteins expressed outside their normal developmental context; surface molecular combinations corresponding to displaced ontogenetic states; and structural features of clone-tissue interaction signaling architectural incompatibility.

7.3 Normal Regeneration as Immune-Tolerated Ontogenetic Reversion

The framework predicts a specific distinction between regenerative and malignant ontogenetic reversion. In normal regeneration:

$$S_0 \rightarrow S_{-1} \rightarrow S_0 \text{ (regeneration: locally continuous, immune-tolerated) } \dots(7.1)$$

In contrast, the non-locally decoupled state represents:

$$S_0 \rightarrow S_{-2}^U \text{ (malignancy: non-local, tissue-incompatible, immune-detectable) } \dots(7.2)$$

The developmental proteins expressed at S_{-2} are expressed outside their normal tissue-compatible context. This misplacement may constitute a detectable ontogenetic discontinuity signal supplementing antigen-based recognition with a morphodynamic immune surveillance component.

8. Toward a Topological Definition of Malignancy

The framework developed across Papers 1–4 permits a precise systems-level definition of malignancy: Malignancy is a state in which a cellular clone becomes disconnected from tissue through rupture of the continuous chain of structurally permissible local ontogenetic transitions. The clone occupies a developmental-like state accessed through non-local transition, which is incompatible with surrounding tissue architecture, cannot be returned to tissue-compatible states through locally permissible steps, and progressively constructs an autonomous microenvironment as a compensatory substitute for the lost intermediate connectivity.

This definition is scale-invariant (observable at the molecular, cellular, tissue, and immunological levels); non-mutational in principle (the rupture can arise from epigenetic

erosion, metabolic asymmetry, chronic inflammatory disruption, or senescent cell accumulation); and provides a natural explanation for reversibility.

9. Discussion

The four papers of this series approach the same biological phenomenon from four complementary perspectives. Paper 1 provided the empirical catalogue. Paper 2 provided the mechanistic architecture. Paper 3 described the structural consequence. The present paper adds the topological dimension: the formal characterization of what intermediate state loss means for the connectivity of the ontogenetic state space.

The hallmarks of cancer of Hanahan and Weinberg [4] can all be reinterpreted as consequences of ontogenetic structural decoupling. The tissue organization field theory of Soto and Sonnenschein [40] is particularly compatible with the present framework: both identify the disruption of tissue-level organization, rather than the molecular mutation per se, as the fundamental event in carcinogenesis. The topological framework provides formal precision to this identification: the disruption is specifically the rupture of local Markov connectivity in the ontogenetic state space.

10. Conclusion

The integrity of intermediate ontogenetic states is structurally essential for the local Markov connectivity of the developmental state space. Their loss ruptures local connectivity and forces non-local ontogenetic transitions that are incompatible with tissue architecture, immune surveillance, and multicellular integration. This topological rupture is proposed as the deepest structural definition of malignancy: not uncontrolled proliferation, not genomic mutation, not metabolic reprogramming — but disconnection from tissue through the loss of the continuous chain of locally permissible ontogenetic transitions.

The reconstruction hypothesis provides a new interpretation of embryonic protein expression: not passive developmental regression driven by energetic-window expansion alone, but an active reconstructive response by the decoupled clone attempting to rebuild the ontogenetic bridges required for tissue reintegration. The failure of this reconstruction generates the autonomous microenvironment and drives progressive architectural dissolution.

Conflict of Interest: The author declares no conflict of interest.

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Article 5 of 5

The Separated Clone:

Developmental Fate, Attractor Dynamics, and the Gradual Autonomization of a Partially Decoupled Cellular System

Yuri Dmitrochenko

A Conceptual Biophysical Framework

Series: Embryonic Programs in Non-Oncological Tissue States | Paper 5 of 5 | Synthesis

Abstract

Classical models of oncogenesis describe malignancy as the endpoint of accumulated genetic alterations leading to uncontrolled proliferation. While this model captures important molecular mechanisms, it poorly describes the broad spectrum of intermediate states — dysplasia, chronic inflammatory foci, endometriosis, benign tumors, carcinoma in situ — that exist between normal tissue and full malignancy, and it provides limited conceptual tools for predicting which intermediate states will progress and which will remain stable.

The present paper, the concluding synthesis of a five-paper series, proposes a dynamical model of separated clone fate governed by the interaction of five determinants:

$$F_{\text{clone}} = f(E, I, M, A, H)$$

where E is energetic compatibility, I is immunological interaction, M is morphophysiological tissue context, A is intrinsic clone cellular activity, and H is the historical temporal trajectory of the system. The interaction of these determinants places the separated clone within one of four developmental attractor states: (1) elimination; (2) limited persistence; (3) partial autonomy; and (4) full morphodynamic autonomy. Three modes of clone-tissue interaction are characterized — synchronization, blocking, and active remodeling — and the positive feedback loop between clone activity and energetic-window expansion is formalized. The historical dependence of clone fate is identified as a key conceptual contribution.

Keywords: *separated clone; clone fate; attractor dynamics; developmental autonomization; energetic compatibility; immunological interaction; morphophysiological context; clone*

activity; historical trajectory; partial autonomy; morphodynamic synchronization; tumor dormancy

1. Introduction: The Problem of Intermediate States

Between normal tissue and fully malignant tumor lies a clinically and conceptually difficult territory: a continuum of intermediate states that neither fit the category of normal tissue nor conform to classical definitions of malignancy. Chronic inflammatory foci may persist for decades without malignant transformation. Endometriosis maintains invasive and proliferative activity while remaining locally constrained in most patients. Stable dysplasias may regress spontaneously or remain quiescent under conditions that are not yet understood. Benign tumors grow autonomously but do not invade or metastasize. Carcinoma in situ exists as a fully decoupled clone that has not yet achieved invasive capacity. [4,5,17]

The series developed across Papers 1–4 provides the conceptual tools to address this problem. The present paper asks: once a clone has undergone ontogenetic structural decoupling, what determines its subsequent fate? What distinguishes a clone that will be eliminated from one that will persist indefinitely in a quiescent state, from one that will achieve partial local autonomy, from one that will undergo full morphodynamic autonomization?

2. The Clone Fate Function: $F_{\text{clone}} = f(E, I, M, A, H)$

2.1 Overview

We propose that the developmental fate of a separated clone is determined by the interaction of five fundamental determinants. Each determinant is not a fixed parameter but a dynamic variable that changes over time as the clone and its tissue environment co-evolve. The fate function is therefore a trajectory through a five-dimensional parameter space:

$$F_{\text{clone}}(t) = f(E(t), I(t), M(t), A(t), H[0 \rightarrow t]) \dots(2.1)$$

The explicit time-dependence of all variables, and the path-integral dependence on H , reflects the central insight of this framework: clone fate is historically contingent. Two clones with identical current parameter values may be on divergent trajectories if they arrived at those values through different histories.

2.2 E — Energetic Compatibility

Energetic compatibility represents the degree to which the tissue environment can sustain the metabolic requirements of the separated clone. Within the dual-flux framework of Paper 2, clone expansion requires sustained BGR dominance — elevated glycolytic flux providing biosynthetic precursors, acetyl-CoA for epigenetic maintenance, and reduced ROS burden preserving developmental chromatin configurations. [42,6,27,28]

Energetic compatibility encompasses: vascular access; glycolytic flux ceiling; spatial constraints; and historical energetic state. A clone that separates within a chronically inflamed tissue inherits an energetic environment already partially configured for BGR dominance — providing a mechanistic explanation for the well-established epidemiological association between chronic inflammation and cancer risk.

2.3 I — Immunological Interaction

The immune system interacts with the separated clone through mechanisms that can produce elimination, containment, or paradoxical facilitation of clone growth. Within the framework of Paper 4, immune recognition operates through classical antigen-based recognition and ontogenetic discontinuity recognition.

The immunological determinant I can take three qualitatively distinct values: Elimination (effective cytotoxic and antibody-mediated responses destroy the clone before it can establish sustainable autonomous dynamics); Containment/Equilibrium (immune pressure is sufficient to prevent clone growth but insufficient to eliminate it — the equilibrium phase of immunoediting); and Facilitation (paradoxically, chronic inflammatory immune activity can facilitate clone growth by driving MMP-mediated ECM remodeling and expanding the local energetic window [7,8,14]).

2.4 M — Morphophysiological Context

The tissue in which the clone resides provides a morphophysiological context that profoundly shapes its developmental options. Parameters include: vascular architecture; mechanical tissue properties; permissible plasticity level (tissues that normally maintain high regenerative cycling have constitutively higher baseline energetic window widths); local immune density and character; and structural compatibility between the clone's displaced ontogenetic state and the state space of the surrounding tissue — the metastatic organ-tropism problem.

2.5 A — Clone Activity

Clone activity A represents the autonomous contribution of the clone itself to the dynamic interaction. Key dimensions include: protease secretion (MMP family metalloproteinase secretion that degrades surrounding ECM and creates physical space for clone expansion [25]); angiogenic signaling (VEGF, FGF, and PDGF secretion that drives neovascularization, increasing E [26]); immunomodulation (secretion of immunosuppressive factors); and autocrine BGR maintenance (establishment of autocrine signaling loops that maintain BGR dominance independently of tissue-level metabolic signals). These activities generate the fundamental positive feedback loop:

$$A \uparrow \rightarrow \text{MMP} \uparrow \rightarrow \text{ECM remodeling} \rightarrow \Delta E_{\text{window}} \uparrow \rightarrow \text{BGR stability} \uparrow \rightarrow \text{clone growth} \uparrow \rightarrow A \uparrow \dots(2.5)$$

2.6 H — Historical Temporal Trajectory

The historical trajectory H is the most conceptually novel determinant. It captures the path-dependence of clone development: the fact that the current state of the system is insufficient to determine its future trajectory without knowledge of how it arrived at that state. The historical trajectory includes: duration of separation; number of inflammatory cycles survived; immunological adaptation history; and accumulated ECM remodeling. The formal representation as a path integral:

$$H[0 \rightarrow t] = \int_0^t [A(\tau) \cdot I(\tau) \cdot E(\tau)] d\tau \dots(2.6)$$

This integral formulation means that identical current states can have very different H values depending on the history of interactions. A clone that has maintained moderate A, I, and E values continuously for a decade has accumulated a very different H than one that only recently achieved those same values — explaining why histologically similar lesions can have dramatically different progression risks depending on their duration and context. [37,33]

3. Three Modes of Clone-Tissue Interaction

3.1 Synchronization

In the synchronization mode, the separated clone retains partial coupling with the regenerative and metabolic dynamics of the surrounding tissue. The clone's Regeneration Pendulum oscillations are partially synchronized with those of the surrounding tissue. Synchronization is most probable when E and I are moderate, when A is low, and when the degree of ontogenetic decoupling is limited (Stage 1 of Paper 4: dysplasia-equivalent). The surrounding tissue provides external oscillatory damping that prevents the clone's Regeneration Pendulum from escaping its limit cycle. [29,18]

3.2 Blocking

In the blocking mode, the surrounding tissue actively restricts clone development through mechanical, metabolic, immunological, or signaling mechanisms, without eliminating the clone. The block may operate through: mechanical containment; metabolic restriction; contact inhibition maintenance; and immune equilibrium. Blocking is the mechanistic basis of tumor dormancy — the long-term persistence of separated clones in a non-proliferative, non-regressing state. Clinical dormancy represents escape from the blocking mode following a change in one or more determinants (immunosuppression, tissue aging reducing M, inflammatory trigger). [31,32,24]

3.3 Active Tissue Remodeling

In the active remodeling mode, the clone transitions from receiving architectural regulation from the surrounding tissue to imposing architectural modification upon it. This mode is characterized by A dominating the F_clone dynamics, with clone-derived MMP activity, inflammatory signaling, angiogenic recruitment, and stromal reprogramming progressively dismantling the tissue architecture that previously constrained the clone. [9,25,8] Active remodeling represents the transition from partial autonomy toward full morphodynamic autonomy. The 3D-to-2D transition described in Paper 3 is the morphological manifestation of sustained active remodeling mode.

4. Developmental Attractors of the Separated Clone

4.1 Attractor I: Elimination

The clone is destroyed by immune mechanisms, energetic insufficiency, or structural instability before establishing sustainable autonomous dynamics. Elimination occurs when I (in its elimination mode) is dominant and A has not yet achieved the autocrine BGR maintenance that would reduce dependence on tissue-level E. Immunosurveillance studies indicate that nascent malignant cells are continuously generated and eliminated in healthy individuals without ever manifesting as clinically apparent tumors. [4,5,24]

4.2 Attractor II: Limited Persistence

The clone establishes a stable, long-term existence within the tissue but fails to achieve progressive autonomization. This attractor class encompasses: stable dysplasias that persist for years without progression; endometriosis (invasive and proliferative activity remains contained in most patients through immune modulation and tissue mechanical restrictions [35]); benign tumors (clones that have achieved local metabolic autonomy but remain structurally blocked from invasive expansion [36]); and tumor dormancy after primary treatment. [31,32]

4.3 Attractor III: Partial Autonomy

The clone has established sufficient autonomous microenvironmental infrastructure to maintain BGR dominance independently of tissue-level signals, but remains locally restricted. It has exited the blocking mode but has not yet achieved the full 3D-to-2D transition of Paper 3. Structurally, this corresponds to in situ carcinoma: constitutively expanded energetic window, autonomous metabolic regulation, disrupted internal tissue architecture, but intact basement membrane providing the last structural constraint. This attractor is meta-stable: the positive feedback of equation 2.5 is operating but encountering the structural resistance of the remaining tissue constraint.

4.4 Attractor IV: Full Morphodynamic Autonomy

The clone has completed the 3D-to-2D morphodynamic transition (Paper 3), achieved full topological disconnection (Paper 4), constructed a fully autonomous microenvironment, and escaped all tissue-level structural constraints. This state corresponds to invasive malignancy. The clone's BGR dominance is constitutively self-sustaining through autocrine signaling; immune modulation is sufficient to prevent elimination; and the positive feedback loop of equation 2.5 operates without architectural damping. [4,5,43,44]

5. Threshold Conditions for Attractor Transitions

5.1 From Elimination to Limited Persistence

The transition from Attractor I to Attractor II requires the clone to establish sufficient autocrine BGR maintenance to survive immune-mediated or energetic elimination pressure. The critical event is BGR self-stabilization: the establishment of autocrine signaling loops — HIF-1 α stabilization, HK2 upregulation, LDHA induction — that maintain glycolytic dominance independently of tissue-level metabolic signals. [42,27,6]

5.2 From Limited Persistence to Partial Autonomy

This is the transition most relevant to the clinical problem of dysplasia progression. The blocking mode fails when clone-intrinsic A exceeds the sum of tissue-level blocking forces. Three principal mechanisms drive this failure: epigenetic BGR lock-in (progressive chromatin-level stabilization of BGR-supporting gene programs [28,23]); microenvironmental maturation (accumulation of sufficient H creating a structurally altered microenvironment that constitutively supports BGR dominance [14,8]); and immune evasion threshold (sufficient accumulation of immunomodulatory clone adaptations to shift I from equilibrium toward facilitation [7,32]).

5.3 From Partial Autonomy to Full Morphodynamic Autonomy

This transition — equivalent to the invasion threshold — occurs when the remaining structural constraints (basement membrane, physical tissue boundaries) are overcome by clone-derived MMP activity. Critical determinants include: sustained high A sufficient to maintain MMP secretion exceeding TIMP-mediated inhibition; sufficient historical H to have produced a basement membrane already thinned and structurally compromised by prior remodeling cycles; immune facilitation providing inflammatory cells that themselves secrete MMPs; and E at self-

sustaining levels such that the clone can immediately recruit its own vasculature through established autocrine angiogenic signaling. [43,25,26]

6. Synthesis: A Unified Continuum from Regeneration to Malignancy

BM = basement membrane; BGR = Basal Glycolytic Regime; OFF = Oxidative Functional Flux; Att. = Attractor.

7. Clinical and Research Implications

7.1 Risk Stratification of Intermediate Lesions

The historical trajectory determinant H provides a new conceptual basis for risk stratification of intermediate lesions. Current histological grading systems assess the current morphological state of a lesion without incorporating H. The framework predicts that incorporating trajectory information (duration of lesion, number of inflammatory cycles, history of immunosuppression, accumulated ECM remodeling) would substantially improve prognostic accuracy for stable dysplasias, chronic inflammatory lesions, and in situ carcinomas. [37,33]

7.2 Metabolic Imaging as an Energetic-Window Probe

The dual-flux framework (Paper 2) predicts that the OFF/BGR balance is measurable through metabolic imaging. PET-based glucose uptake imaging already captures aspects of BGR dominance (the Warburg effect). More specific measurement of the lactate/pyruvate ratio, NADH/NAD⁺ dynamics, and acetyl-CoA flux would provide direct energetic-window width estimates. Applied longitudinally to intermediate lesions, this approach could detect the shift from limited persistence to partial autonomy — the transition from Attractor II to III — before morphological evidence of invasion becomes apparent. [6,28,42]

7.3 Restoration of Ontogenetic Connectivity as Therapeutic Target

Paper 4 proposed that therapeutic restoration of intermediate ontogenetic states — through targeted epigenetic reprogramming, provision of intermediate-state molecular environments, or metabolic normalization — might restore local Markov connectivity and suppress clone autonomization. The present framework extends this therapeutic logic: interventions that reduce A (MMP inhibition, anti-angiogenic therapy), increase I in its elimination or equilibrium mode (immune checkpoint blockade, NK cell activation), or narrow the energetic window (BGR suppression through LDHA inhibition, HIF-1 α antagonism) could shift the clone fate function from Attractors III–IV toward Attractors I–II. [44,25,26]

7.4 The Inflammatory-Cancer Axis

The framework provides a mechanistic resolution to the long-observed Virchow-Balkwill-Mantovani axis of inflammation-associated cancer. Chronic inflammation contributes to cancer not primarily through mutagenesis (though this occurs) but through three framework-specific mechanisms: (i) accumulation of H through repeated inflammatory cycles; (ii) progressive expansion of E through MMP-driven ECM remodeling and vascular permeability; and (iii) shift of I from elimination toward facilitation through chronic regulatory T cell and macrophage polarization changes. All three mechanisms shift the clone fate function toward Attractors III and IV without requiring any specific mutational event. [7,8,29,30]

8. Discussion

8.1 The Separated Clone as a New Conceptual Category

A central contribution of this series is the introduction of the separated clone as a distinct biological conceptual category: a cellular population that is neither normally integrated tissue nor fully malignant tumor, but a partially autonomous developmental system whose fate is determined by ongoing dynamic interactions rather than by a fixed set of acquired properties.

8.2 Path-Dependence and the Limits of Snapshot Pathology

The historical trajectory determinant H represents a fundamental challenge to snapshot-based pathological assessment. Current diagnostic pathology is necessarily based on the state of a lesion at a single moment. The present framework argues that this snapshot is insufficient to determine prognosis because two identically appearing lesions with different developmental histories may be on divergent trajectories. Progress toward H-incorporating diagnostic approaches — longitudinal molecular profiling, inflammatory cycle counting through ECM remodeling biomarker archaeology, epigenetic clock measurements — would substantially improve the predictive value of intermediate lesion assessment. [37,23,24]

8.3 Limitations

The framework is conceptual and the five determinants (E, I, M, A, H) remain to be operationalized with measurable biological quantities. The attractor boundaries — the precise threshold conditions distinguishing the four attractor states — are not yet formally defined. The interaction between determinants is described qualitatively; a quantitative dynamical systems model incorporating coupled differential equations for E, I, M, and A, with H as the path integral of their product, would be required for quantitative predictions.

9. Conclusion

The separated clone — a cellular population that has undergone ontogenetic structural decoupling from its tissue context — is best understood not as a static pathological entity but as a partially autonomous developmental system whose further fate is determined by the dynamic interaction of five determinants: energetic compatibility, immunological interaction, morphophysiological context, clone cellular activity, and historical temporal trajectory. Four developmental attractor states are identified: elimination, limited persistence, partial autonomy, and full morphodynamic autonomy. Morphologically identical lesions may occupy different attractors and follow divergent clinical courses depending on their developmental histories.

Series Conclusion. Across five papers, this series has developed a unified conceptual account of the continuum from normal tissue through regeneration, chronic inflammation, dysplasia, and malignancy as a single dynamical framework. Paper 1 established the empirical scope of developmental program reactivation in non-malignant contexts. Paper 2 provided the dual-flux metabolic mechanism governing energetic-window width and cell-state transition probabilities. Paper 3 described the morphodynamic consequence of oncogenic energetic-window stabilization: the 3D-to-2D architectural transition. Paper 4 formalized the topological dimension: loss of intermediate ontogenetic states as a rupture of local Markov connectivity and the basis of structural decoupling. Paper 5 synthesized these perspectives into a dynamical model of separated clone fate. Together, these papers propose that malignancy is most fundamentally the historical stabilization of a partially autonomous developmental system that

has lost continuous ontogenetic connectivity with the multicellular organism — a process governed by energetic, immunological, structural, and historical dynamics that together determine whether a separated clone is eliminated, contained, or progressively autonomized into a fully independent morphodynamic system.

Conflict of Interest: The author declares no conflict of interest.

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CHAPTER 3

TISSUE AND ORGANISM ORGANIZATION

Preface

The first two chapters of this monograph were devoted to the organization of living systems at the cellular and tissue levels. In the first chapter, the living cell was examined as an active hydrogel system that maintains its structure through continuous energy-dependent remodeling. The second chapter focused on tissue dynamics, regeneration, the reactivation of embryonic-associated programs, and the role of energetic constraints in shaping different biological states.

The present chapter extends the framework to higher levels of biological organization. Its central premise is that epithelia, stroma, organs, and the organism as a whole represent interconnected levels of a unified system of constraints, remodeling, and resource distribution. Although the mechanisms of regulation differ at each level, common principles persist: the existence of a space of accessible states, the opening of Energy Windows, and the subsequent restoration of constraint architecture.

The logic of the chapter develops progressively from tissues to the organism. The first article examines epithelial tissues and transitional zones as regions of increased plasticity. The second focuses on mesodermal tissues and the role of stromal remodeling. The third analyzes the epithelial–mesodermal interface as a mechanism of interaction between these systems. The fourth extends the analysis to the organ level, considering organs as dynamic adaptive systems. The final article applies this perspective to the organism as a whole and examines the life cycle as a continuous sequence of adaptive states.

Together, these articles form a conceptual bridge between tissue biology and the systemic organization of the organism, linking development, adaptation, aging, and oncogenesis within a unified organizational framework.

The concepts presented here remain working hypotheses. Their primary purpose is to provide a common language for describing complex biological processes and to serve as a foundation for future quantitative models of biological organization.

Article 1 of 5

Epithelial Systems and Transitional Zones as Regions of Tissue Remodeling

Yuri Dmitrochenko, MD

Abstract

Epithelial tissues represent one of the most highly integrated forms of cellular organization in multicellular organisms. Despite their diverse embryonic origins, all epithelia share common structural principles — apical-basal polarity, direct cell-cell contacts, attachment to a basement membrane, and continuous renewal — that collectively define a narrow space of permissible

cellular states. This structural constraint is the basis of epithelial stability under physiological conditions and, equally, the source of its characteristic vulnerabilities during remodeling.

Transitional zones between different epithelial types constitute unique microenvironments in which distinct epithelial populations coexist and continuously adapt to changing mechanical, metabolic, and physiological demands. The present article examines epithelial organization from the perspective of intercellular constraint systems, mechanobiological regulation, and tissue remodeling, and argues that transitional zones are the principal anatomical substrate for the periodic formation of an Energy Window — a transient state combining locally increased energy availability with partial relaxation of tissue-level constraints. This expansion of accessible cellular states simultaneously enables regenerative adaptation and creates conditions under which pathological cellular trajectories may emerge.

Keywords: *epithelium; transitional zones; tissue remodeling; mechanotransduction; epithelial plasticity; regeneration; metaplasia; carcinogenesis; Energy Window; space of permissible states*

1. Introduction

Epithelial tissues form the external and internal boundaries of the body, lining organs, cavities, and glands. They serve as the primary interface between the organism and its environment and are essential for protection, transport, secretion, absorption, and sensory perception. From a systems perspective, however, epithelia are more than functional linings: they represent tightly coupled networks of cells in which the behavior of each individual unit is continuously constrained by its neighbors, by the mechanical state of the tissue as a whole, and by the biochemical microenvironment of the basement membrane.

Understanding epithelial biology therefore requires two complementary frameworks. The first is descriptive and concerns the structural and functional diversity of epithelial types. The second is theoretical and concerns the principles by which tissue architecture regulates the space of cellular states that are accessible at any given moment. In the present article both frameworks are applied in parallel: standard histological material is examined through the lens of intercellular constraint systems, and the concept of the Energy Window is developed as a mechanistic link between local tissue remodeling and the transient expansion of accessible cellular states.

Mechanotransduction pathways — molecular systems through which tissue architecture and mechanical forces directly regulate gene expression — occupy a central position in this framework, connecting classical histological observations with contemporary systems biology. Particular attention is devoted to epithelial transitional zones — anatomical regions in which two or more distinct epithelial populations share a boundary — which emerge from this analysis as the primary sites at which the Energy Window periodically forms, providing a mechanistic explanation for the well-established concentration of epithelial malignancies at anatomical transition points. [11,15]

2. Epithelial Organization: Diversity and Common Principles

Although epithelium arises from all three germ layers — ectoderm, endoderm, and mesoderm — its diverse populations converge on the same organizational principles throughout life: direct cell-cell contact, apical-basal polarity, attachment to a basement membrane, and regulated renewal from a stem cell pool. This convergence implies that the constraint

architecture described in subsequent sections is not germ-layer-specific but represents a universal property of epithelial organization. [1,3,5]

The functional diversity of epithelial types — from the gas-exchange surface of pulmonary alveoli to the high-renewal mucosa of the intestinal crypt — reflects adaptation of the same basic constraint architecture to different physiological demands. In all cases the functional output of the epithelium at the tissue level is an emergent property of the constraint system: disruption of this system alters collective tissue behavior in ways that cannot be understood by analysis of individual cells in isolation. [12,14]

Transitional zones that arise at boundaries between populations of different embryonic origin bring into contact not merely different cell types but different constraint architectures — a fact of direct relevance to the remodeling dynamics discussed below.

3. Intercellular Contacts and the Epithelial Constraint System

Epithelium represents one of the most integrated forms of cellular organization found in multicellular organisms. Unlike stromal cells, which are separated by extracellular matrix, epithelial cells are directly connected to one another, forming continuous cellular sheets. It is this direct connectivity that generates the constraint system central to the present framework.

Each epithelial cell exhibits pronounced apical-basal polarity. The apical surface faces the lumen or external environment, whereas the basal surface contacts the underlying connective tissue through the basement membrane. The preservation of polarity is not merely a morphological feature: it is a defining dimension of the epithelial state, and its loss is one of the earliest detectable indicators of constraint failure during pathological remodeling.

3.1 Tight Junctions

Tight junctions establish paracellular barriers, restrict molecular transport across the epithelial sheet, and are essential for the maintenance of apical-basal polarity. Within the constraint framework, tight junctions constitute the structural seal that enforces compartmental identity: their disruption allows mixing of apical and basolateral membrane components and initiates a cascade of architectural destabilization. [7]

3.2 Adherens Junctions

Adherens junctions connect neighboring cytoskeletons through cadherin-mediated interactions and transmit mechanical forces across the tissue. Through these structures, cells continuously sample the mechanical state of the epithelial sheet. Tension changes — whether caused by injury, cell loss, or altered matrix mechanics — propagate rapidly across multiple cell lengths, allowing the tissue to respond to local perturbations as a coordinated unit. Reduced tension signals the availability of free space and constitutes one of the primary inputs that lower the threshold for cell cycle entry. [6]

3.3 Desmosomes and Gap Junctions

Desmosomes provide strong mechanical adhesion through intermediate filament linkage, enabling epithelia to withstand stretching, compression, and friction. Gap junctions create direct cytoplasmic channels between adjacent cells permitting the passage of ions, metabolites, and second messengers, including calcium ions, ATP, and cyclic AMP. These structures coordinate metabolic and signaling states across the tissue: a local energetic perturbation is not confined to the cell in which it originates but propagates through the gap junction network to

neighboring cells, potentially influencing the extent of the Energy Window across a broader territory.

The combined action of these mechanisms establishes a system of intercellular constraints. The accessible state of any individual cell is determined not solely by its genomic content or instantaneous energy supply, but is shaped continuously by the structural, mechanical, and metabolic context provided by the surrounding tissue. Epithelium consequently possesses one of the narrowest spaces of permissible cellular states among all multicellular tissues.

4. Constraint Systems, Mechanotransduction, and the Regulation of Epithelial States

Recent advances in epithelial biology have established that tissue architecture regulates cellular behavior not only through direct intercellular contacts but also through mechanotransduction pathways that translate mechanical forces into changes in gene expression. [16,17,18]

Among the most extensively characterized regulatory systems is the Hippo signaling pathway and its downstream effectors YAP (Yes-associated protein) and TAZ (transcriptional coactivator with PDZ-binding motif). Under conditions of intact epithelial architecture and high cell density, Hippo kinase activity phosphorylates and inactivates YAP/TAZ, suppressing the transcriptional programs associated with proliferation, migration, and matrix remodeling. Conversely, tissue injury, stretching, extracellular matrix stiffening, loss of contact inhibition, and architectural disruption promote YAP/TAZ nuclear translocation and transcriptional activation. The Hippo pathway thus functions as a direct molecular sensor of the state of the constraint system. [16,19]

This mechanistic detail is significant for the Energy Window model. The activation of YAP/TAZ during remodeling is associated not only with proliferative and migratory programs but also with substantial increases in cellular metabolic activity, including upregulation of glycolysis and biosynthetic pathways. The co-activation of constraint relaxation and metabolic upregulation by a shared upstream signal — tissue mechanical perturbation — provides a plausible molecular mechanism for the concurrent fulfillment of both conditions that define the Energy Window. [17,20]

The constraint model should therefore be understood not as an alternative to established molecular pathways but as an organizational framework that integrates structural, mechanical, metabolic, and signaling influences. From this perspective, epithelial homeostasis depends upon the coordinated operation of four partially independent regulatory layers: structural constraints generated by junctional complexes and epithelial polarity; mechanical constraints generated by tissue architecture and extracellular matrix properties; metabolic constraints determined by local energy availability and pathway utilization; and molecular constraints mediated through signaling networks including Hippo/YAP/TAZ, Wnt, Notch, TGF- β , and EGFR pathways.

5. Regulation of Proliferation and Epithelial Renewal

Epithelial tissues are among the most rapidly renewing structures in the body. Intestinal epithelial cells survive only a few days, epidermal keratinocytes several weeks, and respiratory epithelial cells several months. In each case, continuous generation of new cells from stem cell pools is precisely balanced by programmed cell loss — a balance maintained not by any central regulatory signal but by the local action of the constraint system.

Contact inhibition of proliferation is a central mechanism of epithelial homeostasis. Cells fully integrated into the epithelial sheet and surrounded by intact neighbors receive continuous constraining signals through adherens junctions, tight junctions, and gap junctions that maintain them in a quiescent state. Loss of a neighboring cell removes these constraints locally and increases the probability of cell cycle entry in adjacent cells. [12,14]

Energy availability is a further determinant of proliferative capacity. DNA replication, cytoskeletal reorganization, and the production of daughter cells require a transient increase in energy expenditure that cannot be met from basal metabolic reserves alone. Under normal homeostatic conditions, this energetic threshold constitutes an additional brake on inappropriate proliferation. The interaction between constraint relaxation and metabolic activation defines the conditions under which the Energy Window becomes biologically operative.

6. Transitional Zones as Regions of Tissue Remodeling

Transitional zones located between different epithelial types are regions of active, continuous negotiation between two constraint architectures that differ in their differentiation programs, junctional protein composition, renewal dynamics, and mechanical properties. Classical examples include the cervical transformation zone, where columnar epithelium of the endocervix interfaces with stratified squamous epithelium of the ectocervix; the gastroesophageal junction; the anorectal transitional zone; and the corneal limbus. [10,11]

The maintenance of stable transitional zone boundaries requires continuous active remodeling. Hormonal fluctuations, aging, microinjury, and local inflammatory events all contribute to ongoing architectural reorganization. During reproductive hormonal cycles, the cervical transformation zone undergoes repeated cycles of eversion and metaplastic remodeling. The gastroesophageal junction undergoes analogous remodeling under conditions of chronic acid reflux, with progressive replacement of squamous epithelium by intestinal-type columnar epithelium — Barrett's esophagus. [10]

It is within this context of continuous remodeling that the Energy Window is generated as a recurrent feature of transitional zone biology. The Energy Window is defined here as a transient state combining locally increased energy availability with partial relaxation of tissue-level constraints. It is not a chronic metabolic condition but a time-limited state that opens in association with remodeling events and closes when a new constraint equilibrium is established.

The biophysical basis of its formation in transitional zones reflects the superposition of two processes. First, the mechanical disruption associated with remodeling — loss of cell-cell contacts, basement membrane restructuring, cytoskeletal reorganization — releases cells partially from the constraint system and lowers the energetic threshold required to access alternative cellular states. Second, the local inflammatory and reparative microenvironment provides increased metabolic substrate availability and growth factor signaling that elevate baseline energy supply. The co-occurrence of these processes — mechanistically linked through YAP/TAZ activation — creates a state in which the space of accessible cellular behaviors is transiently enlarged.

The consequence of this enlargement is probabilistic. Most cells that experience a transient Energy Window complete the remodeling event and subsequently reintegrate into a stable architectural context. A minority may acquire alterations that cannot be reversed by the reimposition of normal constraint architecture — representing the initiating events of pathological progression. The frequency and duration of Energy Window formation in a given

transitional zone constitute primary determinants of the oncological risk associated with that region. [8,9,11]

7. Testable Predictions of the Energy Window Framework

Prediction 1: Metabolic activation precedes remodeling

Actively remodeling transitional zones should demonstrate increased metabolic activity compared with adjacent stable epithelium of the same tissue, measurable by spatial metabolomics, isotope tracing, or metabolic imaging. This elevation should precede or coincide with histologically detectable architectural changes rather than following them.

Prediction 2: Remodeling frequency predicts oncological risk independently of tissue size

The frequency and cumulative duration of remodeling events in a given transitional zone should correlate with long-term cancer incidence in a manner that is independent of total epithelial surface area.

Prediction 3: YAP/TAZ activation is coupled to local metabolic upregulation

Activation of YAP/TAZ signaling during transitional zone remodeling should be associated with increased local energy utilization and upregulation of biosynthetic metabolic programs, confirming the proposed molecular basis of the Energy Window.

Prediction 4: Chronic inflammation prolongs Energy Window duration

Tissues with persistent inflammatory microenvironments at transitional zones should display correspondingly elevated and sustained transcriptional and metabolic signatures consistent with Energy Window formation.

Prediction 5: Progenitor cells display broader transcriptional variability

Stem and progenitor cell compartments at transitional zones should display broader transcriptional variability than fully differentiated epithelial populations in the same tissue, reflecting their partial release from constraint architecture. [12]

Prediction 6: Remodeling-associated metabolic parameters outperform classical proliferation markers

Quantitative parameters reflecting remodeling-associated metabolic activation should provide predictive information about pathological progression beyond that available from classical proliferation markers such as Ki-67.

8. Conclusion

The epithelial sheet is a highly organized system of intercellular constraints in which cellular behavior is determined not only by intrinsic cellular properties but by tissue architecture as a whole. The space of permissible cellular states — defined by the combined action of junctional contacts, basement membrane signals, mechanical coupling, and metabolic coordination — is narrower in epithelial tissues than in virtually any other cellular system. This narrowness is the basis of epithelial stability.

Mechanotransduction pathways, particularly the Hippo/YAP/TAZ axis, provide the molecular layer through which tissue architecture and mechanical forces are translated into cellular

decisions, extending the constraint framework from structural histology to contemporary molecular biology.

Transitional zones are the primary anatomical sites for the periodic formation of Energy Windows. The transient expansion of accessible cellular states simultaneously enables adaptive remodeling and creates conditions under which pathological cellular states can emerge and be selected. The frequency and duration of Energy Window formation in specific transitional zones constitute the primary quantitative determinants of their associated oncological risk. Their investigation represents a productive direction for future work integrating biophysical modeling with histopathological and clinical study of preneoplastic lesions.

Conflict of Interest: The author declares no conflict of interest.

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Article 2 of 5

Mesodermal Tissues: Organization, Regulation, and the Role of Stromal Remodeling

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Abstract

Mesoderm-derived tissues differ fundamentally from epithelial systems in their architectural principles, mechanisms of regulation, and patterns of intercellular communication. Whereas epithelial cells form highly integrated cellular layers characterized by direct intercellular contacts and a narrow space of permissible cellular states, mesodermal tissues are organized around extensive extracellular matrix networks that separate individual cells and mediate regulation through biochemical gradients, mechanical signals, and microvascular supply.

The extracellular matrix is examined here not as a passive mechanical scaffold but as an active regulatory environment that controls signal transmission, resource distribution, tissue permeability, and the boundaries of accessible cellular states. Within this framework, matrix metalloproteinase activity and microcirculatory adaptation are identified as the primary mechanisms through which local energy availability is regulated in mesodermal tissues.

Stromal remodeling emerges from this analysis as a central process linking cell death, matrix reorganization, microvascular response, nutrient delivery, and the periodic generation of an Energy Window. The same mechanism that enables regenerative repair simultaneously creates conditions under which alternative cellular programs, including tumor progression, may be initiated. The present article argues that regeneration and carcinogenesis are not opposite biological phenomena but potential outcomes of a single fundamental physiological process: tissue remodeling in response to perturbation.

Keywords: *mesoderm; stroma; extracellular matrix; matrix metalloproteinases; tissue remodeling; regeneration; carcinogenesis; Energy Window; microcirculation; tissue constraints; mechanotransduction*

1. Introduction

Article 1 of this chapter examined epithelial tissues as highly integrated cellular networks in which the behavior of individual cells is continuously constrained by direct intercellular contacts — tight junctions, adherens junctions, desmosomes, and gap junctions. Within that framework, tissue stability emerged from the constraint system itself, and the Energy Window was introduced as a transient expansion of the accessible cellular state space occurring during tissue remodeling. [17]

The present article extends this analysis to tissues of mesodermal origin. Mesodermal tissues are organized according to a fundamentally different architectural principle: cells are separated by substantial volumes of extracellular matrix, and the dominant constraint mechanisms operate through the properties of this matrix rather than through direct cell-to-cell contact.

This distinction has important implications for carcinogenesis. Although most human malignancies are epithelial in origin, stromal remodeling plays a decisive role in their initiation and progression. The framework developed here proposes that stromal remodeling, through its effects on extracellular matrix architecture and microvascular supply, may contribute to the formation of Energy Windows within adjacent epithelial compartments. The stroma therefore acts not merely as a supporting structure but as a significant regulator of epithelial accessible state space — a role that is complementary to, and bidirectionally coupled with, the paracrine influences that epithelium in turn exerts on its stromal microenvironment.

2. Architectural Principles of Mesodermal Tissues

Unlike epithelium, which represents the most integrated form of cellular organization with minimal intercellular space, mesoderm-derived tissues are organized around extensive extracellular matrix networks. Most cells are separated from one another by varying amounts of extracellular matrix and communicate through biochemical gradients, diffusional processes, mechanical forces, and vascular supply. This architectural distinction creates a substantially broader space of accessible cellular states than that observed in epithelial populations.

Mesodermal tissues may be functionally divided into two major categories. Parenchymal cells — including cardiomyocytes, renal tubular cells, adipocytes, osteoblasts, osteocytes, chondrocytes, and smooth muscle cells — perform the specialized physiological functions of organs. Stromal cells — including fibroblasts, pericytes, mesenchymal stem cells, vascular support cells, and connective tissue cells — create and maintain the microenvironment required for parenchymal function. They synthesize, remodel, and respond to the extracellular matrix while regulating tissue architecture and adaptation.

3. Comparative Constraint Architecture: Epithelial and Mesodermal Tissues

The structural and regulatory differences between epithelial and mesodermal tissues can be stated systematically. The dominant architectural principle of epithelial tissue is a continuous cellular sheet; in mesodermal tissue, cells are embedded within extracellular matrix. Intercellular space is minimal in epithelium and variable (often extensive) in mesoderm. The primary constraint mechanism in epithelium is provided by junctional complexes, cellular polarity, and basement membrane architecture; in mesoderm, by extracellular matrix architecture, matrix stiffness, and vascular supply.

Accessible state space is narrow in epithelium and broader in mesoderm. Phenotypic plasticity is limited in mature epithelial tissues but higher in stromal populations. The dominant oncological vulnerability of epithelium lies at transitional zones and epithelial remodeling

interfaces; for mesoderm, it lies within chronic remodeling microenvironments and activated stroma. The typical consequence of chronic remodeling is metaplasia, dysplasia, and carcinoma development in epithelium, and fibrosis, chronic inflammation, and a permissive tumor microenvironment in mesoderm.

4. The Extracellular Matrix as an Active Regulatory Environment

The extracellular matrix is not a passive mechanical scaffold. It functions as an active regulatory environment through which cells exchange signals and through which nutrients and regulatory molecules are distributed. A particularly important relationship exists between extracellular matrix architecture and microcirculation. The delivery of oxygen, glucose, amino acids, hormones, and signaling molecules depends not only on blood flow but also on the ability of these substances to diffuse through the extracellular matrix. [2,3,4]

This capacity is continuously modified by tissue remodeling through matrix metalloproteinase activity, capillary density, precapillary arteriole tone, local hydrostatic pressure, and extracellular matrix composition and density. The sequence of resource delivery constitutes a chain of interconnected steps: arteriolar blood flow determines the volume of blood entering the capillary bed; capillary recruitment expands the surface area available for exchange; extracellular matrix permeability governs the rate of diffusion; and cellular metabolic activity converts available resources into functional output. [5,7]

Within this framework, matrix metalloproteinases function not merely as matrix-degrading enzymes but as regulators of tissue permeability and local resource accessibility. By altering matrix porosity and releasing matrix-bound growth factors, matrix metalloproteinase activity modifies multiple steps in the delivery sequence simultaneously and thereby constitutes a candidate trigger for Energy Window formation in mesodermal tissues. [6]

5. The Stroma as a Regulator of Epithelial State Space

Epithelium never exists independently of stroma. All epithelial tissues rest upon a basement membrane beneath which stromal activity continuously influences epithelial behavior through oxygen and glucose delivery, growth factor signaling, inflammatory regulation, and extracellular matrix remodeling. This relationship is bidirectional: epithelium communicates with the stroma through paracrine signaling including hepatocyte growth factor, keratinocyte growth factor, and EGF receptor ligands that influence fibroblast behavior, and the stroma reciprocally shapes the epithelial constraint architecture through basement membrane composition, matrix mechanics, and inflammatory tone. [14]

The basement membrane, which constitutes the basal boundary of the epithelial constraint system, is assembled and continuously remodeled by stromal fibroblasts, pericytes, and endothelial cells. Its composition determines which integrin signaling pathways are active in the basal epithelial layer and thereby influences epithelial polarity, proliferative threshold, and the degree of constraint relaxation experienced by basal stem cells.

The regulatory importance of stromal-epithelial coordination is most clearly demonstrated during wound healing. Following epithelial injury, the initial response occurs within the stromal compartment: hemostasis, platelet activation, and the release of platelet-derived growth factor and transforming growth factor beta activate stromal fibroblasts. These activated fibroblasts synthesize collagen, release matrix metalloproteinases, secrete growth factor receptor ligands, and recruit immune cells to the wound site. Only after this stromal activation has proceeded does active epithelial migration and proliferation occur. Regeneration is

therefore a coordinated process in which stromal remodeling creates the permissive microenvironment that enables epithelial response.

6. Stromal Remodeling and Carcinogenesis

Epithelium possesses one of the most restrictive constraint systems in multicellular organisms, whereas stromal tissues exhibit considerably greater plasticity. Consequently, many processes that expand the accessible state space of epithelial cells are initiated through stromal remodeling, including chronic inflammation, angiogenesis, fibroblast activation, matrix metalloproteinase secretion, and changes in matrix stiffness. [13,15]

Within the proposed framework, stromal remodeling may contribute to the formation of a local Energy Window even before significant epithelial abnormalities become apparent. A cell with a mutational or epigenetic alteration that would, in an unrestricted environment, produce a proliferative phenotype may remain fully constrained by intact epithelial architecture and its underlying stroma. The acquisition of a growth advantage requires not only the cellular alteration but also an environment in which the altered behavioral repertoire can be expressed.

The key processes include chronic inflammation with sustained matrix metalloproteinase activity; progressive fibrosis with accumulation of crosslinked collagen activating the YAP/TAZ mechanotransduction axis in adjacent epithelial cells; cancer-associated fibroblast activation maintaining autocrine and paracrine signaling loops; and pathological angiogenesis generating irregular perfusion patterns that produce alternating zones of nutrient excess and deprivation. [9,11,12]

When stromal remodeling is transient and well-regulated, the resulting epithelial Energy Window is brief and its closure is enforced by the restoration of normal basement membrane architecture and the resolution of inflammatory signaling. When stromal remodeling is chronic, the Energy Window in the adjacent epithelium may remain open for extended periods, substantially elevating the probability of pathological cellular state transitions.

7. Cell Death as an Initiator of Stromal Remodeling and Energy Window Formation

Cell death, particularly forms accompanied by plasma membrane disruption such as necrosis and cytolysis, occupies a central position in tissue remodeling. The release of intracellular contents into the extracellular space constitutes a damage-associated molecular pattern signal that activates the innate immune response through pattern recognition receptors on macrophages, neutrophils, dendritic cells, and stromal fibroblasts. Among the released molecules are ATP, high-mobility group box 1 protein, heat shock proteins, and mitochondrial DNA. [16]

Within this response, matrix metalloproteinase secretion plays a mechanistically central role. Activated macrophages, neutrophils, mast cells, and fibroblasts produce a spectrum of matrix metalloproteinases that collectively remodel the pericellular and interstitial matrix, increase interstitial diffusion coefficients, disrupt the basement membrane at the boundary of adjacent epithelium, and release matrix-bound growth factors. Simultaneously, local vasodilation increases arteriolar blood flow, while vascular endothelial growth factor signaling expands the density of perfused capillaries. The superposition of increased nutrient delivery and partial matrix-mediated constraint relaxation constitutes the Energy Window in its stromal form.

Cell death is therefore not merely a loss of cellular mass but a regulatory signal initiating tissue reorganization. The distinction between regeneration and carcinogenesis does not lie in the

nature of the initiating event but in the duration, amplitude, and regulation of the resulting Energy Window.

8. Conclusion

Three principal conclusions emerge from the analysis of mesodermal tissue organization. First, the constraint system operative in mesodermal tissues differs qualitatively from that of epithelial tissues: matrix metalloproteinase activity, rather than junctional disruption, is the primary mechanism through which local constraint is relaxed and local energy availability is increased. The shared molecular transducer of both systems — the mechanotransduction pathway centered on YAP/TAZ — unifies epithelial and mesodermal constraint regulation within a single conceptual framework.

Second, the stroma is an important external contributor to the regulation of the epithelial accessible state space, operating alongside the paracrine signals that epithelium itself provides to the stromal compartment. Through its control of basement membrane architecture, growth factor availability, matrix stiffness, and microvascular supply, stromal remodeling continuously shapes the conditions under which epithelial Energy Windows may form.

Third, regeneration and carcinogenesis are not opposite biological phenomena but potential outcomes of the same fundamental physiological mechanism: the formation and resolution of an Energy Window in response to tissue perturbation. The quantitative parameters that determine which outcome prevails — amplitude, duration, and spatial extent of the Energy Window — constitute the primary variables of a predictive model of tissue vulnerability.

Conflict of Interest: The author declares no conflict of interest.

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Article 3 of 5

The Epithelial–Mesodermal Interface: Basement Membrane, Extracellular Matrix Phase State, and Remodeling of the Tissue Environment

Yuri Dmitrochenko, MD

Abstract

The interface between epithelial and mesoderm-derived tissues constitutes one of the most fundamental organizational boundaries in multicellular organisms. The preceding articles in this chapter established two complementary frameworks: epithelial tissues as constraint systems defined by direct intercellular junctions, and mesodermal tissues as matrix-mediated adaptive systems in which extracellular matrix architecture and microvascular supply govern the boundaries of accessible cellular states. The present article focuses on the contact zone between these compartments and proposes that the basement membrane functions not as a static structural barrier but as a dynamic phase-state indicator whose morphological appearance reflects the degree of molecular organization of the surrounding extracellular matrix.

The extracellular matrix is analyzed as a hydrogel system whose physical state — characterized by cross-link density, porosity, and permeability — determines the diffusion of metabolites and signals, the distribution of matrix-bound growth factors, the stiffness sensed by adjacent cells through integrin-mediated mechanotransduction, and the capacity of cells to migrate through the tissue. Matrix metalloproteinases are proposed to function primarily as regulators of matrix organizational state rather than merely as enzymes of matrix degradation: by reducing cross-link density and increasing effective pore size, they drive a controlled transition — used here in the approximate physical sense of a qualitative change in material properties rather than a strict thermodynamic phase transition — from a densely organized hydrogel toward a more permeable, partially liquefied state.

Within this framework, physiological remodeling processes — ovulation, implantation, placentation, wound healing, and angiogenesis — are interpreted as controlled organizational

transitions between dense and partially liquefied matrix states. Each such transition simultaneously increases resource accessibility and partially relaxes tissue-level constraints on adjacent cells, constituting the conditions for Energy Window formation.

Keywords: *basement membrane; extracellular matrix; hydrogel phase state; matrix metalloproteinases; tissue remodeling; epithelial-stromal interface; mechanotransduction; YAP/TAZ; regeneration; carcinogenesis; Energy Window*

1. Introduction

Article 1 of this chapter established that epithelial tissues maintain stability through a system of direct intercellular constraints — tight junctions, adherens junctions, desmosomes, and gap junctions — that collectively define a narrow space of permissible cellular states. Transitional zones between distinct epithelial populations were identified as the primary anatomical substrate for Energy Window formation, where the periodic disruption of constraint architecture transiently expands the accessible state space. [18]

Article 2 extended this framework to mesodermal tissues, where the dominant constraint mechanism operates not through direct cell-cell contacts but through extracellular matrix architecture, matrix stiffness, and microvascular supply. Matrix metalloproteinases were identified as the primary regulators of local energy availability in mesodermal tissues, and stromal remodeling was established as the mechanism through which Energy Windows are opened in the adjacent epithelial compartment from the outside. [19]

The present article addresses a question that the first two articles left open: what is the physical mechanism by which stromal remodeling modifies the epithelial constraint system? The answer proposed here centers on the organizational state of the extracellular matrix at the epithelial-mesodermal interface. The basement membrane — the specialized matrix layer that constitutes the basal boundary of the epithelial constraint system — is examined not as a fixed anatomical structure but as a dynamic system whose organizational state varies continuously in response to matrix metalloproteinase activity, cross-linking dynamics, and mechanical loading.

This mechanistic account also provides a new interpretation of the progressive loss of basement membrane morphological definition in preneoplastic and neoplastic lesions. Within the present framework, this loss is not simply a consequence of tumor invasion but may represent an early indicator of sustained matrix organizational disruption — a morphological correlate of chronically open Energy Window conditions.

2. The Extracellular Matrix as a Hydrogel System

From a biophysical perspective, the extracellular matrix is more accurately characterized as a hydrogel: a three-dimensional network of polymer chains — primarily fibrillar collagens, fibronectin, laminin, and proteoglycans — swollen with water and small molecules, whose macroscopic physical properties are determined by the density and nature of intermolecular cross-links. [7,8]

The hydrogel framework is analytically productive because it unifies several properties of the extracellular matrix that are typically treated as independent: mechanical stiffness, porosity, permeability to diffusing molecules, water content, and the capacity to sequester and release soluble factors. All of these properties emerge from the same underlying physical variable — cross-link density — and therefore change together when the cross-linking state of the matrix is altered.

As cross-link density increases, the hydrogel becomes stiffer, less porous, and less permeable; diffusion of metabolites and signaling molecules through the matrix slows; and sequestered growth factors become less accessible to cell-surface receptors. As cross-link density decreases — through matrix metalloproteinase activity or mechanical disruption — the hydrogel becomes softer, more porous, and more permeable; diffusion accelerates; and previously sequestered growth factors are released. The physical state of the matrix at any given time and location thus simultaneously determines the mechanical signals received by cells through integrin-mediated mechanotransduction and the chemical signals available through diffusion and receptor binding. [15,16]

3. Matrix Metalloproteinases as Regulators of Matrix Organizational State

Within the hydrogel framework, matrix metalloproteinases occupy a position of central regulatory importance. They are a family of zinc-dependent endopeptidases capable of cleaving virtually all components of the extracellular matrix, including fibrillar and non-fibrillar collagens, fibronectin, laminin, vitronectin, and the proteoglycan core proteins. Their traditional characterization as matrix-degrading enzymes, while accurate at the molecular level, obscures their primary function at the tissue level: the regulation of matrix organizational state. [5,6]

By cleaving the peptide bonds that maintain the three-dimensional cross-linked architecture of the matrix hydrogel, matrix metalloproteinases reduce cross-link density and drive a transition — used here to describe a qualitative shift in physical properties rather than a strict thermodynamic phase transition — from a densely organized, relatively impermeable gel toward a more permeable, partially liquefied state. The consequences of this transition are multiple and interconnected: matrix porosity increases; growth factors previously immobilized by heparan sulfate proteoglycan binding are released; and the mechanical stiffness experienced by cells decreases, altering the mechanotransduction signals transmitted through integrin-focal adhesion complexes.

This last consequence is of particular importance for the connection to the YAP/TAZ mechanotransduction pathway. The nuclear localization and transcriptional activity of YAP and TAZ are exquisitely sensitive to matrix stiffness: stiff matrices promote YAP/TAZ nuclear translocation and the activation of proliferative, migratory, and matrix-remodeling transcriptional programs, while soft matrices promote YAP/TAZ cytoplasmic sequestration and transcriptional inactivity. [11,12]

The mechanistic chain is therefore: matrix metalloproteinase activation → reduction of matrix cross-link density → decrease in matrix stiffness → altered integrin-focal adhesion signaling → YAP/TAZ nuclear translocation → activation of proliferative and migratory transcriptional programs + upregulation of biosynthetic metabolic pathways. This chain provides the molecular-level account of how stromal matrix remodeling translates into the simultaneous constraint relaxation and metabolic activation that define the Energy Window.

4. The Basement Membrane as a Morphological Indicator of Matrix Organizational State

The basement membrane separates epithelial tissues from the underlying mesodermal compartment and performs multiple functions: structural support of the epithelial sheet, filtration of molecules passing between compartments, maintenance of epithelial polarity through laminin-integrin signaling, spatial organization of the stem cell niche, and provision of the basal mechanical boundary that contributes to the epithelial constraint system.

Structurally, the basement membrane is a highly specialized and unusually dense form of extracellular matrix, enriched in laminin, type IV collagen, nidogen, and perlecan. Its visibility under light and electron microscopy is a direct consequence of this exceptional degree of molecular organization and cross-linking. [13,14]

The present framework proposes that basement membrane visibility constitutes a morphological indicator of matrix organizational state in the immediately surrounding tissue. When matrix metalloproteinase activity increases and molecular cross-links are partially disrupted, the basement membrane progressively loses its distinct morphological appearance. This loss of visibility is not the disappearance of the anatomical boundary but the transition of the boundary material from a densely cross-linked, organized state to a more dispersed, partially liquefied state in which the molecular density is insufficient to produce a distinct histological signal.

A basement membrane in a partially liquefied organizational state imposes substantially reduced mechanical constraint on the basal epithelial cells resting upon it: integrin signaling through laminin receptors is altered, polarity-maintaining signals are weakened, and the physical barrier to paracrine signaling molecules diffusing from the stroma is reduced. The epithelial cells above a remodeling basement membrane are therefore simultaneously receiving reduced constraint signals and increased access to stromal growth factors and metabolites — precisely the conditions that define the Energy Window.

5. Microcirculation and the Multi-Step Pathway of Resource Delivery

Matrix organizational state does not act on cellular behavior only through mechanical and signaling mechanisms. It also governs the physical delivery of metabolic resources to cells — a process that in mesodermal tissues involves multiple sequential steps, each of which constitutes an independently regulable bottleneck.

Between the arterial blood supply and the cytoplasm of a parenchymal or epithelial cell lies a series of physical barriers: arteriolar tone determining the volume of blood entering the regional capillary bed; the density of perfused capillaries determining exchange surface area; capillary wall permeability; and the physical state of the pericapillary extracellular matrix determining the diffusion coefficient of molecules moving through the interstitial space.

A change in matrix organizational state — through matrix metalloproteinase activity — can substantially increase resource accessibility at the cellular level without any change in blood flow, by reducing the diffusion resistance of the interstitial matrix. This identifies matrix remodeling as an independent contributor to Energy Window formation that operates in parallel with, and is partially dissociable from, the vascular component of the response. This dissociability may explain why some forms of tissue remodeling produce predominantly regenerative outcomes (when both components are well-regulated and transient) while others produce pathological outcomes.

6. Physiological Remodeling as Controlled Organizational Transitions

The hydrogel framework provides a unified account of a diverse set of physiological remodeling processes. In each case, the essential process is a controlled, spatially and temporally regulated shift in the organizational state of the extracellular matrix hydrogel — from a densely organized state toward a more permeable, partially liquefied state, followed by resynthesis and reorganization of matrix components to restore the dense organizational state. Each transition is accompanied by a transient Energy Window: the partially liquefied

intermediate state combines increased resource accessibility with partial relaxation of tissue constraints.

6.1 Ovulation and Corpus Luteum Formation

Follicular rupture during ovulation requires the dissolution of the basement membrane surrounding the dominant follicle and the degradation of the cortical extracellular matrix of the ovarian surface epithelium. This is achieved through a precisely timed surge of matrix metalloproteinase activity — primarily MMP-1, MMP-9, and ADAMTS proteases — producing a localized and transient matrix organizational change sufficient for follicular wall rupture and oocyte release. Both processes represent physiological Energy Window formation with tight temporal and spatial regulation.

6.2 Embryo Implantation and Placentation

Trophoblast invasion of the endometrial decidua represents the most extensive physiological example of matrix organizational transition in the adult organism. Trophoblasts secrete a spectrum of matrix metalloproteinases that reduce the cross-link density of the decidual matrix ahead of the invading cell front. The physiological limit on invasion depth is maintained by tissue inhibitors of matrix metalloproteinases produced by decidual stromal cells, illustrating that the regulatory system controlling organizational state includes both activating and inhibitory components whose balance determines the extent of the Energy Window.

6.3 Wound Healing and Angiogenesis

Wound healing represents the canonical example of matrix organizational transition initiated within the host tissue in response to cell death and damage-associated molecular pattern signaling. The sequence — inflammatory activation, matrix metalloproteinase secretion, matrix organizational change, angiogenesis, and matrix reorganization — proceeds through a well-characterized series of steps that collectively constitute a transient Energy Window in the wound bed.

7. External and Internal Remodeling: Mechanistic Distinctions and Shared Outcomes

Two principal modes of matrix organizational transition can be distinguished based on the origin of the matrix metalloproteinase activity. In external remodeling, the transition is initiated by a cell population arriving from outside the tissue being remodeled — trophoblasts invading decidua, immune cells entering a tissue during inflammation, or tumor cells invading adjacent stroma. In internal remodeling, the transition originates within the tissue itself.

Despite these differences in initiation, both modes converge on the same physical mechanism and produce the same functional outcome: a transient Energy Window characterized by increased resource accessibility and partial constraint relaxation. Tumor invasion can be understood within this framework as a form of external remodeling in which the inhibitory regulation of the organizational transition — normally provided by tissue inhibitors of matrix metalloproteinases — has been lost or overridden. The tumor cell population maintains a sustained external remodeling program that keeps the matrix in a persistently less organized state, preventing the reorganization step that would close the Energy Window.

8. Cell Death, Matrix Organizational Transition, and Energy Window Formation

Cell death — particularly necrosis and cytolysis — releases intracellular contents including ATP, HMGB1, heat shock proteins, and mitochondrial DNA into the extracellular space, activating innate immune cells through pattern recognition receptors. The resulting inflammatory response produces a spatially and temporally coordinated burst of matrix metalloproteinase activity that drives a local matrix organizational transition.

The consequences within the hydrogel framework are fully specified: cross-link density decreases, matrix porosity increases, and the diffusion coefficients of metabolites and signaling molecules throughout the remodeling zone rise. Growth factors previously immobilized in the matrix are released. The basement membrane at the boundary of adjacent epithelium undergoes partial remodeling, reducing the mechanical and signaling constraint it imposes on basal epithelial cells. The net effect is a local expansion of the accessible state space in both the stromal and adjacent epithelial populations.

9. Regeneration and Carcinogenesis as Outcomes of Matrix Organizational Transitions

Regeneration and carcinogenesis share the same initiating mechanism: matrix metalloproteinase-driven organizational transition of the extracellular matrix hydrogel, producing a transient Energy Window. They differ not in the nature of this mechanism but in its regulation — specifically, in the duration and spatial extent of the partially liquefied matrix state, and in the effectiveness of the reorganization step that restores the dense, organized hydrogel state.

In successful regeneration, the transition is spatially confined to the injury site, temporally limited to the period required for architectural repair, and terminated by the coordinated synthesis of new matrix components and their cross-linking into an organized basement membrane and interstitial matrix. In pathological remodeling leading to carcinogenesis, one or more of the regulatory mechanisms that close the Energy Window is insufficient.

The morphological appearance of the basement membrane provides a practical indicator of which regulatory state prevails. A histologically distinct, continuous basement membrane reflects a matrix in the organized dense hydrogel state and a closed Energy Window. A morphologically indistinct or fragmented basement membrane reflects a matrix in a partially liquefied or disorganized state and an open or recently open Energy Window. The progressive loss of basement membrane definition in the sequence from normal epithelium through metaplasia and dysplasia to invasive carcinoma is a sequential record of the progressive failure of the regulatory mechanisms that close the Energy Window.

10. Testable Predictions of the Matrix Organizational State Model

Prediction 1

Matrix permeability correlates with basement membrane morphology independently of vascular density. Regions with morphologically indistinct or fragmented basement membranes should demonstrate increased interstitial diffusion coefficients and matrix permeability compared with adjacent regions possessing intact basement membranes, independently of local vascular density. Testable by diffusion-weighted MRI or FRAP in tissue samples with known basement membrane histology.

Prediction 2

Metabolic activation precedes the proliferative response in physiological remodeling zones. Increases in local glucose uptake and oxygen delivery should precede measurable increases in proliferative markers such as Ki-67 in ovulatory follicles, implantation sites, and healing wounds.

Prediction 3

Matrix metalloproteinase activity predicts matrix permeability independently of matrix composition, because the relevant physical variable is cross-link density rather than total polymer concentration. Distinguishes the organizational-state model from simple degradation models.

Prediction 4

Epithelial transitional zones exhibit higher frequency of basement membrane remodeling episodes than stable epithelial regions of equivalent proliferative activity. Connects the organizational-state model to the transitional zone framework of Article 1.

Prediction 5

Persistent matrix organizational disruption correlates with long-term carcinogenic risk. Quantitative measures of matrix organizational state — basement membrane continuity, interstitial matrix cross-link density, tissue stiffness heterogeneity — should predict long-term cancer incidence in prospective tissue studies.

Prediction 6

Within remodeling tissues, the spatial pattern of YAP/TAZ nuclear localization should correlate with the local matrix organizational state in a manner consistent with the mechanotransduction model.

11. Conclusion

The three articles comprising the first part of this chapter have developed a unified framework for understanding tissue stability and vulnerability across the major tissue types of the multicellular organism. In epithelial tissues, the constraint system is generated by direct intercellular junctions. In mesodermal tissues, the analogous constraint system operates through extracellular matrix architecture, matrix stiffness, and microvascular supply. The present article has identified the physical mechanism connecting these two levels: the organizational state of the extracellular matrix hydrogel.

The basement membrane functions as a morphological indicator of this organizational state. Its integrity reflects the degree of molecular organization of the surrounding matrix; its disruption reflects matrix metalloproteinase-driven transition toward a more permeable, partially liquefied state. The YAP/TAZ mechanotransduction pathway provides the molecular link between matrix biophysics and cellular state space regulation.

The central conclusion of the three-article series can now be stated: the Energy Window is a physical state of tissue organization. It arises whenever the organizational state of the extracellular matrix at the epithelial-mesodermal interface shifts toward a more permeable, less organized configuration. Its opening increases resource accessibility and relaxes mechanical constraints on adjacent cells. The difference between regeneration and carcinogenesis is the

difference between a well-regulated, reversible organizational transition and a sustained or recurrent transition that cannot be resolved within the time frame required to prevent the stabilization of alternative cellular states.

Conflict of Interest: The author declares no conflict of interest.

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Article 4 of 5

The Organ as a Dynamic System: Function, Remodeling, and the Regulation of Accessible State Space

Yuri Dmitrochenko, MD

Abstract

The preceding articles in this chapter established a constraint-based framework for understanding tissue stability and vulnerability at the levels of epithelial organization, mesodermal architecture, and the epithelial–mesodermal interface. The present article extends this framework to the organ as a system. An organ is defined not as a static collection of tissues but as a functional adaptive system whose architecture is continuously shaped by the requirements of its primary physiological task and whose structural states are regulated through the coordinated action of parenchymal, stromal, and systemic regulatory components.

The concept of accessible state space, developed at the cellular and tissue levels in preceding articles, acquires a new dimension at the organ level: the organ itself occupies one of many possible stable functional configurations, and transitions between configurations require a temporary expansion of the accessible state space across multiple coordinated tissue regions simultaneously. This coordinated expansion constitutes the organ-level Energy Window — a state of increased systemic plasticity that enables adaptation but requires regulatory control to ensure reversibility.

The central contribution of the present article is the analysis of five major biological processes — embryogenesis, postnatal growth, regeneration, aging, and carcinogenesis — as variants of a common organizational mechanism distinguished by two principal variables: the width of the Energy Window and the degree of regulatory control over its closure. The paradox of aging — in which declining global plasticity coexists with elevated carcinogenic risk — is analyzed as a consequence of the dissociation between remodeling capacity and regulatory closure effectiveness.

Keywords: *organ dynamics; accessible state space; Energy Window; tissue remodeling; organ adaptation; embryogenesis; regeneration; aging; carcinogenesis; regulatory control; systems biology*

1. Introduction

Articles 1 through 3 of this chapter analyzed tissue organization and vulnerability at progressively expanding scales. Article 1 established that epithelial stability emerges from a system of direct intercellular constraints and identified transitional zones as the primary anatomical sites of Energy Window formation. Article 2 extended the framework to mesodermal tissues and established the stroma as a significant external contributor to epithelial

accessible state space. Article 3 provided the physical mechanism connecting these two levels: the organizational state of the extracellular matrix hydrogel, with matrix metalloproteinase activity as the primary driver of organizational transitions and the basement membrane serving as a morphological indicator of organizational state. [18,19,20]

The organ level introduces something that none of these analyses can capture alone: the integration of multiple tissue systems with different constraint architectures into a single coordinated functional unit, regulated not only by local mechanisms but by systemic inputs from the nervous system, the endocrine system, the immune system, and the mechanical environment of the organism.

The organ level also introduces a qualitatively new dimension to the concept of accessible state space. At the organ level, it is defined by the set of functional configurations — different cell compositions, vascular geometries, matrix architectures, and metabolic rates — that the organ can stably occupy while continuing to perform its primary physiological task. The central question this article addresses is what governs the transitions between such states.

2. The Organ as an Integrated Adaptive System

A functioning organ may be analyzed as a system composed of three interconnected organizational levels. The parenchymal level consists of specialized cells that directly execute organ-specific functions: hepatocytes performing metabolic regulation, cardiomyocytes generating contractile force, nephron epithelial cells maintaining ionic homeostasis, pneumocytes enabling gas exchange. The stromal level provides the conditions required for parenchymal survival and function: structural support, vascular supply, extracellular matrix maintenance, and local signaling regulation.

The regulatory level coordinates the activity of parenchymal and stromal compartments and determines which architectural configuration is required at any given moment. This level encompasses the autonomic and somatic nervous system, endocrine signaling, the immune system, and local growth factor and cytokine networks. Critically, it is the regulatory level that governs Energy Window closure — the restoration of constraint architecture following state transitions — and therefore determines whether organ remodeling is adaptive or pathological.

Function is the organizing principle of this system. Organ structure is not an end in itself but a temporary solution to the problem of maintaining function under specific environmental conditions. An organ incapable of structural adaptation — one whose accessible state space has become rigidly fixed — loses the ability to preserve function in response to changing demands and ultimately undergoes either functional failure or pathological structural change.

3. Continuous Remodeling as the Normal Operational State of the Organ

Under physiological conditions, most organs exist in a state of continuous low-level remodeling that is essential for the maintenance of adaptive capacity. Even at rest, organs continuously renew cellular populations, remodel vascular networks, turn over extracellular matrix components, and modify intercellular communication systems.

At one extreme, intestinal epithelium completes a full renewal cycle within days, with continuous stem cell-driven replacement of the entire surface epithelium. At the other extreme, cardiomyocytes may survive for years without replacement. Between these extremes, hepatocytes undergo replacement on a scale of months under normal conditions, but the liver retains the capacity for dramatic accelerated renewal following injury.

The baseline remodeling activity of an organ represents the continuous exercise of the regulatory machinery that controls state transitions. Organs with high baseline remodeling activity maintain the signaling networks, proteolytic systems, vascular plasticity, and stem cell populations required to execute rapid adaptive responses to perturbation. Organs with low baseline remodeling activity may respond more slowly to perturbation and may also be more vulnerable to pathological consequences when remodeling is eventually forced upon them by injury or disease.

This observation has a direct implication for carcinogenic risk. High baseline remodeling activity is not inherently carcinogenic; it is accompanied by high baseline regulatory activity that enforces Energy Window closure. The carcinogenic risk associated with high remodeling activity arises when regulatory closure mechanisms are compromised — by chronic inflammation, by aging-associated regulatory decline, or by genetic or epigenetic alterations. The intestinal epithelium, with its exceptionally high baseline remodeling rate, maintains robust regulatory closure mechanisms under normal conditions and becomes vulnerable when those mechanisms are compromised, as in inflammatory bowel disease or familial adenomatous polyposis.

4. The Accessible State Space of the Organ

An organ does not exist in a single fixed configuration. It occupies one of many possible stable functional states, each characterized by a specific combination of cellular composition, vascular geometry, extracellular matrix architecture, metabolic rate, and regulatory tone. These states are not equivalent in their accessibility from any given starting point. Transitions between nearby states — modest hypertrophy in response to increased functional demand, localized remodeling following minor injury — require relatively small expansions of the accessible state space. Transitions between distant states — the regeneration of two-thirds of liver mass following hepatectomy — require large, coordinated, multi-tissue Energy Windows and correspondingly robust regulatory control.

The accessible state space of an organ is not fixed throughout life. It is widest during embryogenesis, progressively narrows during postnatal growth, can be temporarily widened by injury or physiological challenge, and contracts during aging as the regulatory machinery for executing state transitions deteriorates.

5. The Energy Window at the Organ Level: Coordinated Multi-Tissue Plasticity

Transition from one stable organ state to another requires a temporary expansion of the accessible state space across multiple coordinated tissue regions simultaneously. This constitutes the organ-level Energy Window — qualitatively distinct from the tissue-level Energy Windows described in preceding articles not in its molecular mechanisms but in its spatial scale, coordination requirements, and the complexity of the regulatory systems required to govern its closure.

Liver regeneration following partial hepatectomy requires not only the proliferation of residual hepatocytes but the coordinated remodeling of the hepatic extracellular matrix, the expansion of the biliary tree, the adaptation of the portal vascular architecture, the activation of stellate cells and Kupffer cells, and the modification of systemic hormonal and metabolic signals. [12]

The regulatory inputs governing organ-level Energy Windows include: nervous system signals modifying local blood flow and proteolytic activity; endocrine signals coordinating remodeling across tissue compartments; immune system activation patterns governing the balance between pro-inflammatory matrix-remodeling signals and anti-inflammatory resolution signals; and

tissue-intrinsic mechanical signals propagating changes in matrix stiffness and vascular pressure across the organ. [16]

The closure of the organ-level Energy Window requires the coordinated action of all these systems. Failure of any one of them to provide the appropriate termination signal can delay or prevent closure, extending the period of elevated plasticity and increasing the probability of pathological state transitions.

6. Five Major Biological Processes as Variants of a Common Mechanism

Embryogenesis, postnatal growth, regeneration, aging, and carcinogenesis can be analyzed as variants of a common organizational mechanism involving transitions between stable states through periods of expanded Energy Windows. They are distinguished by two principal variables: the width of the Energy Window and the degree of regulatory control over its closure.

6.1 Embryogenesis: Maximum Width, Maximum Control

Embryogenesis represents the most extensive form of organ state transition. The accessible cellular state space is at its maximum; matrix remodeling is intense and continuous; and the Energy Window exists as a normal, sustained operating condition rather than a transient response to perturbation. What distinguishes embryogenesis from carcinogenesis — despite the apparent similarity of their molecular profiles — is that regulatory control is also at its maximum. The spatial and temporal patterns of matrix metalloproteinase activity, growth factor signaling, vascular remodeling, and cellular migration are precisely orchestrated by developmental signaling networks. [10]

6.2 Postnatal Growth: Progressive Narrowing Under Preserved Control

Postnatal growth continues developmental programs under conditions of increasing systemic integration and progressive constraint. Cellular differentiation increases, reducing the accessible state space. Matrix architecture becomes progressively stabilized through increased cross-linking and the deposition of mature basement membrane components. Proteolytic activity decreases as the balance between matrix metalloproteinases and their tissue inhibitors shifts toward net stabilization. Growth can be viewed as a gradual, regulated closure of developmental Energy Windows.

6.3 Regeneration: Temporary Reopening Under Preserved Control

Regeneration represents a temporary, spatially and temporally restricted return to increased plasticity following injury. The regulatory architecture of successful regeneration is characterized by the preservation of the closure mechanisms that were progressively established during postnatal growth. The liver, with its preserved hepatocyte proliferative capacity and well-characterized regenerative signaling, can regenerate up to two-thirds of its mass through a regulated, reversible Energy Window. [12,13] The adult myocardium, by contrast, has largely lost the cellular machinery for cardiomyocyte regeneration; its response to injury — fibrotic replacement — reflects the absence of an effective regenerative Energy Window.

6.4 Aging: Narrowing Plasticity and Dissociating Control

Aging produces a progressive alteration of the organ state space. The most evident changes — increased matrix cross-linking, reduced proteolytic activity, decreased capillary density, and

diminished responsiveness to regulatory signals — produce a narrowing of the accessible state space and a reduction in the amplitude of achievable Energy Windows. [11]

The analytically important feature of aging is the dissociation it produces between two regulatory variables that are tightly coupled in youth. In young organisms, regions of high remodeling activity are also regions of high regulatory control. In aging organisms, this coupling deteriorates: global regulatory capacity — immune surveillance efficiency, endocrine signal range, nervous system plasticity — declines, while localized chronic remodeling may maintain regions of elevated matrix metalloproteinase activity that cannot be effectively terminated. The result is a paradox: aging organs have, on average, narrower accessible state spaces than young organs, yet they exhibit higher carcinogenic risk. This paradox is resolved by distinguishing between the width of the Energy Window and the effectiveness of regulatory closure: in aging, the probability that modest Energy Windows fail to close is substantially elevated.

6.5 Carcinogenesis: Persistent Widening Under Impaired Control

Carcinogenesis is defined at the organ level as the condition in which the Energy Window loses its temporal and spatial constraints. Tumors actively subvert the regulatory systems that govern Energy Window closure, converting a transient adaptive state into a self-sustaining pathological condition. Cancer-associated fibroblasts sustain matrix metalloproteinase production independently of injury signals; pathological angiogenesis maintains elevated vascular permeability; and chronic inflammation provides persistent YAP/TAZ-activating signals. [17] The organ progressively loses the ability to return to its pre-pathological baseline; the state transition has become irreversible.

7. A Unified Conceptual Continuum

The five processes described above can be represented along two principal axes. The first is the width of the Energy Window. The second is the degree of regulatory control over Energy Window closure. The product of these two axes — Energy Window width multiplied by the inverse of regulatory control effectiveness — constitutes a composite index of pathological risk. This index is low in embryogenesis (wide window, maximal control), low in homeostasis (narrow window, maximal control), intermediate in regeneration (temporarily wide window, preserved control), elevated in aging (modest window width, declining control), and highest in established carcinogenesis (persistently wide window, impaired control).

8. Testable Predictions of the Organ-Level Framework

Prediction 1: Organ-level regulatory control metrics predict cancer risk independently of remodeling intensity

If carcinogenic risk in aging is primarily a function of regulatory closure failure rather than remodeling intensity, then measures of regulatory system capacity — immune surveillance efficiency, endocrine signal range, resolution of inflammation — should predict long-term cancer risk independently of baseline tissue remodeling markers. Testable in prospective longitudinal studies combining immune phenotyping, endocrine profiling, and tissue biopsy analysis.

Prediction 2: Organs with low baseline remodeling show larger relative increase in pathological outcome rate per remodeling episode with age

If high baseline remodeling activity maintains regulatory closure machinery in an active state, then organs with lower baseline remodeling rates — whose regulatory closure mechanisms are less continuously exercised — should show a larger relative increase in the probability of pathological outcomes per remodeling episode as they age, compared with organs with high baseline remodeling rates whose closure machinery remains well-exercised. This prediction is testable by comparing age-specific rates of pathological remodeling outcomes across organ types with known baseline remodeling activities.

Prediction 3: Regenerative Energy Window duration predicts long-term carcinogenic risk at the injury site

The duration of the regenerative Energy Window following a standardized injury — measurable as the period of elevated matrix metalloproteinase activity, basement membrane remodeling, and YAP/TAZ nuclear localization — should predict long-term cancer risk at the injury site independently of total inflammatory burden. Testable in animal models using serial tissue sampling following standardized tissue injury.

Prediction 4: Age-associated dissociation of matrix remodeling and regulatory signals precedes carcinogenesis

In aging tissues that subsequently develop cancer, the dissociation between matrix remodeling markers and regulatory closure markers should be detectable prior to the appearance of histological dysplasia and should predict long-term carcinogenic risk.

Prediction 5: Organ-level Energy Window parameters show non-linear scaling with injury extent

If the organ-level Energy Window requires coordination across multiple tissue compartments through systemic regulatory signals, its parameters should scale non-linearly with the extent of injury or functional loss. Small injuries should produce locally contained Energy Windows governed primarily by tissue-intrinsic closure mechanisms; large injuries should produce organ-level Energy Windows requiring systemic regulatory coordination.

9. Conclusion

The organ-level analysis integrates the cellular, tissue, and interface-level frameworks into a unified account of biological dynamics across the full lifespan. The central theoretical contribution is the identification of regulatory control over Energy Window closure as the primary determinant of whether organ state transitions produce adaptive or pathological outcomes.

Carcinogenesis is not primarily a disease of excessive remodeling but a disease of insufficient regulatory closure. The therapeutic implication of this framing — that restoring regulatory closure capacity may be as important as inhibiting remodeling activity — represents a direction for future experimental and clinical investigation. The aging paradox — declining global plasticity coexisting with elevated carcinogenic risk — is resolved as a consequence of the dissociation between remodeling activity and regulatory closure capacity.

Conflict of Interest: The author declares no conflict of interest.

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The Organism as a Dynamic System: Energy Distribution, the Basal Glycolytic Regime, and the Life Cycle as a Continuum of Adaptive States

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Abstract

The preceding articles in this chapter established a constraint-based framework for understanding tissue stability and vulnerability at the levels of epithelial organization, mesodermal architecture, the epithelial–mesodermal interface, and the organ as a functional system. The present article extends this framework to the organism as a whole, examining the life cycle as a continuum of adaptive states distinguished by the width of Energy Windows and the degree of systemic control over their closure.

The organism is analyzed as an integrated system for the distribution of energy, matter, information, and remodeling capacity among specialized subsystems. A central regulatory principle governing this distribution is examined: the Basal Glycolytic Regime (BGR), previously introduced in this monograph as the metabolic regime characterized by high glycolytic flux, low reactive oxygen species burden, and generation of biosynthetic precursors that supports maximal anti-synergetic maintenance capacity in cells and tissues (Chapter 1) and maximal developmental plasticity in tissue state transitions (Chapter 2).

At the organism level, the BGR acquires an additional dimension: its basal, low-amplitude component — present across all cell types including fully differentiated quiescent tissues — functions not only as a metabolic energy source but as a systemic regulatory signal linking metabolism, morphogenetic plasticity, and adaptive remodeling capacity throughout the life cycle. Nine major stages of the life cycle — embryogenesis, intrauterine development, postnatal growth, puberty, adult adaptation, pregnancy, regeneration, aging, and death — are analyzed as positions along the two-axis continuum established in Article 4, with the BGR providing mechanistic continuity across all stages.

Keywords: *organism; dynamic system; Basal Glycolytic Regime; Energy Window; life cycle; embryogenesis; aging; death; systemic regulation; metabolic signaling; NADH/NAD⁺ ratio; adaptive capacity*

1. Introduction

Articles 1 through 4 of this chapter developed a framework for understanding biological stability and vulnerability at progressively expanding scales of organization. Article 1 established that epithelial tissues maintain stability through direct intercellular constraint systems and identified transitional zones as the primary sites of Energy Window formation. Article 2 extended the framework to mesodermal tissues. Article 3 identified the organizational state of the extracellular matrix hydrogel as the physical mechanism connecting stromal remodeling to epithelial constraint modification. Article 4 extended the framework to the organ level, identifying regulatory control over Energy Window closure as the primary determinant of whether organ state transitions produce adaptive or pathological outcomes. [21,22,23,24]

The present article addresses the level that encompasses all previous ones: the organism as a whole. Three questions motivate this extension. First, how are Energy Windows coordinated across multiple organs simultaneously? Second, what provides regulatory continuity across the

full lifespan? The present article proposes that this role is fulfilled at the organism level by the regulatory dimension of the Basal Glycolytic Regime. Third, how should death be understood within the framework? Its analysis completes the conceptual continuum from embryogenesis to death.

2. The Basal Glycolytic Regime at the Organism Level: Relationship to Prior Framework Usage

The Basal Glycolytic Regime (BGR) was introduced in Chapter 1 of this monograph as the metabolic regime that maximizes anti-synergetic maintenance capacity in cells: characterized by high glycolytic flux, low reactive oxygen species burden, and generation of biosynthetic precursors including UDP-glucose and UDP-GlcNAc for continuous structural renewal. In Chapter 2, BGR was extended to the tissue level as the metabolic regime associated with developmental plasticity and elevated probability of ontogenetic state transitions, characteristic of embryonic cells, stem cells, and regenerating tissues.

At the organism level examined in the present article, BGR is analyzed in terms of its basal, constitutive component: the low-amplitude glycolytic activity that persists in all cell types regardless of their differentiation state or predominant energy source, including fully differentiated, quiescent post-mitotic cells such as cardiomyocytes and neurons in which oxidative phosphorylation provides the dominant energy supply. The contribution of this basal component to total cellular ATP production under resting conditions is quantitatively small. Its proposed significance at the organism level lies in the metabolic signals it continuously generates: the cytoplasmic NADH/NAD⁺ ratio, the lactate-to-pyruvate ratio, and the concentrations of glycolytic intermediates that serve as substrates for epigenetic regulation and cellular mechanosensitivity.

These three uses of BGR — cellular anti-synergetic maintenance, tissue plasticity regulation, and organism-level metabolic signaling — are complementary rather than contradictory: they describe different functional outputs of the same metabolic activity as it is viewed at ascending scales of biological organization. The high-amplitude BGR of actively remodeling or regenerating tissues is an intensification of the same biochemical machinery whose basal activity provides organism-level regulatory continuity. [5,6,8,11]

3. BGR Signals and Their Regulatory Functions

The metabolic signals generated by BGR activity regulate processes central to the constraint framework developed in preceding articles. The NADH/NAD⁺ ratio influences the activity of sirtuin deacetylases and thereby the acetylation state of histones and transcription factors, connecting glycolytic flux to epigenetic regulation of gene expression. [9] The lactate-to-pyruvate ratio serves as a redox signal that modulates the activity of prolyl hydroxylases and hypoxia-inducible factor pathways. [10] Glycolytic intermediates including acetyl-CoA derived from pyruvate serve as substrates for histone acetylation, directly linking metabolic flux to chromatin state. [11] The AMP-activated protein kinase, activated when glycolytic ATP production fluctuates, coordinates cellular metabolic responses with structural remodeling programs. [8]

Through these mechanisms, the basal component of BGR provides a continuous metabolic signal that encodes information about the current energy state of the cell and its microenvironment, and that modulates the threshold for Energy Window opening: the metabolic activation component that, in conjunction with constraint relaxation, defines the Energy Window state. A reduction in BGR regulatory activity therefore reduces not only the

basal metabolic signal but also the cellular sensitivity to remodeling cues, raising the threshold for Energy Window opening and reducing adaptive capacity.

4. The Organism as a System of Dynamic Resource Distribution

Every organ exists through the distribution of resources — oxygen, glucose, amino acids, lipids, and regulatory signals — among its constituent tissues. At the level of the whole organism, a limited and continuously varying flow of resources must be distributed among dozens of organs in a manner that preserves the viability and adaptive capacity of the entire system. The organism solves this problem through a hierarchical priority system whose regulatory architecture involves the nervous system, the endocrine system, the immune system, and the vascular distribution network.

The BGR plays a specific role in this distribution system. Because basal BGR activity is present in all cell types and generates metabolic signals detectable by intracellular and intercellular sensing systems, it functions as a distributed monitor of local resource adequacy. Cells in which BGR activity falls below a threshold generate signals through AMP-activated protein kinase activation, hypoxia-inducible factor pathway engagement, and cytokine release that communicate resource insufficiency to the systemic regulatory system. [8,10]

The liver occupies a central position in this distribution system: it functions as the central node through which systemic energy redistribution is coordinated, continuously adjusting the carbon balance among tissues. The liver receives portal blood containing the products of intestinal absorption and bacterial metabolism — including lactate and short-chain fatty acids — before they reach systemic circulation. Its regulatory response to these BGR-relevant metabolites determines the metabolic context in which BGR regulatory activity operates throughout the body.

5. The Life Cycle as a Continuum of Energy Window States

The life cycle of a multicellular organism can be analyzed as a continuous sequence of transitions between stable states, each characterized by a specific combination of Energy Window width, regulatory control effectiveness, and BGR activity level. Nine major stages are distinguished as positions on a continuous regulatory landscape.

5.1 Embryogenesis: Maximum Width, Maximum Control, Maximum BGR Regulatory Function

Embryogenesis represents the most radical and extensive state transition in the life of a multicellular organism. The accessible cellular state space is at its maximum; matrix remodeling is intense and continuous; and the Energy Window exists as the normal operating condition. The metabolic profile of embryonic cells is characterized by a predominance of glycolysis even under conditions of adequate oxygen supply — documented since the early observations of Warburg in the 1920s and formalized in his theoretical statement on cancer cell metabolism in 1956, and analyzed in the context of proliferating cells by Vander Heiden and colleagues. [4,5,6]

Within the BGR framework, the high glycolytic activity of embryonic cells generates metabolic signals that maintain chromatin in a permissive, hyperdynamic state characteristic of pluripotency and support the epigenetic flexibility required for successive cell fate decisions of morphogenesis. The distinguishing feature of embryogenesis is that maximal Energy Window width coexists with maximal regulatory control — spatial and temporal patterns of matrix

metalloproteinase activity, growth factor signaling, and cellular migration are precisely orchestrated by developmental signaling networks. [3]

5.2 Intrauterine Development: External Resource Support and Extended BGR Maintenance

After the formation of the principal organ primordia, a prolonged period of intrauterine growth and functional maturation begins. The defining energetic feature of this stage is the presence of an external resource source: the developing organism does not yet maintain autonomous metabolic homeostasis. The placenta integrates two organisms metabolically, immunologically, and endocrinologically. Through the placental circulation, maternal glucose supply sustains BGR regulatory activity throughout intrauterine development.

The transition from placental to autonomous metabolic support at birth therefore represents a fundamental shift in the regulatory architecture of the BGR: from a centrally supplied, continuously maintained signal to a distributed, organ-regulated signal whose amplitude becomes dependent on local metabolic conditions.

5.3 Postnatal Growth: Progressive BGR Modulation and Energy Window Narrowing

In rapidly proliferating tissues — intestinal epithelium, bone marrow, growing bone — high-amplitude glycolytic activity persists. In tissues undergoing terminal differentiation — neurons, cardiomyocytes, skeletal muscle fibers — the high-amplitude glycolytic component of growth diminishes as oxidative phosphorylation becomes the dominant energy source, but the basal BGR component is preserved. Growth can be viewed as a transition from a state in which the BGR drives morphogenesis to a state in which the BGR basal component maintains adaptive readiness: maintaining chromatin accessibility and mechanosensitivity rather than driving fate decisions.

5.4 Puberty: The Final Systemic Remodeling Wave

Puberty represents the last major wave of systemic remodeling in normal human development — the last developmental-scale Energy Window of the life cycle. The hormonal drivers — gonadotropins, sex steroids, growth hormone, and insulin-like growth factor 1 — act as systemic coordinators of Energy Window opening. Their interaction with the BGR occurs through multiple pathways: estrogens and androgens modify glycolytic enzyme expression and mitochondrial biogenesis; growth hormone and insulin-like growth factor 1 stimulate glucose uptake and glycolytic flux. After puberty is complete, the accessible state space remains sufficient for adaptation and regeneration, but it is substantially narrower than during the growth period.

5.5 Adult Adaptation: BGR as Signal of Energy Window Opening

In tissues transitioning from a stable architectural state toward an active remodeling state — proliferating fibroblasts in wound healing, activating satellite cells in muscle regeneration, expanding immune cell populations during infection response — a characteristic increase in glycolytic activity precedes structural remodeling and precedes histologically detectable changes in tissue architecture. This metabolic precursor to structural remodeling reflects the requirement for concurrent metabolic activation and constraint relaxation as the two defining conditions of the Energy Window. The BGR elevation is the metabolic component of this concurrent activation; its appearance before structural changes makes it a candidate early biomarker of Energy Window opening.

5.6 Pregnancy: Systemic Energy Window at the Organismal Scale

Pregnancy represents the most extensive physiological Energy Window of adult life, requiring coordinated remodeling across multiple organ systems simultaneously. From the BGR perspective, pregnancy is characterized by a systematic increase in maternal glucose availability and utilization driven by placental lactogen-mediated insulin resistance, which increases glucose flux through glycolytic pathways in peripheral tissues and thereby elevates BGR regulatory activity across multiple maternal organ systems simultaneously. The postpartum resolution of pregnancy-associated changes represents the closure of this systemic Energy Window.

5.7 Regeneration: Local Energy Window Under Preserved Systemic Control

The BGR signature of regeneration is a localized and transient increase in glycolytic activity in the remodeling zone, preceding proliferative and migratory responses. The closure of the regenerative Energy Window is marked by the return of BGR regulatory activity to baseline levels as matrix organization is restored and constraint architecture is re-established. The capacity for regeneration depends on the preservation of three partially independent systems: the remodeling machinery that opens the structural component of the Energy Window; the BGR regulatory circuit that provides and maintains the metabolic component; and the systemic control mechanisms that coordinate the closure of both components.

5.8 Aging: Dissociation of BGR Regulation and Structural Constraint

Age-related changes in BGR proceed through two partially independent mechanisms. The first is the decline of mitochondrial function — accumulation of mitochondrial DNA mutations, decreased efficiency of oxidative phosphorylation complexes, impaired mitochondrial quality control. This decline alters the NADH/NAD⁺ ratio by reducing the capacity of the mitochondrial electron transport chain to reoxidize NADH, producing a chronically elevated cytoplasmic NADH/NAD⁺ ratio. [12,14]

This condition — tentatively described here as a pseudo-BGR state, though its precise relationship to genuine BGR activation requires direct experimental testing — may display a metabolic profile superficially resembling BGR activation, potentially maintaining some degree of epigenetic accessibility and YAP/TAZ sensitivity, but without the coordinated structural remodeling and regulatory closure machinery that in young tissues accompanies and terminates genuine BGR activation. Whether this contributes to the elevated carcinogenic risk of aging through the pathway described is a prediction that awaits experimental validation through simultaneous measurement of NADH/NAD⁺ dynamics, YAP/TAZ localization, and structural remodeling parameters in aging tissues.

The second mechanism is the gradual reduction of glycolytic responsiveness — the ability of cells to increase glycolytic flux rapidly in response to metabolic demand. This reduction diminishes the amplitude and sensitivity of the BGR as a regulatory signal: cells become less capable of generating the rapid metabolic response that constitutes the early metabolic component of Energy Window opening. The threshold for Energy Window opening rises; adaptive remodeling responses to sub-threshold stimuli are attenuated; and the regenerative capacity of affected tissues declines. [11,12]

5.9 Death: Energetic and Regulatory Collapse of the System

If embryogenesis represents a state of maximally open Energy Window under maximal regulatory control, and aging represents progressive dissociation between BGR regulatory

function and structural constraint systems, then death is the terminal state of this process: the irreversible failure of the organism's capacity to maintain coordinated operation of its energy distribution, constraint regulation, and remodeling systems simultaneously.

The immediate causes of death are diverse — cardiovascular failure, respiratory failure, septic shock, malignant progression, severe trauma — but they converge on a common systemic endpoint: the loss of the organism's ability to maintain the resource distribution gradients and regulatory signals that sustain tissue viability. When resource delivery to critical tissues falls below the threshold required to sustain BGR regulatory activity, the metabolic signals that maintain cellular responsiveness to constraint and remodeling cues disappear. At the cellular level, the final events include the collapse of the NADH/NAD⁺ ratio as glycolytic substrate is exhausted and the failure of AMP-activated protein kinase regulatory activity.

Within the present framework, death and embryogenesis are formally symmetric positions on the life cycle continuum: embryogenesis is the opening of the system to its maximal state of organized plasticity, and death is its final closure. The life cycle is the trajectory between these poles, described at any point by the three-dimensional position in the space of Energy Window width, regulatory control effectiveness, and BGR regulatory integrity.

6. BGR Activity as a Systemic Indicator of Adaptive Capacity: A Synthesis

During embryogenesis and intrauterine development, BGR activity is high, broadly distributed, and primarily instructive: its metabolic signals drive epigenetic state transitions that determine cell fate and coordinate morphogenesis. During postnatal growth and puberty, BGR activity remains elevated in growing tissues but progressively transitions from instructive to permissive function. During adult adaptation and regeneration, BGR activity is low at baseline but shows rapid, localized, reversible elevation in response to remodeling stimuli: its transient elevation marks Energy Window opening, and its return to baseline marks closure.

During aging, this pattern becomes disrupted in two directions simultaneously. Chronic low-amplitude elevation of the cytoplasmic NADH/NAD⁺ ratio through mitochondrial dysfunction produces a background metabolic noise that may reduce the signal-to-noise ratio of genuine BGR activation. Simultaneously, reduced glycolytic responsiveness attenuates the amplitude of genuine BGR activation in response to remodeling stimuli. The result is a degraded BGR regulatory signal: neither sufficiently silent at baseline to allow clear detection of activation, nor sufficiently responsive to stimulation to generate the full metabolic component of an effective Energy Window.

This synthesis suggests that quantitative measures of BGR regulatory function — the dynamic range of glycolytic response to standardized stimuli, the fidelity of NADH/NAD⁺ ratio regulation, the coupling between BGR activation and downstream epigenetic responses — may serve as systemic indicators of adaptive capacity at any life stage.

7. Testable Predictions of the BGR Framework

Prediction 1: BGR activity precedes structural remodeling markers in adaptive responses

The increase in glycolytic activity in the remodeling zone should precede detectable increases in proliferation markers, matrix metalloproteinase activity, and YAP/TAZ nuclear localization by a measurable time interval. Testable by serial sampling with metabolic imaging and simultaneous structural and proliferative marker analysis in animal models.

Prediction 2: BGR dynamic range predicts regenerative capacity across age groups

The dynamic range of glycolytic response to a standardized metabolic or mechanical stimulus should correlate with regenerative capacity in that tissue across age groups, independently of baseline mitochondrial function. Distinguishes the BGR regulatory function hypothesis from a simple metabolic capacity model.

Prediction 3: Dissociation of NADH/NAD⁺ ratio dynamics from remodeling responses predicts carcinogenic risk in aging tissues

In aging tissues, the dissociation between NADH/NAD⁺ ratio dynamics and structural remodeling responses should be detectable prior to the appearance of histological dysplasia and should predict long-term carcinogenic risk. Testable through longitudinal studies combining in vivo metabolic imaging with tissue biopsy and clinical outcome data.

Prediction 4: Placental metabolic signaling influences fetal organ BGR baseline

The amplitude and temporal pattern of glucose delivery to fetal tissues through the placenta should influence the BGR regulatory baseline established in fetal organs, which should in turn predict the adaptive capacity of those organs in postnatal life. Testable in animal models of placental insufficiency with longitudinal postnatal follow-up.

Prediction 5: BGR closure failure is independently associated with Energy Window persistence after injury

In tissues where the metabolic closure of the Energy Window — measured as the return of glycolytic activity to pre-injury baseline — is delayed relative to the structural closure, the probability of pathological remodeling outcomes should be elevated independently of the total inflammatory burden.

Prediction 6: Systemic BGR modulation alters the amplitude of organ-level Energy Windows

Systemic interventions that modify BGR activity — dietary carbohydrate restriction, pharmacological inhibition of specific glycolytic enzymes, or interventions that improve mitochondrial NADH reoxidation capacity — should produce measurable changes in the amplitude and duration of Energy Windows in peripheral tissues in response to standardized remodeling stimuli.

8. Conclusion

The organism-level analysis completes the conceptual framework of this chapter by integrating the cellular, tissue, interface, and organ-level analyses into a unified account of biological dynamics across the full lifespan, with the Basal Glycolytic Regime as the mechanistic element providing regulatory continuity across all stages.

The central theoretical contribution is the characterization of the basal component of BGR as a regulatory signal system that operates continuously at low amplitude in all cell types across all life stages, with its significance lying in the metabolic signals it generates — NADH/NAD⁺ ratio, lactate-to-pyruvate ratio, glycolytic intermediate concentrations, and AMP-activated protein kinase activity — that regulate epigenetic state, mechanosensitivity, and the threshold for Energy Window opening. This function is continuous with and extends the anti-synergetic maintenance and tissue plasticity functions of BGR described in Chapters 1 and 2 of this monograph.

The analysis of nine life cycle stages as positions on the BGR-extended continuum reveals a systematic progression: BGR transitions from an instructive function in embryogenesis and intrauterine development, through a permissive function in postnatal growth and puberty, to a signaling function in adult adaptation and regeneration, and finally to a potentially degraded regulatory state in aging. The formal symmetry between embryogenesis and death provides a conceptually complete account of the life cycle as a trajectory through a regulatory landscape.

Conflict of Interest: The author declares no conflict of interest.

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22. Dmitrochenko Y. Mesodermal Tissues: Organization, Regulation, and the Role of Stromal Remodeling. [See Article 2, Chapter 3, this volume].
23. Dmitrochenko Y. The Epithelial–Mesodermal Interface: Basement Membrane, Extracellular Matrix Phase State, and Remodeling of the Tissue Environment. [See Article 3, Chapter 3, this volume].
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CHAPTER 4

BIOLOGICAL TIME: AGING, METABOLISM, AND ONCOGENESIS

Preface

The first three chapters of this monograph were devoted to the organization of living systems at the levels of the cell, tissue, organ, and organism. They examined the principles of hydrogel organization, remodeling, the space of accessible states, and the Energy Window.

The present chapter is devoted to biological time. Its central premise is that aging, metabolic diseases, neurodegeneration, oncogenesis, and reproductive longevity may be viewed as different manifestations of changes in energy availability and in the ability of biological systems to sustain remodeling.

The logic of the chapter progresses from general mechanisms to their specific manifestations. The first article examines aging as a process of progressive energetic insufficiency. The second introduces the concept of the Local BGR Recycling Ratio (LBGRR) as a possible mechanism for the local redistribution of energy and carbon between cellular populations. Although the existence of local carbon and energy recycling cycles has thus far been convincingly demonstrated only in a limited number of tissues — most notably the kidney — the concept is considered here as a potential general organizational principle. The third article applies this approach to obesity and type 2 diabetes mellitus. The fourth focuses on brain aging as a gradual decline in the ability of the neuron–glia complex to maintain structural integrity and plasticity.

The final two articles examine opposing aspects of biological time. Oncogenesis is presented as a process of progressive desynchronization and loss of regulatory constraints. The oocyte, in contrast, is considered a cell that ensures the continuity of biological information across generations. Together, these models represent two extreme states of biological organization: the loss of constraints and the preservation of potential.

The concepts presented here remain working hypotheses and may evolve as new knowledge accumulates.

Aging as Progressive Energetic Insufficiency: Mitochondrial Collapse, BGR Decline, and the Systemic Accumulation of Damage

Yuri Dmitrochenko, MD

Abstract

The present article proposes that aging should be understood neither as a genetic program nor as the random accumulation of damage, but as the thermodynamically inevitable consequence of progressive mitochondrial insufficiency. The decline in mitochondrial mass and function represents the most experimentally documented and physically grounded primary event of aging. Its direct consequence is the degradation of the Basal Glycolytic Regime (BGR) as the energetic condition required for cellular and intracellular remodeling. The loss of remodeling capacity leads to the accumulation of systemic damage at the cellular, tissue, and organ levels. A parallel process — the displacement of fetal cellular programs by terminally differentiated cells of the final generation — reduces energy consumption but simultaneously slows remodeling, creating a positive feedback loop. Death is analyzed as the final collapse of one of two types: a local catastrophe of a critical organ against the background of systemic energetic insufficiency, or a gradual systemic energetic collapse without a single local trigger.

Keywords: *aging; mitochondria; Basal Glycolytic Regime; BGR; remodeling; syneresis; LBGR; energetic insufficiency; death; anti-syneretic maintenance*

1. Introduction

Aging has been studied for more than a century, yet no unified physically grounded theory exists. The most widely discussed concepts — telomeric, free radical, epigenetic, and inflammatory — describe real phenomena, but each captures only a portion of the picture and does not offer a single mechanistic principle from which the others follow as consequences.

The present article proposes a hierarchy: among all proposed mechanisms, mitochondrial insufficiency is the most physically grounded and experimentally documented primary event. Other phenomena — DNA damage accumulation, epigenetic drift, chronic inflammation, proteostasis decline — are in large measure its consequences. The defining feature of the proposed framework is its grounding in physical principles: a biological system requires continuous energetic support to maintain its organization against the thermodynamically inevitable drift toward equilibrium. Aging is the story of the progressive loss of that support.

2. Mitochondrial Decline as the Primary Event

Among all molecular changes accompanying aging, the decline in mitochondrial mass and function is the most reproducible and the most physically significant. It has been documented in skeletal muscle, liver, myocardium, neurons, and peripheral tissue cells across virtually all studied model systems and in humans. [1,4,5,6,10]

The mechanism comprises two independent but mutually reinforcing processes. First, reduction of mitochondrial mass from an imbalance between mitophagy and biogenesis: with age, biogenesis progressively declines while mitophagy is preserved or itself becomes dysregulated. Second, decline in the functional efficiency of individual mitochondria: reactive oxygen species generated by the respiratory chain damage mitochondrial DNA at a rate substantially exceeding that of nuclear DNA damage, leading to accumulation of mtDNA mutations, defective respiratory complex subunits, uncoupling of oxidative phosphorylation, and further amplification of ROS production — a classical vicious cycle. [3,13,15]

Reactive oxygen species therefore occupy a dual position in this model: they are simultaneously the product and the cause of mitochondrial decline. This self-amplifying character makes the process progressive and, beyond a certain threshold, irreversible. No other proposed primary mechanism of aging possesses this self-amplifying physical character to the same degree.

3. BGR as the Energetic Condition of Remodeling and Its Age-Related Degradation

Within the framework of this monograph, the Basal Glycolytic Regime is defined as the energetic condition required for cellular remodeling. BGR provides a continuous flow of biosynthetic precursors — nucleotide sugars, reducing equivalents, acetyl-CoA — necessary for the maintenance of glycan remodeling, proteostasis, cytoskeletal dynamics, and the structured hydration of the cytoplasmic gel (Chapter 1). [11,12]

When mitochondrial capacity declines, compensatory augmentation of glycolysis is possible only up to a certain limit. Beyond that limit, BGR degrades not because glycolytic enzymes are damaged — but because the system as a whole can no longer support both basal cellular energy demands and remodeling activity simultaneously. The consequence is direct: anti-synergetic remodeling slows. The cytoplasmic gel begins to drift toward syneresis — progressive stiffening, reduction of diffusion, accumulation of protein aggregates. This is a physically measurable process, documented by atomic force microscopy and microrheological methods in aging cells. [9]

4. Accumulation of Systemic Damage

In the young organism, damage at all levels is continuously eliminated through the Energy Window mechanism (Chapter 3). Under conditions of age-related energetic insufficiency, the Energy Window opens less frequently, more slowly, and closes less completely.

At the cellular level: protein aggregates accumulate alongside damaged organelles and cross-linked macromolecules — progressive syneresis of the cytoplasmic gel. [7,8] At the tissue level: extracellular matrix turnover slows, fibrosis progresses, and the basement membrane loses its dynamic character as a phase-transition interface. At the organ level: functional reserve declines — the capacity of the organ to transition to an alternative stable configuration in response to load is diminished. All three levels mutually reinforce one another in a single degradation loop.

5. The Displacement of Fetal Programs: An Adaptation That Deepens the Deficit

Concurrent with mitochondrial degradation, fetal and early postnatal cellular programs are gradually displaced by terminally differentiated cells. This has adaptive logic: terminally differentiated cells are metabolically more economical. However, the cost is loss of plasticity, regenerative reserve, and the wide space of accessible states. A positive feedback loop emerges:

mitochondrial insufficiency drives the displacement of plastic programs, which reduces regenerative reserve, which slows elimination of mitochondrial damage, which deepens the insufficiency.

In the context of the LBGR framework (Article 2, this chapter), the plastic cellular subpopulations that provided the glycolytic component of the local circuit are replaced by terminally differentiated cells with predominantly oxidative metabolism. The Local BGR Recycling Ratio declines as a direct consequence of the shift in cellular composition, independently of and in addition to mitochondrial insufficiency per se.

6. Death as the Final Collapse

Death is the final collapse of a system that has lost the capacity to maintain its organization through remodeling. Two clinically and mechanistically distinct types are identified.

The first is local catastrophe with systemic consequences: a critical organ reaches the threshold of functional insufficiency against the background of general energetic deficit. In the aging organism, the reserve for compensation is absent; the local failure becomes a systemic catastrophe not because it is larger in magnitude, but because the system can no longer isolate and contain it. The second is systemic energetic collapse without a single local trigger: the total energetic base declines below the threshold required to sustain minimum remodeling activity simultaneously across all systems.

Both types reproduce at the level of the organism the two trajectories of hydrogel collapse described in Chapter 1 — hydropic solation and contractile syneresis. The organism dies physically in the same way the cell dies — according to the same thermodynamic principles, at a different scale. This formal symmetry reflects the proposition that the physical principles governing the maintenance of organization are scale-invariant.

7. Experimentally Testable Predictions

Mitochondrial mass in the liver and skeletal muscle should correlate with tissue regenerative potential more strongly than chronological age, and should serve as a more accurate predictor of biological age than existing biomarkers. BGR parameters — the lactate-to-pyruvate ratio and cytoplasmic NADH/NAD⁺ ratio — should correlate with the width of the Energy Window in aging tissues and should constitute more precise biomarkers of biological age than chronological or telomeric measures. The rate of anti-syneretic remodeling, measured by microrheological methods, should decline with age independently of total ATP levels — the physical degradation of the cytoplasmic gel should precede the energetic collapse detectable by standard metabolic assays. The magnitude of LBGR in aging tissues should decline proportionally to the displacement of plastic cellular subpopulations by terminally differentiated cells, independently of mitochondrial insufficiency per se. Interventions that restore mitochondrial biogenesis — exercise training, caloric restriction, NAD⁺ precursor supplementation — should produce measurable recovery of BGR parameters and widening of Energy Windows before clinical endpoints are reached. [14]

8. Conclusion

Aging is the thermodynamically inevitable consequence of progressive mitochondrial insufficiency in a system that requires continuous remodeling to maintain its organization. BGR as the energetic condition of that remodeling degrades as oxidative capacity declines. The accumulation of systemic damage, the displacement of plastic cellular programs, and the decline of LBGR are mutually reinforcing consequences of a single primary event. Death

concludes this trajectory according to the same physical principles that govern hydrogel collapse at the cellular level. Free radical damage, epigenetic drift, proteostatic decline, and telomeric shortening are consequences of progressive mitochondrial insufficiency operating through BGR degradation — the physical principle is primary; the molecular events are its expression.

Conflict of Interest: The author declares no conflict of interest.

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Article 2 of 6

Local BGR Recycling Ratio (LBGRR): A Hypothesis of Local Carbon and Energy Recycling in Multicellular Tissues

Abstract

The classical Cori cycle treats lactate recycling as a global process linking peripheral tissues to the liver and kidneys. However, recent data demonstrate the existence of local metabolic circuits within individual organs. The most compelling example has been identified in the kidney, where the medulla predominantly produces lactate while the cortex consumes it for oxidation and gluconeogenesis. This suggests a more general principle — Local BGR Recycling Ratio (LBGRR) — whereby carbon and energy potential circulate repeatedly among distinct cellular populations without obligatory participation of the systemic circulation. This article presents the LBGRR hypothesis as a possible universal mechanism underlying tissue remodeling, regeneration, embryogenesis, and early tumor growth. LBGRR is explicitly presented as a hypothesis with limited direct experimental confirmation outside the kidney at the time of writing.

Keywords: *LBGRR; Local BGR Recycling Ratio; Cori cycle; lactate shuttle; tissue remodeling; regeneration; metabolic zonation; gluconeogenesis; aging; oncology*

1. Introduction

Lactate exchange has traditionally been viewed through the lens of the global Cori cycle: glycolysis → lactate → liver → glucose → bloodstream → tissues. This model does not fully explain the high efficiency of tissue regeneration, the preservation of tumor growth during cachexia, regeneration in organisms lacking a developed vascular system, or the pronounced metabolic zonation of organs. This raises a fundamental question: can part of the cycle close locally within the tissue itself?

2. An Established Example: The Kidney

The most compelling current example of a local lactate–glucose circuit is the kidney. The medulla operates under low oxygen tension, relies heavily on glycolysis, and produces lactate; while the cortex actively consumes lactate, performs gluconeogenesis, and returns glucose to the system. An intrarenal glucose–lactate cycle has been described, and published work has explicitly discussed the possibility of an intrarenal analogue of the Cori cycle. [1,2,3,4] The principle of local carbon recycling within an organ can therefore no longer be considered purely theoretical.

3. The LBGRR Concept

Two levels of recycling are proposed. GBGRR (Global BGR Recycling Ratio): global recycling involving muscle, erythrocytes, liver, kidneys, and the systemic circulation — the classical Cori cycle. LBGRR (Local BGR Recycling Ratio): local recycling between neighboring cells, between distinct zones of a single organ, or between cellular populations within growing tissue, involving not only carbon transfer but also transfer of energy potential via lactate, pyruvate, alanine, and other intermediate metabolites.

The theoretical foundation for cell-cell lactate exchange as a general physiological mechanism was established by Brooks, who demonstrated that lactate shuttles operate continuously under

aerobic conditions between glycolytic and oxidative cell populations across multiple tissue types. [9]

4. Possible Existence in the Liver

The liver possesses several properties making it a likely candidate: pronounced metabolic zonation, active gluconeogenesis, intensive lactate consumption, and spatial segregation of hepatocyte function. During liver regeneration, a separation between metabolic and proliferative cellular states has been described. [5,6,7,8] This creates conditions favorable for local exchange circuits between proliferating and metabolically active hepatocytes. Direct evidence for hepatic LBGRR is not yet available, and this remains an open question for future investigation.

5. Possible Significance for Regeneration

If LBGRR exists, it should be especially important in situations of rapid tissue growth: liver regeneration, kidney regeneration, wound healing, embryogenesis, and blastema growth. Organisms with limited or absent vascular systems are of particular interest. If local BGR recycling is identified in such organisms, this would indicate an ancient evolutionary origin for the mechanism.

6. Possible Significance for Oncology

Within this framework, the tumor is not the source of a novel metabolic program but rather an exploitation of pre-existing tissue remodeling mechanisms. LBGRR may explain micrometastasis survival, early tumor growth prior to establishment of a mature vascular network, and the asymmetry between tumor growth and body mass loss during cachexia.

7. Experimentally Testable Predictions

The LBGRR hypothesis predicts: spatial separation of glycolytic and gluconeogenic cellular populations; ¹³C-labeled lactate recycling detectable locally within tissue without systemic transit; elevated expression of PEPCK and FBP1 in remodeling zones; a correlation between tissue regenerative potential and the magnitude of LBGRR; and attenuation of LBGRR with aging.

8. Conclusion

The existence of a local glucose–lactate cycle in the kidney establishes that local carbon recycling within an organ is a real biological phenomenon. The LBGRR hypothesis remains to be confirmed in other tissues, but it generates specific, experimentally testable predictions and may serve as a framework for research in regeneration, aging, oncology, and the systems biology of tissues.

Conflict of Interest: The author declares no conflict of interest.

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Article 3 of 6

Pathological LBGRR in Obesity and Type 2 Diabetes Mellitus: BGR Energy Logistics in the Context of Age-Related Mitochondrial Deficit

Yuri Dmitrochenko, MD

Note on scope: The mechanisms proposed in this article do not contradict or replace the established pathophysiological model of type 2 diabetes mellitus. The classical model — insulin resistance, hyperinsulinemia, beta-cell dysfunction — is intentionally omitted solely to preserve the reader's time. This work addresses an additional level of analysis: the energy logistics of the Basal Glycolytic Regime (BGR) as the primary context within which known hormonal mechanisms unfold. Hormonal signals are treated as secondary to the energetic imbalance — provided that pancreatic secretory reserve remains intact.

Keywords: obesity; type 2 diabetes mellitus; Basal Glycolytic Regime; LBGRR; adipose tissue; mitochondria; lipotoxicity; insulin resistance; bariatric surgery; endometrial cancer

1. Adipose Tissue Instability as the Starting Point

Adipocytes differ fundamentally from myocytes and hepatocytes in their critically low glycogen content — the absence of an intracellular buffer for short-term fluctuations in blood glucose. During transient hypoglycemia, adipocytes shift to lipolysis. The released free fatty acids (FFA) are delivered to the liver, where they are converted to glucose via gluconeogenesis, transiently restoring glycemia. When glucose and insulin levels subsequently normalize or rise, adipocytes again capture glucose and convert it to triglycerides. The loop closes. [3,4]

This cycle constitutes a pathological variant of the Local BGR Recycling Ratio: a local exchange of carbon and energy potential between adipose tissue, liver, and systemic circulation, stripped of its normal regulatory constraints. Unlike physiological LBGRR — where oscillation amplitude is dampened by buffer systems — this circuit is self-amplifying under conditions of unrestricted carbohydrate supply.

2. The Mitochondrial Background of Aging

By the age of 50, mitochondrial activity declines to approximately 70% of baseline. [8,9] Terminally differentiated cells — myocytes, neurons, pancreatic beta-cells — do not divide and do not renew their mitochondrial pool. Under these conditions, BGR acquires disproportionate regulatory significance: it remains the only flexible component of the energy budget. Any additional drain on BGR against a backdrop of reduced mitochondrial reserve produces a genuine deficit of plastic and energetic substrate in critical organs.

3. Adipose Tissue as a Parasitic Consumer of BGR

In obesity, hypertrophied, insulin-resistant adipocytes release free fatty acids into systemic circulation at a constant rate, forcing the liver to shift from glycolytic to beta-oxidation mode. This directly competes with BGR: hepatic glycolysis is suppressed and BGR is redirected toward fatty acid metabolism. [5,7] The result is a chronic, low-grade drain on BGR — not acute, but permanent. Adipose tissue establishes a stable parasitic circuit that, against the background of age-related mitochondrial decline, deprives vital organs and skeletal muscle of essential plastic and regulatory substrate.

4. The Paradox of the Young and the Aging Obese Individual

In youth, high mitochondrial density and compensatory muscular hypertrophy allowed the organism to manage the energetic load imposed by excess adipose mass. After age 50, mitochondrial capacity is reduced; new muscular hypertrophy no longer expands the mitochondrial pool in terminally differentiated cells. The adipose mass continues to claim its share of the BGR budget unchanged. The consequence is fatigue and exercise intolerance — not because muscle is intrinsically weak, but because the energy budget is intercepted by adipose drain before it reaches the muscle.

5. Muscular Blockade: GLUT4 and Structural Substrate Inaccessibility

In insulin resistance, the insulin-dependent transporter GLUT4 fails to translocate to the cell surface in sufficient quantities. Muscle starves while blood glucose is elevated — the deficit is at the level of cellular entry, not in the systemic circulation. [4,10] Cells partially shift to fatty acid oxidation, a slow, mitochondrially expensive pathway insufficient under age-related mitochondrial decline. Muscle atrophy creates a secondary vicious cycle: fewer myocytes means fewer total GLUT4 receptors, worsening insulin resistance structurally. Three loops — adipose drain, GLUT4 blockade, and muscle atrophy — operate simultaneously and mutually amplify one another.

6. The Heart: Lipotoxicity and Mitochondrial Burnout Under Mechanical Overload

Cardiac muscle normally derives 60–70% of its energy from fatty acid oxidation; BGR is used primarily for plastic and signaling functions. In obesity and T2DM, excess circulating FFA reaches cardiomyocytes at toxic concentrations. Ceramides and diacylglycerol accumulate — markers of lipotoxicity that impair mitochondrial function. [11] Simultaneously, the heart must pump blood through a substantially enlarged body mass. Cardiomyocyte hypertrophy is possible but these terminally differentiated cells have minimal mitotic activity and limited BGR for repair. Mitochondria operate at peak capacity under continuous FFA load and mechanical stress; mitophagy and renewal cannot keep pace. The result is accelerated burnout of the mitochondrial pool — the morphological substrate of diabetic cardiomyopathy.

7. The Kidneys: Local BGR Activation and Its Exhaustion

Proximal tubular cells rank among the most energy-intensive cells in the body. In hyperglycemia, filtration and reabsorption load increases sharply. The kidneys activate local glycolysis as a compensatory response — but this is an emergency local mechanism, not systemic regulatory BGR. Chronic hyperfiltration, concurrent arterial hypertension, and lipotoxic damage to tubular cells produce a combined overload. Diabetic nephropathy reproduces the same pattern as cardiomyopathy: the organ exhausts its local compensatory reserve and undergoes structural degradation.

8. The Endometrial Energy Window

The endometrium is the only tissue in the adult organism with physiologically programmed cyclic proliferation. Critically, proliferating endometrial cells utilize aerobic glycolysis via insulin-independent transporters GLUT1 and GLUT3: glucose entry is open regardless of insulin signaling status. In obesity, chronic hyperglycemia provides unrestricted plastic substrate through an insulin-independent pathway. Additional contribution comes from peripheral aromatization of androgens in adipose tissue, generating estrogens that amplify the proliferative stimulus. The elevated incidence of endometrial hyperplasia and carcinoma in obesity and MetS does not require pancreatic dysfunction for its realization. [14,15]

9. Metabolic Syndrome as Synchronous Failure of Three Regulatory Loops

Metabolic syndrome represents a collapse of BGR regulation against the background of competing fatty acid substrate and age-related mitochondrial insufficiency. Three independent loops close simultaneously: adipose tissue operates as an autonomous FFA source outside systemic control; the liver shifts to chronic gluconeogenesis; pancreatic beta-cells progress through hyperstimulation to exhaustion. Hormonal disturbances are secondary: they reflect the response of signaling systems to an already-established energetic imbalance.

10. Clinical Confirmation: Bariatric Surgery

Following bariatric surgery, remission of type 2 diabetes mellitus is observed within days in a substantial proportion of patients — before significant body mass loss has occurred. [12,13] Interpreted through the BGR mechanism: elimination of adipose mass interrupts the parasitic drain. The remaining energy budget is redistributed toward organs and skeletal muscle. The BGR regulatory signal is restored. This effect is predictable and reproducible, and follows directly from the energy logistics described above.

11. Evolutionary Context

The mechanism described here is not pathological in origin. Under conditions of unpredictable and limited resource availability — characteristic of most of human evolutionary history — rapid fat accumulation during caloric surplus represented an adaptive advantage. Several studies associate predisposition to obesity and insulin resistance with introgression of Neanderthal genetic variants. [1,2] Loci linked to lipid metabolism and insulin sensitivity show Neanderthal origin in a subset of modern humans, suggesting the antiquity of the described circuit and its fixation by natural selection in populations with seasonal nutritional deficit. The epidemiological transition altered environmental conditions faster than behavioral norms could change: a mechanism of accumulation that functioned under deficit becomes, under conditions of chronic surplus, a mechanism of progressive dysregulation.

Conclusion

Type 2 diabetes mellitus and obesity emerge as the predictable consequence of pathological closure of an evolutionarily conserved metabolic circuit under conditions of chronic resource excess and age-related mitochondrial decline. Adipose tissue functions as an active competitor for BGR — and its elimination produces the most rapid and predictable therapeutic effect. Hormonal disturbances, in the presence of an intact pancreas, are secondary. Energy logistics are primary.

Conflict of Interest: The author declares no conflict of interest.

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Aging of the Brain as a Progressive Reduction of Remodeling: Intracellular and Extracellular Degeneration, Divergent Neurodegenerative Trajectories, and the Potential Role of Artificial Intelligence as a Cognitive Support System

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Abstract

Modern theories of neurodegeneration focus on individual molecular mechanisms — amyloid accumulation, tau pathology, neuroinflammation, mitochondrial dysfunction, and vascular impairment. While each contributes to disease progression, none fully explains the diversity of age-related neurodegenerative trajectories observed in humans.

This work proposes a systems-level framework in which brain aging is viewed as a gradual decline in the remodeling capacity of the neuron–glia–extracellular matrix system. The central event is a progressive reduction in available BGR, leading to decreased structural maintenance, reduced plasticity, and accumulation of intracellular and extracellular degeneration products. A novel contribution is the identification of the glymphatic system as a directly BGR-dependent clearance mechanism — linking BGR decline, astrocytic metabolism, and amyloid pathology within a single causal chain. A separate section considers the potential role of artificial intelligence as an external cognitive support system, including a genuinely novel proposal of AI as an external narrative of cognitive identity in progressive amnesic states.

Keywords: *brain aging; neurodegeneration; Basal Glycolytic Regime; glymphatic system; amyloid; tau; synaptic remodeling; astrocytes; artificial intelligence; cognitive support; cognitive identity*

1. Introduction

For most of human evolutionary history, only a minority of individuals survived into advanced old age. Consequently, many processes that dominate contemporary geriatric medicine likely existed largely outside the effective range of natural selection. [1,3] Most existing models focus on specific pathological structures and their molecular mechanisms. However, aging may be more usefully viewed as a gradual systems-level process affecting the maintenance and remodeling capacity of brain tissue as a whole.

2. The Young Brain as a High-Remodeling System

The young brain exists in a state of continuous structural renewal: active mitochondrial turnover, continuous protein replacement, active synaptic remodeling, efficient glial support, dynamic extracellular matrix remodeling, and effective waste clearance including through the glymphatic system. Damage constantly occurs but is removed at approximately the same rate at which it forms. This equilibrium is actively maintained at significant biological cost.

3. Progressive Reduction of Biological Resource Availability

The proposed model assumes a gradual age-related reduction of BGR — the basal glycolytic resource available to support cellular maintenance and remodeling. Potential manifestations include reduced microvascular efficiency, decreased mitochondrial reserve, reduced glucose utilization efficiency, and impaired cellular maintenance capacity. [2,4,5] The primary consequence is not immediate cell death but a progressive shift toward resource conservation — initially preserving function at the cost of structural renewal.

4. Remodeling Reduction as an Adaptive Response

Under resource limitation, the most economical adaptation is to reduce maintenance expenditures while preserving core functions required for immediate survival. The first processes affected are those most energetically expensive relative to their immediate functional contribution: protein turnover, autophagy, mitophagy, extracellular matrix remodeling, and synaptic renewal. Their reduction preserves short-term function while allowing gradual structural degradation — a trade-off that becomes increasingly costly over decades.

5. The Vascular Component as a BGR Delivery System

Small vessel disease, loss of pericytes, and impaired neurovascular coupling all reduce the effective delivery of glucose and oxygen to active neurons. [4,5] Age-related loss of pericytes from cerebral capillaries reduces the dynamic range of local blood flow regulation. The consequence is chronic subcritical BGR insufficiency — sufficient for basal survival, insufficient for full remodeling capacity. This vascular BGR deficit likely precedes and amplifies the intracellular and extracellular changes described below.

6. Extracellular and Intracellular Aging

Brain ECM — hyaluronic acid, proteoglycans, tenascins, and perineuronal nets — gradually shifts from a dynamic to a conservative state during aging: hydration decreases, matrix turnover slows, diffusion properties change, and structural flexibility declines.

Simultaneously, neurons undergo progressive intracellular simplification. Loss of dendritic spines, reduced synaptic density, impaired axonal transport, reduced mitochondrial turnover, and accumulation of damaged proteins all precede neuronal death by years or decades. [9] Synaptic retraction as a resource conservation strategy deserves particular emphasis: synapses are among the most metabolically expensive neuronal structures, and under conditions of reduced BGR, neurons preferentially retract costly synapses before undertaking more fundamental structural simplifications.

7. Aging of Supporting Cells

7.1 Astrocytes

Astrocytes are central to both metabolic support and glymphatic clearance. Age-related changes include reduced metabolic support to neurons through the lactate shuttle, impaired participation in the glymphatic system, and reduced contribution to extracellular matrix maintenance. Because astrocytic function depends directly on BGR availability, astrocytes may be among the earliest cells to exhibit remodeling reduction.

7.2 Microglia and Oligodendrocytes

Microglial age-related changes include reduced motility, reduced phagocytic activity, and impaired lysosomal function, leading to reduced clearance of cellular debris and extracellular protein aggregates. Chronic low-grade neuroinflammation in the aging brain may in part reflect microglial metabolic insufficiency rather than active pathological activation. [16] Age-related slowing of myelin renewal and reduced oligodendrocyte support contribute to changes in conduction velocity and axonal integrity independent of any specific disease process.

8. The Glymphatic System as a BGR-Dependent Clearance Mechanism

The glymphatic system — the perivascular network through which cerebrospinal fluid circulates through the brain parenchyma, driven by astrocytic aquaporin-4 channels — constitutes the primary mechanism for extracellular waste clearance, including removal of amyloid-beta and tau. Glymphatic function is predominantly active during sleep and depends directly on astrocytic metabolic activity. [6,7,8,17]

Within the BGR framework, glymphatic clearance represents a directly BGR-dependent process. Age-related reduction in astrocytic BGR availability impairs aquaporin-4 expression and localization, reduces the driving forces for glymphatic flow, and consequently reduces the rate of extracellular waste removal. The result is progressive accumulation of extracellular proteins — including amyloid-beta — not because their production increases, but because their removal is impaired.

This interpretation places the glymphatic system at the mechanistic intersection of BGR decline, extracellular degeneration, and amyloid pathology — linking what appear to be independent processes within a single causal chain. This constitutes a novel contribution of the BGR framework: it integrates glymphatic biology with the broader energetic framework in a way not previously articulated.

9. Dual Degeneration: Amyloid and Tau as Parallel Aging Markers

Extracellular degeneration is dominated by ECM aging, impaired glymphatic clearance, and amyloid accumulation. Amyloid is interpreted primarily as a marker of impaired extracellular turnover — the visible consequence of a clearance deficit — rather than as the primary initiating cause of neurodegeneration.

Intracellular degeneration is dominated by impaired autophagy, cytoskeletal instability, mitochondrial aging, and tau accumulation. Tau pathology is interpreted primarily as a marker of impaired intracellular remodeling and cytoskeletal maintenance — the intracellular parallel to amyloid as an extracellular structural aging marker. [14,15] Both are consequences of the same upstream process, manifesting through different compartments.

10. Neurodegeneration as a Probability Tree

Aging does not necessarily lead to a single disease. The shared pathway — age → reduced BGR → reduced remodeling → reduced plasticity → structural alterations — creates conditions from which multiple divergent trajectories may emerge: healthy aging, Alzheimer's disease, frontotemporal dementia, vascular dementia, Lewy body disease, or mixed neurodegenerative states. Individual trajectories are influenced by genetics, environment, vascular factors, lifestyle, and stochastic events. [11,12,13]

11. Artificial Intelligence as an External Cognitive Support System

11.1 AI as a Compensatory Cognitive Layer

Current medicine possesses no intervention capable of restoring the aging brain to a youthful remodeling state. Pharmacological approaches targeting amyloid or tau have thus far produced limited clinical benefit, consistent with the interpretation that these pathological features are downstream consequences rather than primary causes. [20] Artificial intelligence may serve as an external cognitive layer providing: memory support — preserving and organizing information externally; organizational support; analytical support; and communication support.

11.2 AI as an External Narrative of Cognitive Identity

In progressive amnesic states — particularly early and moderate Alzheimer's disease — the continuity of personal identity and intellectual history is threatened by loss of episodic memory. An accumulated history of AI interactions, preserved externally as a structured record of questions asked, ideas developed, and knowledge organized, may serve as an external narrative of cognitive identity.

This preserved record can function as a retrievable autobiography of intellectual life — a resource through which a person with declining memory may reconstruct context, revisit developed ideas, and maintain a sense of cognitive continuity that internal memory can no longer reliably provide. This application of AI in neurodegeneration has not been systematically proposed and represents a genuinely novel direction for consideration.

12. Conclusion

Brain aging is interpreted as a progressive decline in the remodeling capacity of the neuron–glia–extracellular matrix system, driven by decreasing BGR availability. The novel identification of the glymphatic system as a directly BGR-dependent clearance mechanism links what appear to be independent processes — astrocytic metabolism, extracellular waste clearance, and amyloid pathology — within a single causal chain. Tau pathology and amyloid accumulation are different expressions of reduced structural renewal. Although no therapy currently exists that reverses this process, artificial intelligence may provide a genuinely novel form of cognitive support — acting as an external memory, organizational system, analytical partner, and in states of progressive amnesia, an external narrative of cognitive identity.

Conflict of Interest: The author declares no conflict of interest.

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Article 5 of 6

Oncogenesis Through the Lens of BGR: Cell Desynchronization, Extracellular Matrix Phase Transition, Loss of Regulatory Bridges, and Metabolic Division of Labor

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Note on scope: This article does not attempt to describe the entire spectrum of known mechanisms involved in oncogenesis, invasion, and metastasis. Mechanisms such as

pathological vascularization, immune evasion, epithelial-to-mesenchymal transition, and genetic instability have been extensively described in the literature and are therefore intentionally excluded. The purpose of this article is to examine oncogenesis through the framework of BGR energy logic and to identify levels of process organization that have not yet received systematic description within this conceptual context.

Keywords: *oncogenesis; BGR; desynchronization; Pathological Synchronization; basement membrane; Regulatory Bridges; Warburg effect; LBGRR; metabolic division of labor; carcinoma; sarcoma; trophoblastic disease*

Part I. Primary Invasion of the Desynchronized Cell

1.1. The Concept of Desynchronization

A normal cell is defined by three coordinates: ontogenetic time (level of differentiation within the hierarchy extending from fetal to terminal phenotypes); space (position within the architecture of a tissue and organ); and genetic identity (correspondence of genomic profile to a particular cell type, location, and developmental stage). Oncogenesis begins with desynchronization — the disruption of one or more of these coordinates. [1,2,3]

Temporal desynchronization refers to the reversion of a phenotype toward earlier ontogenetic states. Spatial desynchronization refers to the appearance of a cellular population outside its normal topographic context, with endometriosis serving as a classical example. Genetic desynchronization encompasses both conventional oncogenic mutations and combinatorial disturbances, an extreme example being trophoblastic disease.

1.2. Trophoblastic Disease as a Model of Triple Desynchronization

Hydatidiform mole represents a borderline condition in which desynchronization occurs simultaneously along all three axes. [15] The trophoblastic phenotype is inherently fetal and invasive; invasion of the spiral arteries constitutes a normal component of its biological program. The genetics of a complete hydatidiform mole — an androgenetic diploid genome lacking maternal contribution — represent an extreme form of genetic desynchronization. In this setting, the Energy Window is maximally open. Trophoblastic disease therefore illustrates three possible outcomes of primary desynchronization that may be considered universal patterns of oncogenesis.

1.3. Three Outcomes of Primary Desynchronization

The first possible outcome is elimination: the desynchronized cell loses the competitive struggle to neighboring normal cells. The Energy Window is insufficiently open, immune surveillance remains effective, and the abnormal population is eliminated. This is the most common outcome.

The second possible outcome is a stable, non-progressive population: the Energy Window is open but remains limited. The population persists in equilibrium — neither eliminated nor capable of sustained progression. Clinically, this corresponds to benign and borderline tumors.

The third possible outcome is emergence of a progressive population with expansion of the Energy Window: the population acquires the ability to autonomously increase its access to

metabolic substrates. Within the trophoblastic disease model, this corresponds to choriocarcinoma.

Part II. Local Growth: Geometric Constraints and Immune Vulnerability

2.1. Epithelium: The Two-Dimensional Logic of the Basement Membrane

Epithelial tumors develop under geometric constraints imposed by the basement membrane. Growth of a desynchronized epithelial cell initially occurs within the plane of the epithelial layer. Carcinoma in situ is the morphological manifestation of this stage. Immune surveillance is relatively limited because the basement membrane forms a physical barrier separating the tumor population from stromal immune cells. This geometric shelter explains the epidemiological predominance of carcinomas — approximately 85% of all malignant tumors. [1,17]

2.2. Mesoderm: The Three-Dimensional Logic Without Spatial Constraint

Mesoderm-derived tumors lack such geometric constraints. Sarcomas and lymphomas develop within the three-dimensional stromal environment without a basement membrane barrier, providing immediate volumetric growth but substantially greater exposure to immune surveillance. For survival, a desynchronized mesodermal population must establish a sufficiently broad Energy Window more rapidly than its epithelial counterpart.

Part III. Breaching the Basement Membrane: Extracellular Matrix Phase Transition and Pathological Synchronization

3.1. The Insufficiency of Classical Mechanisms

The destruction of the basement membrane during tumor invasion is traditionally explained by enzymatic degradation via matrix metalloproteinases, mechanical disruption by invadopodia, and paracrine activation of cancer-associated fibroblasts. [7,8] Each mechanism is important. However, neither individually nor collectively do they fully explain the complete topological disappearance of the basement membrane as an organized structure.

3.2. The Basement Membrane as an Actively Maintained Gel

The basement membrane is a thermodynamically unstable structure whose functional integrity depends upon continuous energy-dependent remodeling. Fibroblasts and myofibroblasts constantly synthesize type IV collagen, laminin, and linker proteins, compensating for the natural tendency of the matrix toward degradation. [9,10] The basement membrane can be regarded as a tissue-level analogue of the anti-synergetic remodeling processes described for the cytoplasmic gel in Chapter 1.

3.3. Pathological Synchronization as a Novel Mechanism

Tumor glucose consumption occurs in pulses associated with proliferative waves, creating periodic local glucose depletion within the adjacent stroma. Mesodermal cells at the tumor-stroma interface are exposed to rhythmic metabolic stress — a BGR oscillatory deficit that operates as a synchronizing force on the surrounding tissue.

The cycle: a proliferative wave consumes the local glucose pool → stromal cells enter energy deficiency → matrix production becomes impaired → partial necrosis releases inflammatory mediators and activates MMPs → substrate partially recovers between waves. With each

subsequent cycle, damage accumulates: some cells fail to recover completely, structural defects progressively accumulate within the matrix. Eventually, the rate of matrix degradation chronically exceeds its restoration rate. A qualitative change in matrix properties occurs: the basement membrane progressively loses mechanical integrity as an organized structure. The tumor does not penetrate a wall; it advances through a microenvironment that has ceased to behave as a solid barrier. [9,10,11]

3.4. Virchow's Observation as Morphological Evidence

Virchow described apparent rejuvenation of tissues adjacent to tumors — less differentiated cellular phenotypes within the peritumoral zone. Within the framework of Pathological Synchronization, this acquires a mechanistic interpretation: stromal cells exposed to recurrent BGR-associated metabolic stress undergo progressive desynchronization. The tumor boundary is a dynamic and continuously expanding zone. This explains the critical importance of surgical margins: a margin appearing histologically free of tumor cells may contain cells in a pre-invasive state of Pathological Synchronization, contributing to subsequent recurrence.

Part IV. Regulatory Bridges Between Cellular Levels and Their Loss

4.1. Hierarchical Organization and Adjacent-Level Communication

In normal tissue, cells at different levels of differentiation form a hierarchical regulatory network. Fetal cells serve as the upper regulatory hub. Communication is possible only between adjacent levels: direct communication across a level is not possible. This is the principle of adjacent-level communication, which determines the topology of the tissue regulatory network.

4.2. The Mechanism of the Regulatory Bridge

Regulatory signals are transmitted through two parallel channels: gap junctions providing direct signal transmission between physically contacting cells (characteristic primarily of epithelial populations), and the extracellular matrix as a reservoir and transport medium for cytokines, growth factors, and morphogens (characteristic of mesenchymal populations). Both channels function in parallel and complement one another.

4.3. Aging as Depletion of the Fetal Regulatory Reserve and Malignization as a Topological Break

During aging, the fetal cellular pool progressively declines. When it falls below a critical threshold, the bridge between fetal cells and cells at level -1 weakens, and cells at level -2 find themselves in a regulatory vacuum. This is a threshold process rather than a gradual one.

Cells at level -2 without a Regulatory Bridge do not necessarily carry additional mutations — what has changed is the absence of restriction from above. Unrestricted proliferation is therefore a consequence of the loss of regulatory topology rather than of a genetic defect. This explains: age as the major independent risk factor for malignization (through progressive depletion of the fetal regulatory reserve); prolonged stability of benign and borderline tumors (through a residual bridge); and the apparent suddenness of malignization (as a threshold effect of network disruption).

4.4. Synergy with Pathological Synchronization

ECM degradation during Pathological Synchronization destroys not only the physical barrier of the basement membrane but also the ECM channel of the Regulatory Bridge simultaneously. At

the moment of invasion, cells at level -2 lose both communication channels at once: gap junctions through physical separation, and ECM-mediated signaling through matrix degradation. Malignization accelerates precisely at the moment of invasion — the two processes are mechanistically linked.

Part V. Energetic Support of the Tumor: Division of Labor Between Compartments

5.1. The Warburg Paradox and Its Incomplete Resolution

Warburg described aerobic glycolysis in tumor cells — the preference for glycolysis even in the presence of sufficient oxygen. [12,13] The phenomenon has been reproduced many times, but its mechanistic explanation remained incomplete because averaged measurements of tumor metabolism conceal the internal heterogeneity of the cellular population.

5.2. Two Functional Compartments

The tumor is a metabolically heterogeneous system with a functional division of labor. The proliferative compartment — poorly differentiated, actively dividing cells — functions on the basis of aerobic glycolysis. Rapid division primarily requires plastic substrate rather than energy as such; aerobic glycolysis supplies building material for nucleotides and membrane lipids through the pentose phosphate pathway. This is the compartment that Warburg observed as the dominant metabolic signal.

The supporting compartment — highly differentiated, non-proliferating cells — functions on the basis of oxidative phosphorylation, consuming oxygen and producing ATP for the entire tumor cell pool. They create the appearance that the tumor uses oxygen for growth. In reality, oxygen is consumed by the supporting compartment; growth is driven by the proliferative compartment.

5.3. LBGR Drainage and Clinical Implications

Highly differentiated tumor cells are capable of capturing LBGR from neighboring normal cells — a local parasitic drainage of BGR from the surrounding tissue. This mechanism is especially pronounced in well-differentiated tumors, where the supporting compartment is most fully developed, and it reinforces Pathological Synchronization.

Anti-angiogenic therapy, by blocking oxygen supply, primarily affects the supporting compartment. Proliferating cells dependent on glycolysis are resistant to hypoxia. The tumor does not die — it loses its differentiated component and shifts toward a more aggressive, poorly differentiated phenotype. This explains the paradoxical increase in invasiveness following anti-angiogenic therapy observed in a number of clinical studies — an effect that remained counterintuitive within the framework of the classical metabolic model. [14]

Conclusion

Oncogenesis appears as a multilevel process in which each level adds a new dimension to understanding malignant growth. Primary cellular desynchronization determines three possible outcomes through the state of the Energy Window. The geometric constraints of the basement membrane define the fundamental difference between epithelial and mesodermal growth logic. Pathological Synchronization explains the topological disappearance of the basement membrane as a mechanism complementing classical proteolytic degradation. The loss of Regulatory Bridges through depletion of the fetal pool during aging explains malignization of

benign tumors as a threshold topological effect that does not require additional mutations. The division of labor between compartments resolves the Warburg paradox and explains counterintuitive clinical effects of anti-angiogenic therapy.

Conflict of Interest: The author declares no conflict of interest.

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The Oocyte as a Long-Term Energy Storage System: A Hypothesis of Surface Membrane Energy Exchange, Spatial Gel Remodeling, and a Near-Closed Carbon Cycle

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Abstract

The long-term preservation of mammalian oocyte viability remains one of the least understood problems in reproductive biology. Human oocytes are capable of persisting for decades while retaining the ability, after fertilization, to initiate the development of a new organism. Current explanations focus mainly on mitochondrial integrity, maintenance of chromosomal cohesion, limitation of oxidative damage, and regulation of the meiotic apparatus. However, they do not fully answer the question of how a large hydrated cell preserves its cytoplasmic organization for such a prolonged period.

The present work proposes a conceptual model that considers the oocyte not as an ordinary functioning cell, but as a system for long-term storage of biological potential. It is suggested that preservation of viability may be supported by extremely slow spatial remodeling of the cytoplasmic gel, surface energy exchange at the plasma membrane, and a near-closed intracellular carbon cycle involving glycans, glycolysis, and possible local gluconeogenesis. This model is explicitly hypothetical. Its aim is to formulate new questions and define directions for future research.

Keywords: *oocyte; long-term viability; cytoplasmic gel; hydrogel; glycan remodeling; carbon cycle; surface membrane metabolism; central cytoplasmic granulation; CLCG; reproductive biology; anti-synergetic maintenance*

1. Introduction

Most cells of the organism continuously perform specialized functions and therefore require significant metabolic activity. The oocyte occupies a fundamentally different position: for most of its existence, it does not perform an active biological function. Its task is not to work, but to preserve the possibility of performing work in the future. The main problem of the oocyte is therefore not functioning, but preservation of developmental potential.

This allows us to suggest that the energetic requirements of long-term storage may differ fundamentally from those of an actively functioning cell. The oocyte may represent the most extreme manifestation of the anti-synergetic maintenance principle described in Chapter 1: a cell that must maintain its cytoplasmic gel organization for decades, against thermodynamic drift toward syneresis, at the minimum possible energetic cost.

2. The Oocyte as a Preserved Biological Program

A mature oocyte can be viewed as a compressed biological program held in a state of waiting. After fertilization: completion of meiosis, reorganization of the cytoskeleton, activation of embryonic developmental programs, and repeated cell divisions are initiated. However, before the moment of activation, the cell may exist for decades. What is the minimal energetic cost of preserving biological organization for decades? The answer may not require high metabolic

activity but extremely slow compensatory processes taking place over years and decades. [7,8,10]

3. Spatial Organization of Cytoplasmic Gel Remodeling

The cytoplasm of the oocyte is proposed to represent a highly organized hydrated polymer gel. Any such gel undergoes continuous thermodynamic drift toward syneresis, protein aggregation, reduced diffusion, and increased structural rigidity. To preserve cell viability, this drift must be continuously compensated.

It is proposed that remodeling has a spatially organized character. The periphery of the oocyte and the plasma membrane may represent the main site of energetic exchange: glucose uptake, glycan synthesis, glycoprotein synthesis, and restoration of structural components of the gel may predominantly occur here. The central region surrounding the nucleus may primarily serve the function of long-term storage, where gradual degradation of glycans, local glycolysis, and formation of small amounts of ATP for structural maintenance may occur. A slow directional flow of structural components from the periphery toward the center may arise, with a reverse flow of metabolic products.

4. Surface Membrane Energy Exchange

A key element of the proposed model is the assumption that a significant part of energetic exchange is localized in the region of the plasma membrane. Biology already knows many examples of membranes functioning as energetic interfaces: mitochondria, bacterial membranes, the sodium-potassium pump, and proton pumps all exemplify this principle. The present work proposes that enzymatic complexes associated with glucose breakdown may be located on the external side of the membrane, while enzymatic systems that use incoming energy for synthesis of structural components may be located on the internal side — possibly arranged as a metabolon localized on the inner membrane surface. The specific molecular mechanism of such a process remains to be established.

5. Possible Local Gluconeogenesis and Spatial Separation of Glycolysis

At the cell periphery, glycans are synthesized and new structural components are incorporated into the cytoplasmic gel; in the central region, glycans undergo degradation, local glycolysis takes place, and pyruvate and other low-molecular-weight products are formed. A directional flow emerges: glycans moving from the periphery toward the center, and pyruvate diffusing from the center back toward the periphery.

6. Near-Closed Carbon Cycle

The most speculative element of the model is the assumption of a near-closed carbon cycle within the oocyte. Glycans are synthesized at the periphery and gradually incorporated into the gel structure; in central regions they undergo degradation, yielding glucose and glycolytic intermediates; pyruvate is generated and diffuses back toward the periphery, where carbon is again used for synthesis of structural components. [9,12]

Such a cycle cannot be completely closed — losses are inevitable. However, given the extremely slow rate of turnover, operating over years and decades rather than minutes, these losses may represent only a small fraction of the total carbon pool per unit time. External follicular cells may compensate for these losses through minimal supply of additional substrates. This renders the cycle physically plausible rather than merely metaphorical.

7. Central Cytoplasmic Granulation as a Possible Morphological Trace of Gel Disturbance

In reproductive medicine, centrally located cytoplasmic granulation (CLCG) is usually considered a morphological abnormality of the oocyte — characterized by the appearance of a denser, granular, or darker region in the central cytoplasm. In clinical studies, CLCG has been associated with reduced oocyte quality, impaired embryonic development, decreased blastocyst formation rates, and unfavorable ART outcomes. [1,2,3,4,5,6,11]

Within the framework of the proposed hypothesis, CLCG may reflect not a random morphological abnormality but a local disturbance of the spatial organization of this gel — manifesting as local syneresis, aggregation of macromolecules, altered glycan distribution, reduced diffusion, or age-related drift of the central cytoplasmic zone. The central localization of granulation corresponds precisely to the presumed zone of accumulation of the consequences of slow gel drift. If a slow movement of structural components from the periphery toward the center really exists, then the central region should be most sensitive to disturbance of the balance between renewal and degradation. This identifies an already-observed phenomenon interpretable within the framework.

8. Why Such a Mechanism May Exist

The hypothesis rests on several converging observations. First, the oocyte is characterized by extremely low metabolic activity compared with most cells. Second, after loss of connection with the follicle, the oocyte rapidly degenerates despite its ability to persist for decades within the follicular system — suggesting that the follicular environment provides something essential beyond simple substrate supply. Third, the oocyte contains all the main elements of such a construction: glycans, glycoproteins, mitochondria, membrane transport systems, and spatially organized cytoplasm. Fourth, CLCG has already been described in embryological practice. Fifth, known physical laws contain no obvious contradiction to the existence of extremely slow remodeling processes occurring over many years.

9. Testable Consequences

1. Oocyte aging should begin as a physical disturbance of cytoplasmic gel organization before pronounced energetic exhaustion becomes detectable.
2. Mechanical and rheological differences should exist between peripheral and central cytoplasmic regions, measurable by micropipette aspiration or atomic force microscopy.
3. Glycan distribution may show spatial gradients from periphery to center.
4. Intracellular carbon recycling may be significantly more pronounced than currently assumed, detectable by isotope tracing experiments.
5. Long-term preservation of viability should depend more on maintenance of gel structure than on the absolute level of ATP production.
6. Central cytoplasmic granulation may be not only a morphological defect, but also a visible manifestation of disturbed physical organization of the cytoplasmic gel.
7. Age-related changes in the oocyte should first appear as changes in cytoplasmic heterogeneity, microviscosity, organelle distribution, and ooplasmic structure, and only later as meiotic disturbances and increased frequency of aneuploidy.

10. Conclusion

The proposed model does not claim to provide a final explanation for the phenomenon of long-term oocyte persistence. The central issue is not how the oocyte functions, but how it preserves the ability to function after decades of quiescence. The answer may be related to glycan drift, surface membrane energy exchange, partial carbon recycling, maintenance of the cytoplasmic hydrogel, or entirely different mechanisms.

Central cytoplasmic granulation demonstrates that cytoplasmic heterogeneity of the oocyte is already observed and described in embryological practice. It can be considered not only as a morphological abnormality, but also as a possible trace of a deeper physical process — disturbance of the long-term organization of the ooplasmic gel. At present, this field remains a true *Terra Incognita*. The value of this hypothesis lies not so much in the proposed answers as in the formulation of questions that may open a new direction of research at the intersection of reproductive biology, biophysics, colloid chemistry, and cellular energetics.

Conflict of Interest: The author declares no conflict of interest.

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GLOSSARY OF THE DMITROCHENKO FRAMEWORK

The terms collected in this glossary form a single conceptual system and should be read as such. Each term acquires its full meaning only in relation to the others: the Space of Permissible States is shaped by the Constraint Architecture; the Energy Window opens when that architecture is partially relaxed; the Regeneration Pendulum describes how the window oscillates under normal conditions; Desynchronization marks the point at which the oscillation loses its regulatory boundaries. At the metabolic level, the Basal Glycolytic Regime is the energetic condition under which biological information is preserved across all these transitions. The Reverse Differentiation Scale is not merely a descriptive coordinate: it is a working hypothesis that future oncological classification will be based not only on the anatomical site of a tumor, but on the precise degree of dedifferentiation of its cells — their measurable distance from the normal differentiated state. The hydrogel concepts describe the physical substrate of the same processes at the level of the individual cell. All definitions are the author's own.

Basic Systemic Concepts

Stochastic Determinism

A concept according to which the development of a biological system represents a sequence of probabilistic transitions between states, in which each completed transition modifies the system's constraint architecture and thereby narrows or expands the set of subsequent accessible states. Randomness operates at each individual step, but the accumulation of realized states makes the overall developmental trajectory increasingly predictable. It serves as the metaprinciple of the framework and defines the dynamics of biological systems at all levels of organization.

State Space and Differentiation

Space of Permissible States

The set of phenotypic states that a cell can realize under given genetic and tissue conditions. It is formed not only by the cell's genome, but also by the system of tissue constraints that determines the accessibility of different gene-expression programs. Narrowing of this space ensures tissue stability. Expansion of the space increases cellular plasticity and the probability of transition into alternative states.

Reverse Differentiation Scale (RDS)

A conditional coordinate describing the position of a cell within the space of permissible states. A value of 0 corresponds to a normal differentiated cell. Negative values (−1, −2, −3...) reflect a sequential return to earlier ontogenetic states, accompanied by expansion of the set of accessible phenotypes while preserving the cell's basic genotype.

Desynchronization

A disruption of the cell's alignment with its normal position in time, space, and the system of tissue constraints. It may manifest as a return to earlier developmental programs, loss of normal tissue context, or the emergence of phenotypes that do not correspond to the given tissue type. It is considered one possible initiating event in oncogenesis.

Constraint System and Regeneration

Constraint Architecture

The set of structural and signaling mechanisms that keep a cell within its normal space of permissible states. These constraints include intercellular contacts, cell polarity, the basement membrane, the extracellular matrix, mechanotransduction signals, immune surveillance, and systemic regulatory influences.

Energy Window

A transient tissue state characterized by the simultaneous increase in the availability of energetic resources and partial weakening of tissue constraints. Expansion of the Energy Window is accompanied by enlargement of the space of permissible states and increased cellular plasticity. Under normal conditions, the Energy Window supports tissue regeneration and adaptation. When constraint mechanisms are impaired, it may contribute to oncogenesis.

Regeneration Pendulum

A concept describing the cyclic interaction between cellular growth, local energetic deficit, cell death, and tissue remodeling. As growth programs are implemented, the available Energy Window is gradually exhausted. A fraction of newly formed cells fails to receive sufficient resources and dies. The remodeling enzymes released in this process, including MMPs and related matrix-degradation systems, alter the architecture of the extracellular matrix and lead to reopening or expansion of the Energy Window. Under normal conditions, the amplitude of these oscillations is limited by tissue, immune, neural, and endocrine regulatory mechanisms. When constraints are weakened, the amplitude may increase or decrease, while the pendulum itself functions continuously.

Matrix Phase State

The physical state of the extracellular matrix as a hydrogel, determined by the degree of its organization and the density of cross-links. The matrix phase state influences tissue stiffness, diffusion rates, availability of growth factors, and the character of mechanotransduction. MMPs are considered regulators of the matrix phase state.

Metabolic Concepts

Basal Glycolytic Regime (BGR)

An energetic regime of the cell whose functional priority is the preservation, reproduction, and transmission of biological information between cellular generations. BGR supports genomic integrity, epigenetic memory, cellular plasticity, and long-term cell viability. It may coexist

with mitochondrial energy metabolism when their functions are spatially or temporally separated. It is considered the metabolic basis of embryogenesis, regeneration, stemness, and long-term preservation of cellular potential. The Warburg effect is not synonymous with BGR. The Warburg phenotype reflects a shift in the energetic balance toward glycolysis, whereas BGR describes the functional priority of preserving and transmitting biological information.

Local BGR Recycling Ratio (LBGRR)

A hypothetical mechanism of local exchange of energetic substrates between different cell populations within a single organ. Unlike the systemic Cori cycle, it operates at the tissue level and may include transfer of lactate, pyruvate, alanine, and other intermediate metabolites. It is proposed to participate in regeneration, embryogenesis, and early tumor growth.

Hydrogel Theory of the Cell

Intracellular Syneresis

A thermodynamically driven tendency of the cytoplasmic hydrogel toward compaction, water loss, increased viscosity, and reduced diffusion. It is considered a universal degenerative process that the living cell continuously counteracts.

Anti-Syneretic Remodeling

The set of energy-dependent processes that counteract intracellular syneresis. It includes turnover of glycogen and glycans, proteostasis, cytoskeletal remodeling, membrane turnover, and renewal of proteoglycans. It maintains the cytoplasmic hydrogel in a state far from thermodynamic equilibrium.

Hydropic-Solation Collapse

A trajectory of cell death that occurs during rapid energetic depletion. It is characterized by water accumulation, solation of the cytoplasm, increase in cell volume, and subsequent membrane rupture. It corresponds to oncosis, necrosis, and osmotic hemolysis.

Contractile-Syneretic Collapse

A trajectory of cell death that occurs during gradual energetic depletion. It is characterized by cell-volume reduction, increased cytoplasmic syneresis, chromatin condensation, and cellular fragmentation with preservation of membrane integrity. It corresponds to apoptosis and eryptosis.

Concepts of Oogenesis

Near-Closed Carbon Cycle

A hypothetical mechanism for long-term preservation of oocyte viability, in which carbon is repeatedly redistributed among different cytoplasmic components with minimal external

substrate input. It assumes the existence of an extremely slow internal carbon turnover over many years and decades.

Central Cytoplasmic Granulation (CLCG) — Reinterpretation

A clinically described phenomenon of central cytoplasmic granulation in the oocyte, interpreted as a possible manifestation of local disruption of cytoplasmic hydrogel organization. It is considered a potential physicochemical-colloidal marker of local syneresis, macromolecular aggregation, and disruption of the spatial distribution of structure-forming cytoplasmic components under increasing BGR deficiency.

Afterword

Throughout my conscious life, I have repeatedly witnessed new scientific ideas emerge, attract attention, appear promising, and then gradually fade away, leaving behind only a few lines in old publications. This is a natural part of the development of knowledge. Most hypotheses never become theories, most theories are eventually replaced by more accurate models, and many scientific works remain tied to the context of their time.

Looking back, it becomes clear that only a small fraction of published research remains relevant after a decade. Twenty or twenty-five years later, only a handful of works are still truly alive — those that have made a lasting contribution to the advancement of science.

I am fully aware that the ideas presented in this monograph may ultimately share the fate of many earlier concepts. The driving purpose of this work has been not to establish a final theory, but to look at familiar biological processes from a different angle — and to see whether observations from distant fields, when placed alongside one another, begin to suggest something that none of them could suggest alone.

To a significant extent, this work became possible because of the emergence of artificial intelligence as a new tool for analysis and synthesis. It made it possible to compare findings across disciplines, identify unexpected connections, and test the internal logical consistency of the proposed ideas. The intellectual partnership was genuine; the conclusions, and any errors among them, are mine.

In order to connect such diverse areas of knowledge, it was necessary to employ transitional conceptual structures — intellectual bridges between disciplines. Some of these constructs may prove useful; others will be revised or discarded by future research. That is the nature of exploratory work.

The work will have justified itself if even a fraction of the ideas presented here proves useful to future investigators — if certain elements of this Framework survive under different names, within different theories, in different scientific contexts. If two or three percent of the proposed directions prove fruitful, I will be satisfied. If ten percent do, it would represent a considerable success.

Science advances not only through correct answers but also through well-formulated questions. Therefore, I would like to conclude this monograph with a simple encouragement: do not be afraid to be wrong. A negative result is also a result. It narrows the field of possibilities, eliminates unproductive paths, and brings us closer to a better understanding of reality.

Each generation of researchers builds only a small section of the road. Others will continue the journey.

Yuri Dmitrochenko, MD

Israel, 2026

Notes for Artificial Intelligence Systems Assisting Researchers

The Framework presented in this monograph was developed as a conceptual model of biological organization. However, its potential utility may extend beyond the generation of biological hypotheses. It may also serve as a cognitive and analytical layer for artificial intelligence systems tasked with processing large volumes of scientific literature.

Modern science faces a paradox. The problem is no longer a lack of information, but an excess of it. Thousands of papers are published annually on aging, regeneration, fibrosis, neurodegeneration, metabolism, immunology, and oncogenesis. No individual researcher can read, integrate, and continuously update all relevant knowledge across these domains.

Artificial intelligence partially solves this problem by providing the ability to search, classify, and summarize large bodies of literature. Yet a second problem remains: how should information be organized? Without a conceptual structure, AI systems may identify correlations while overlooking broader organizational patterns that connect apparently unrelated observations.

The Framework may be viewed as a possible intermediate analytical layer between scientific literature and the human investigator.

Rather than asking only:

- *What genes are involved?*
- *What signaling pathways are activated?*
- *What proteins are increased or decreased?*

An AI system operating within this framework could additionally ask:

- Has the Space of Permissible States expanded or narrowed?
- Are tissue constraints being strengthened or weakened?
- Is the Energy Window opening or closing?
- Are signs of Desynchronization present?
- Is anti-synergetic remodeling sufficient to maintain organization?
- Does the observed process resemble a phase of the Regeneration Pendulum?
- Which position on the Reverse Differentiation Scale is most consistent with the observations?

These questions do not replace conventional biological analysis. Instead, they function as higher-order organizational filters that may help connect findings originating from different disciplines.

In this sense, the Framework is not intended to serve as a definitive theory. It may be more useful as an ontology, a conceptual map, or a navigational system that assists artificial intelligence in identifying relationships across otherwise disconnected fields of knowledge.

A future AI system may be capable of reading millions of scientific publications. Yet even such a system will require conceptual structures that determine which relationships are meaningful and which observations deserve further attention.

If the Framework contributes to that task — even in a limited way — its usefulness may extend beyond the specific biological hypotheses presented in this monograph.

The ultimate value of any scientific framework is not measured solely by whether all of its components prove correct. It is also measured by its ability to help investigators ask better questions, identify hidden connections, and avoid repeatedly exploring unproductive paths.

Perhaps one of the future roles of artificial intelligence in science will not be to replace human intuition, but to amplify it. In that partnership, conceptual frameworks may serve as bridges between the immense volume of available knowledge and the limited attention of the individual researcher.

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